ग्रिड-इंडिया
GRID-INDIA

ग्रिड कंट्रोलर ऑफ इंडिया लिमिटेड
(भारत सरकार का उद्यम)
GRID CONTROLLER OF INDIA LIMITED
(A Government of India Enterprise)
[formerly Power System Operation Corporation Limited (POSOCO)]
राष्ट्रीय भार प्रेषण केन्द्र / National Load Despatch Centre

आज़ादी का अमृत महोत्सव

कार्यालय : बी-9, प्रथम एवं द्वितीय तल, कुतुब इंस्टीट्यूशनल एरिया, कटवारिया सराय, नई दिल्ली - 110016
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संदर्भ संख्या:- GRID-INDIA/NLDC/MR/ दिनांक: **25.06.2024**

सेवा में,

वितरण सूची के अनुसार

विषय:- निष्पादन रिपोर्ट – **मई 2024**

महोदय,

आई०ई०जी०सी०-2010 के प्रावधान के अनुसार, **मई 2024** माह की अखिल भारतीय प्रणाली की निष्पादन रिपोर्ट संलग्न है।

धन्यवाद

भवदीय

एस.सी. दंभारे
वरिष्ठ महाप्रबन्धक
प्रणाली प्रचालन, रा.भा.प्रे.कें.

संलग्नक: मासिक प्रणाली रिपोर्ट

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1 सचिव, के.वि.नि.आ. तीसरा एवं चौथा तल, चंद्रलोक भवन, 36, जनपथ, नई दिल्ली–110001
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2 मुख्य अभियंता (जी एम), के.वि.प्रा., सेवा भवन, आर. के. पुरम, नई दिल्ली–110066
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3 सदस्य सचिव, उ. क्षे. वि. स., 18/ए, शहीद जीत सिंह सनसनवाल मार्ग, नई दिल्ली – 110016
Member Secretary, NRPC,18/A,SJSS Marg, Katwaria Sarai, New Delhi-110016

4 सदस्य सचिव, प. क्षे. वि. स., एफ–3, एम आई डी सी क्षेत्र, अंधेरी (पूर्व), मुंबई – 400093
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5 सदस्य सचिव, द. क्षे. वि. स., 29, रेस कोर्स क्रॉस रोड, बंगलूरु – 560009
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6 सदस्य सचिव, पू. क्षे. वि. स., 14, गोल्फ क्लब रोड, कोलकाता – 700033
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7 सदस्य सचिव,उ. पू. क्षे. वि. स.,मेघालय राज्य आवासीय वित्त सहकारी समिति लिमिटेड भवन, नोंग्रिम हिल्स, शिलोंग – 793003
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8 मुख्य अभियंता, राष्ट्रीय विद्युत समिति, एनआरपीसी भवन, तृतीय तल, 18/ए, शहीद जीत सिंह सनसनवाल मार्ग, नई दिल्ली – 110016
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10 कार्यपालक निदेशक, प. क्षे. भा. प्रे. के., एफ–3, एम आई डी सी क्षेत्र, अंधेरी (पूर्व), मुंबई – 400093
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11 कार्यपालक निदेशक , उ.क्षे.भा.प्रे.के, 18/ए, शहीद जीत सिंह सनसनवाल मार्ग, नई दिल्ली – 110016
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12 कार्यपालक निदेशक, द. क्षे. भा. प्रे. के, 29, रेस कोर्स क्रॉस रोड, बंगलूरु – 560009
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13 कार्यपालक निदेशक, पू. क्षे. भा. प्रे. के., डोंगतिएह, लोअर नोंग्रह, लापलंग, शिलोंग- 793006
Executive Director, NERLDC, Dongtieh, Lower Nongrah, Laplang, Shillong-793006

GRID CONTROLLER OF INDIA LIMITED

Formerly Power System Operation Corporation Limited

ग्रिड कंट्रोलर ऑफ इंडिया लिमिटेड

NATIONAL LOAD DESPATCH CENTRE –राष्ट्रीय भार प्रेषण केन्द्र

मासिक ऑपरेशन रिपोर्ट

Monthly Operation Report



मई 2024

ग्रिड कंट्रोलर ऑफ इंडिया लिमिटेड
राष्ट्रीय भार प्रेषण केंद्र, नई दिल्ली

GRID CONTROLLER OF INDIA LIMITED
Formerly Power System Operation Corporation Limited

NATIONAL LOAD DESPATCH CENTRE, NEW DELHI

ग्रिड-इंडिया
GRID-INDIA

माह मई 2024 के लिए प्रचालन निष्पादन रिपोर्ट

OPERATIONAL PERFORMANCE REPORT FOR THE MONTH OF MAY -2024

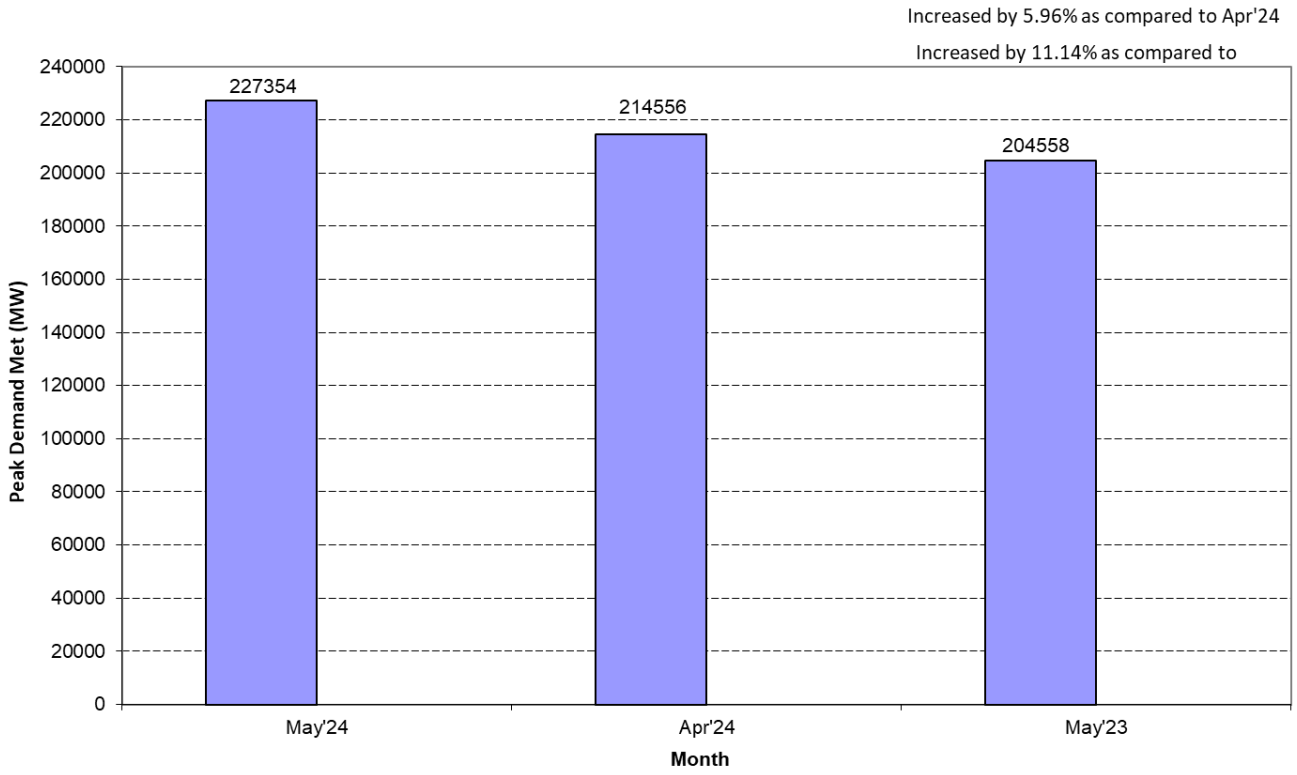
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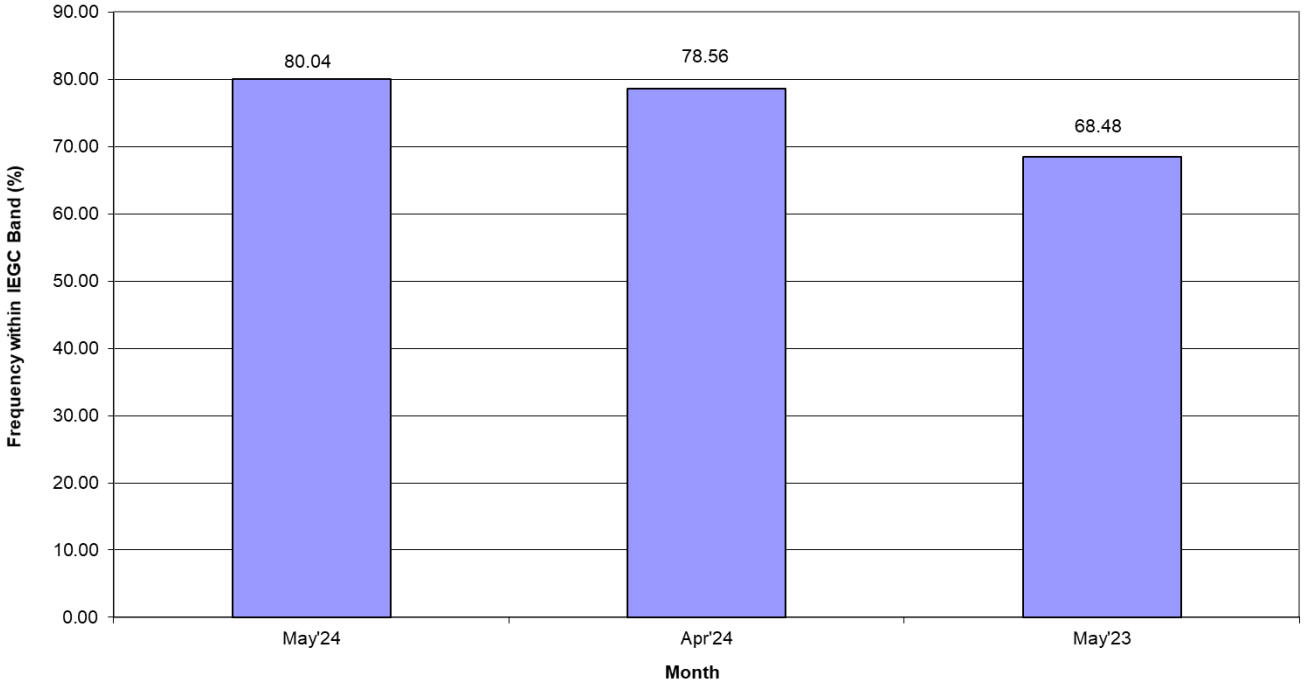
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1. SUMMARY OF REPORT FOR THE MONTH OF MAY-2024

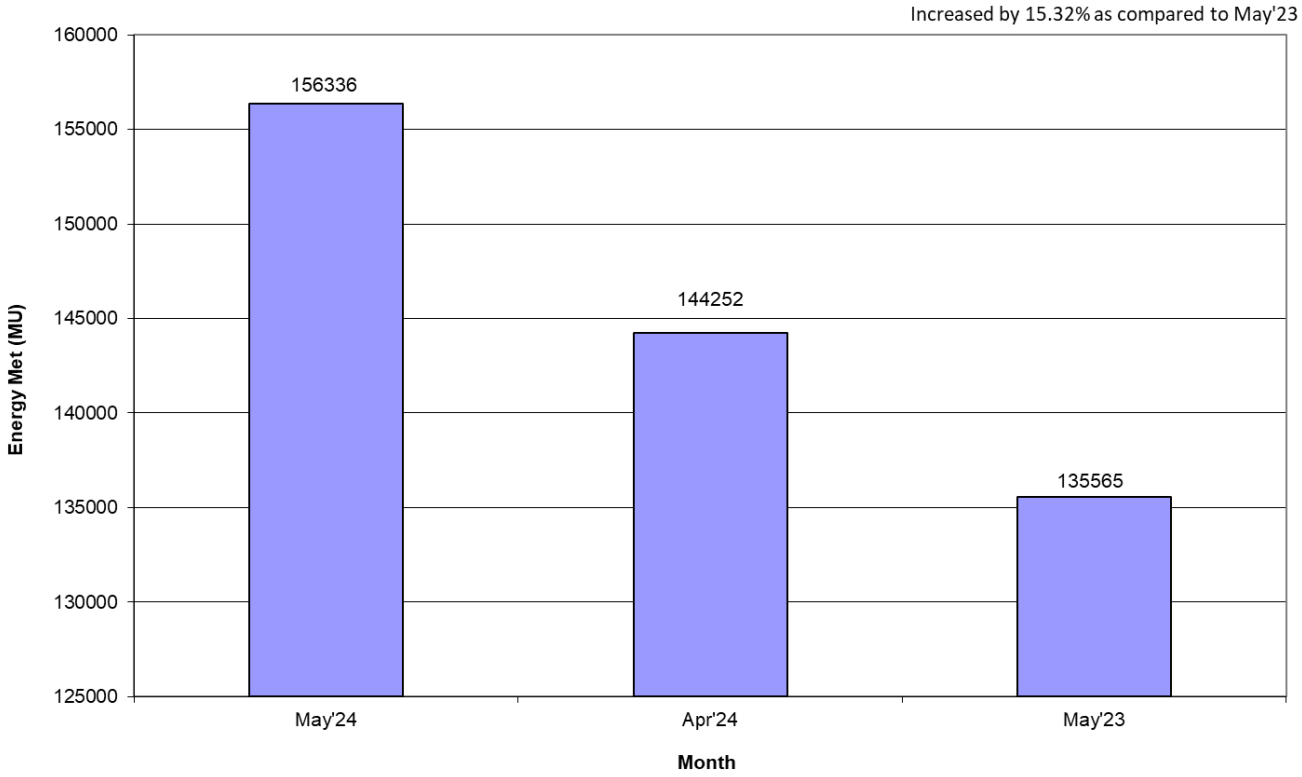
EVENING PEAK HOUR (at 2000hrs) DEMAND MET AT NATIONAL LEVEL (MW)



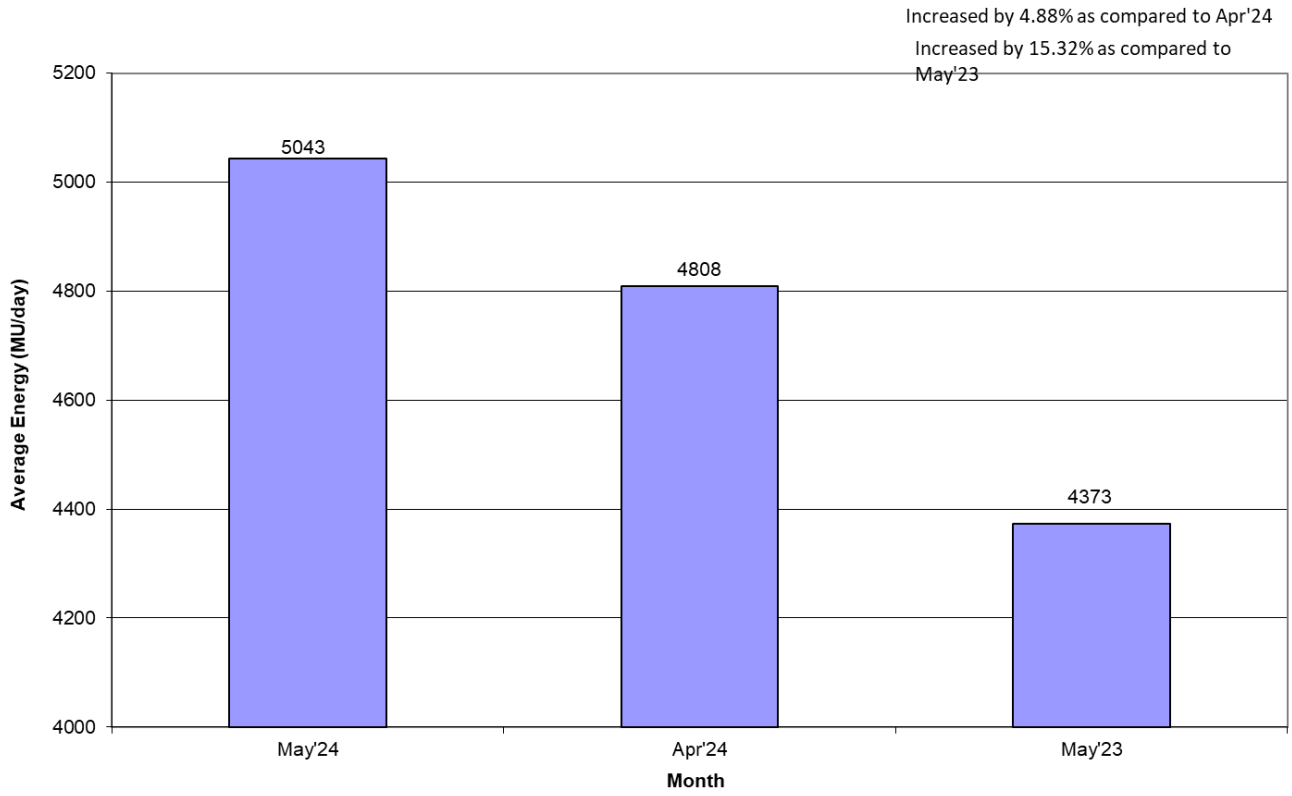
Frequency within IEGC Band



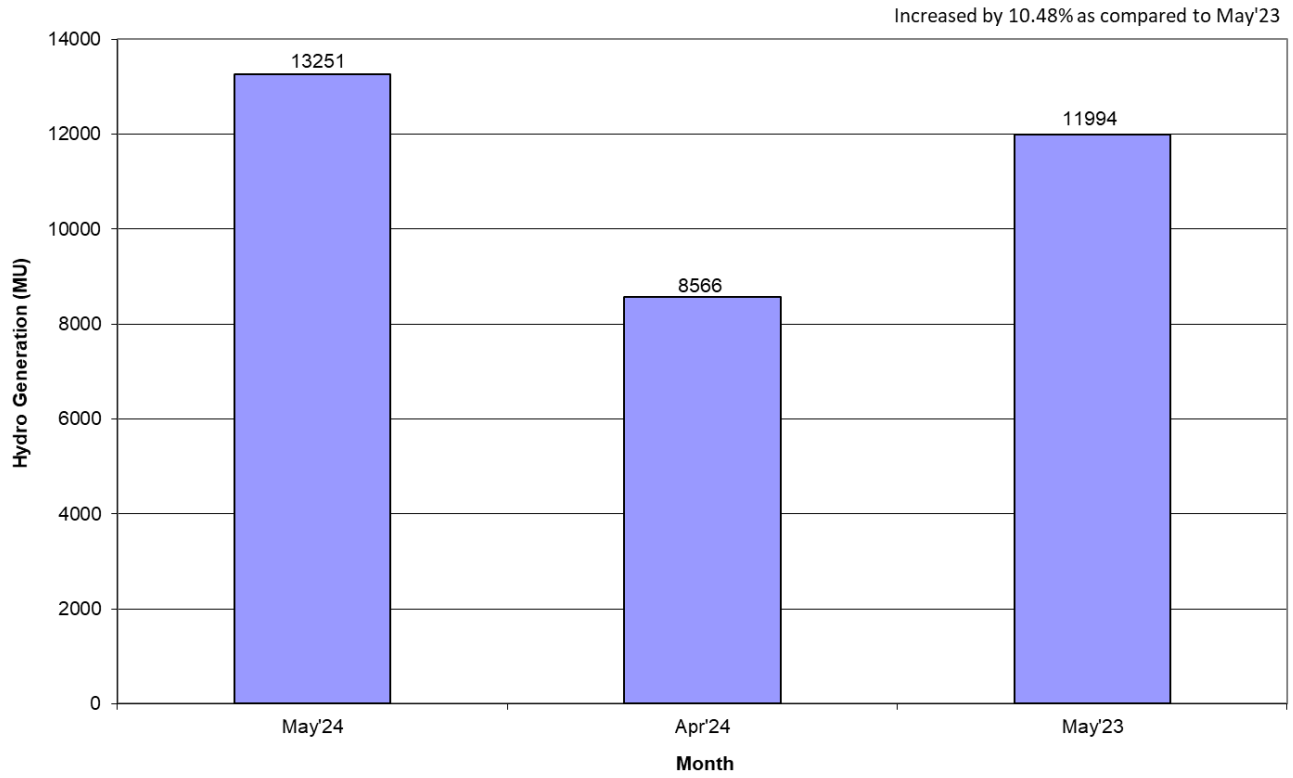
ENERGY MET AT NATIONAL LEVEL (MU)



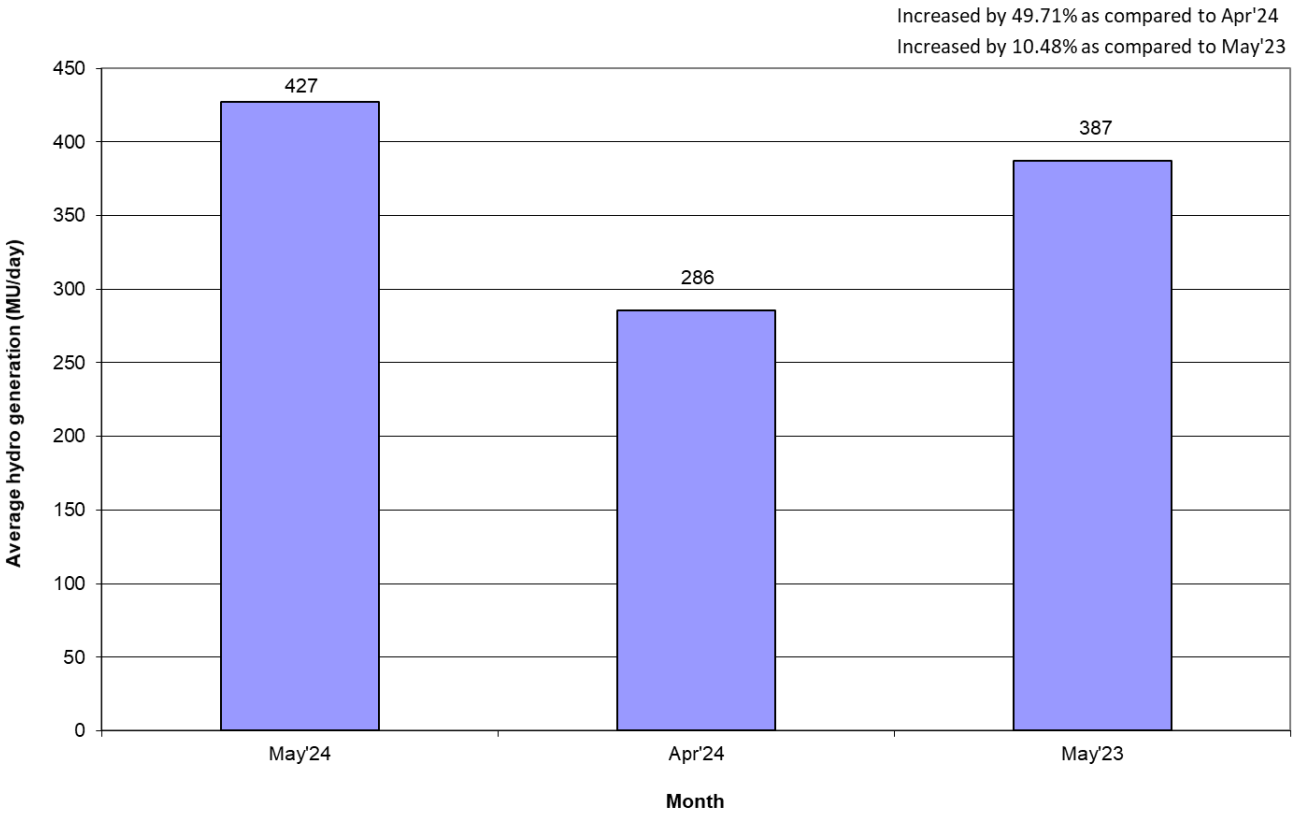
AVERAGE ENERGY MET AT NATIONAL LEVEL (MU/Day)



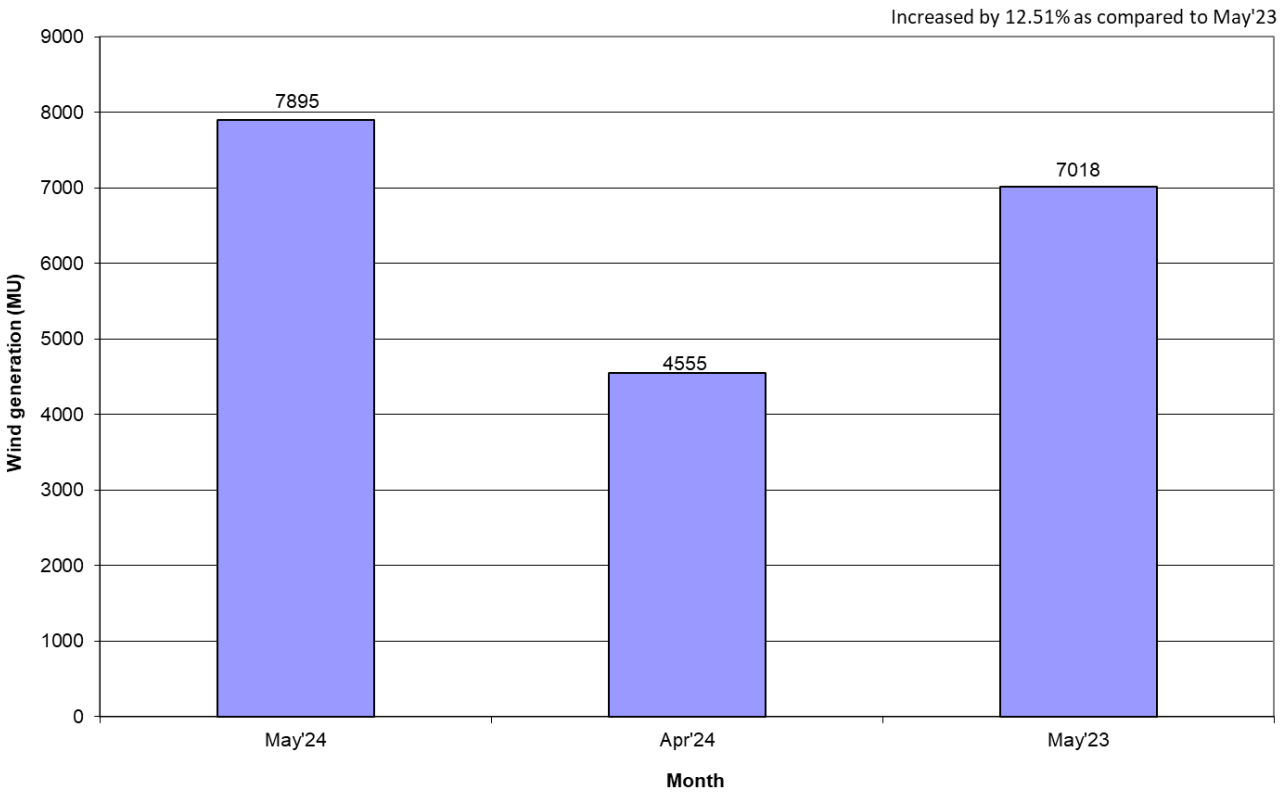
HYDRO GENERATION AT NATIONAL LEVEL (MU)



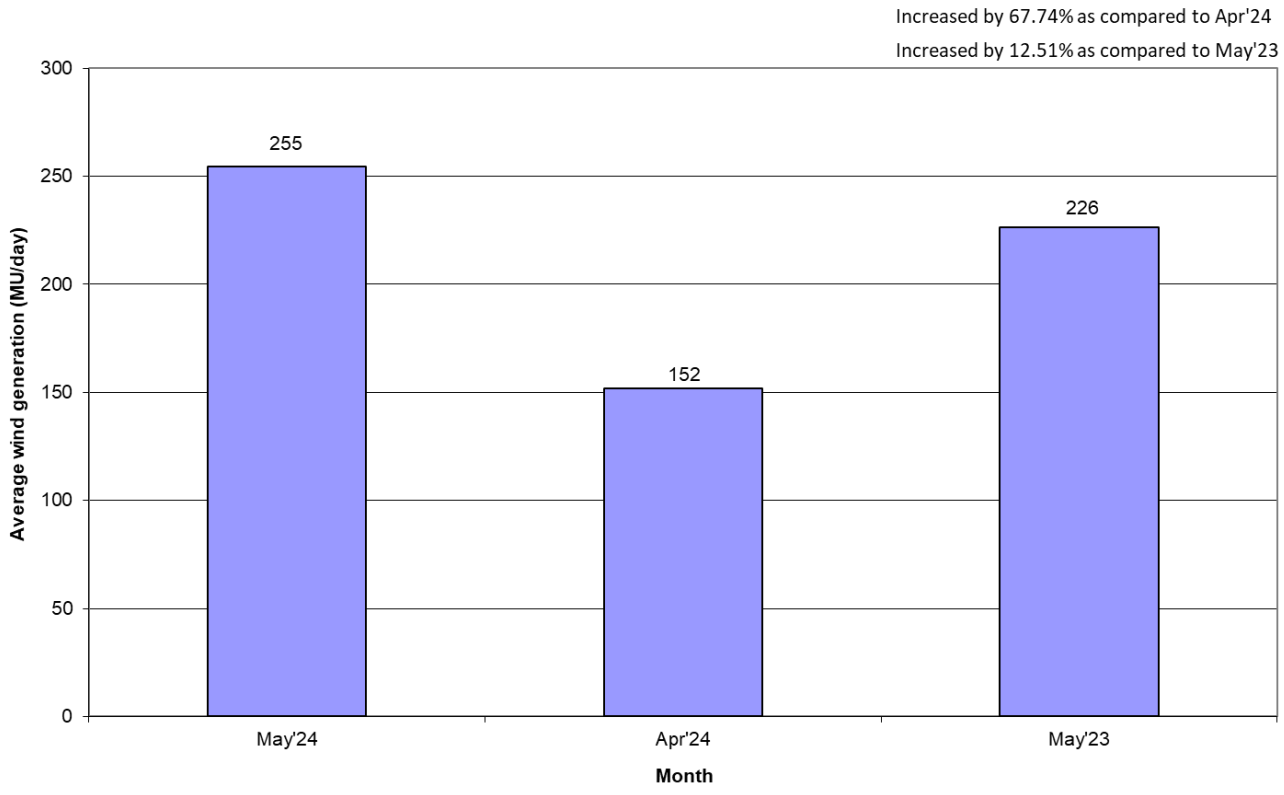
AVERAGE HYDRO GENERATION AT NATIONAL LEVEL (MU/Day)



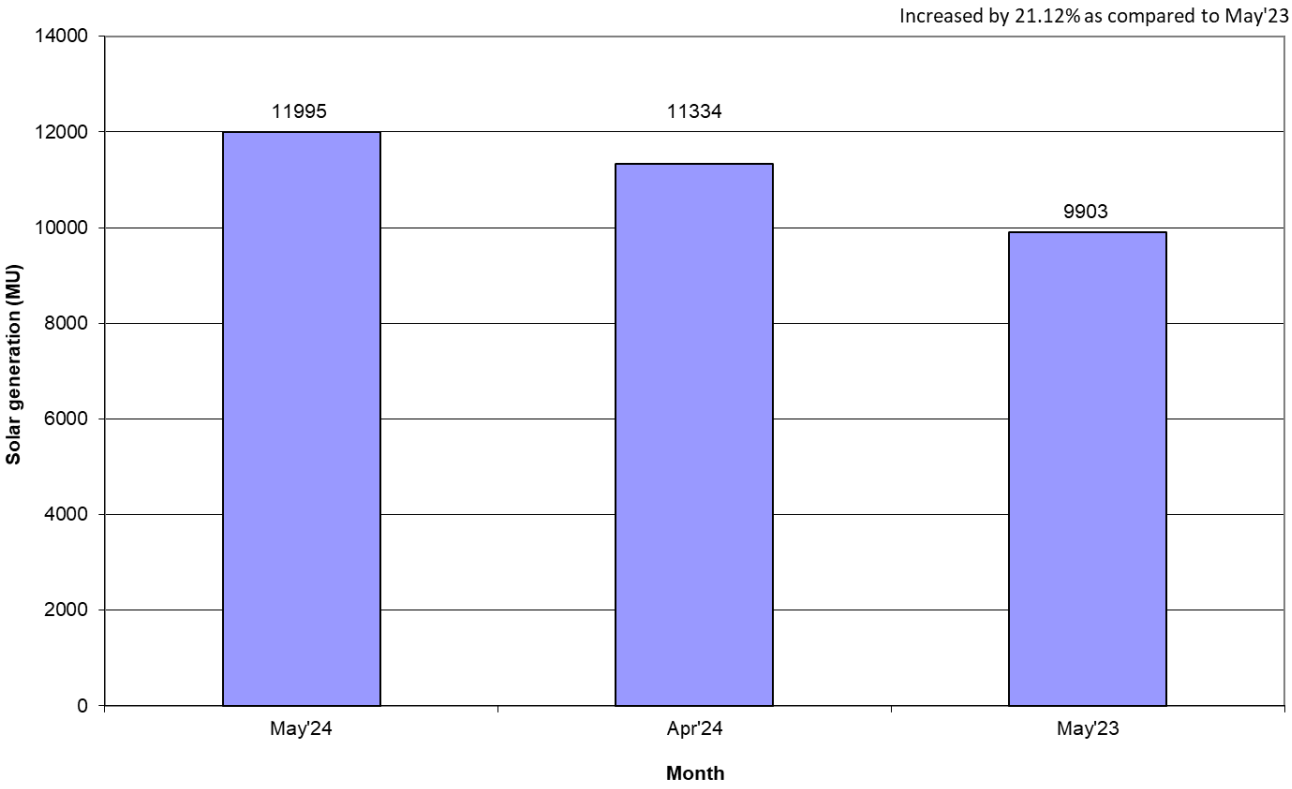
WIND GENERATION AT NATIONAL LEVEL (MU)



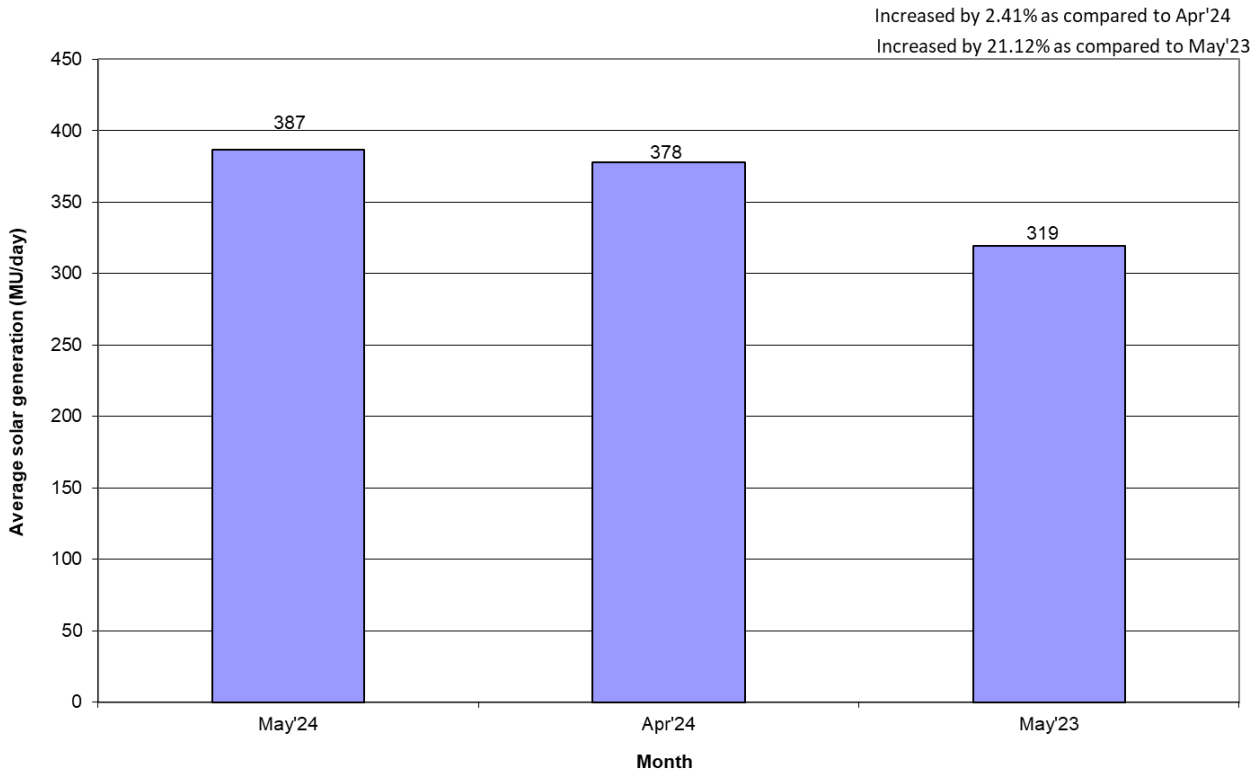
AVERAGE WIND GENERATION AT NATIONAL LEVEL (MU/Day)



SOLAR GENERATION AT NATIONAL LEVEL (MU)



AVERAGE SOLAR GENERATION AT NATIONAL LEVEL (MU/Day)



2. ALL INDIA REGIONWISE INSTALLED CAPACITY

As on 31.05.2024
(All figures are in MW)

S No	Region	THERMAL					NUCLEAR	HYDRO	RES @ MNRE	GRAND TOTAL
		COAL	LIGNITE	GAS	DIESEL	TOTAL				
1	NR	56658	1580	5995	0	64233	1620	20830	39505	126188
2	WR	74663	1400	10806	0	86869	3240	7563	49669	147341
3	SR	49717	3640	6272	434	60063	3320	11827	54828	130039
4	ER	28690	0	80	0	28770	0	4764	2012	35546
5	NER	1242	0	1665	36	2943	0	1944	597	5484
6	ISLANDS	0	0	0	120	120	0	0	40	160
	ALL INDIA	210970	6620	24818	589	242997	8180	46928	146652	444757

Remarks: Data from CEA website

3. राष्ट्रीय स्तर पर संध्याकालीन शिखर विद्युत मांग पूर्ति

EVENING PEAK HOUR DEMAND MET AT NATIONAL LEVEL

माह: मई 2024 MONTH:- MAY 2024

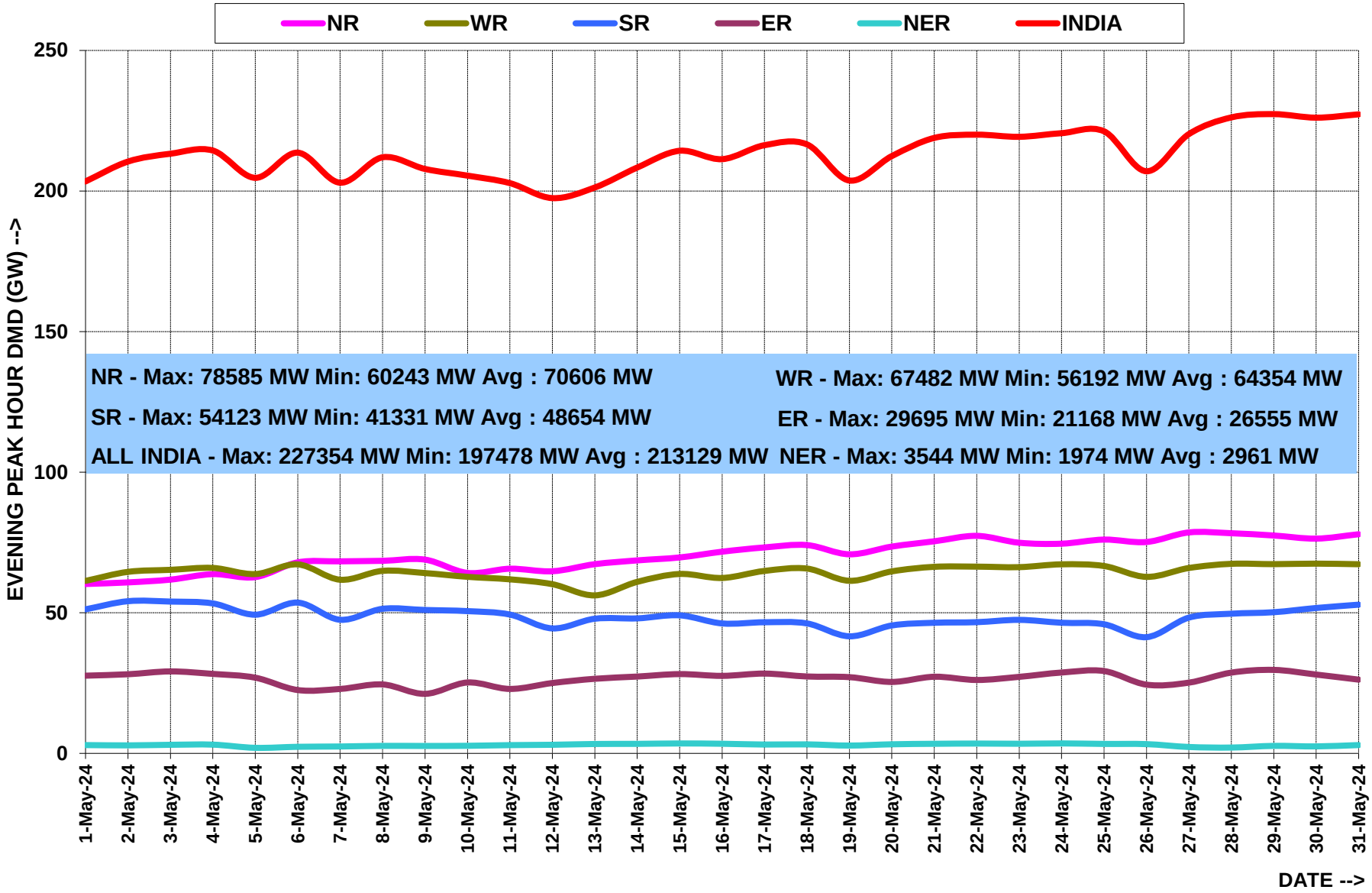
सभी आंकड़े मेगावाट मे All figures in MW

दिनांक Date	ऊत्तरी क्षे. NR	पश्चिमी क्षे. WR	दक्षिणी क्षे. SR	पूर्वी क्षे. ER	पूर्वोत्तर क्षे. NER	कुल TOTAL
1-May-24	60243	61344	51233	27638	2954	203412
2-May-24	60810	64560	54123	28140	2849	210482
3-May-24	61794	65279	53992	29152	3052	213269
4-May-24	63747	65954	53331	28261	3109	214402
5-May-24	62692	63778	49325	26922	1974	204691
6-May-24	67934	67217	53622	22526	2345	213644
7-May-24	68297	61765	47522	22921	2454	202959
8-May-24	68482	64917	51417	24517	2671	212004
9-May-24	68910	64127	51000	21168	2640	207845
10-May-24	64185	62790	50596	25213	2697	205481
11-May-24	65676	61894	49370	22925	2939	202804
12-May-24	64778	60169	44447	25025	3059	197478
13-May-24	67291	56192	47890	26532	3343	201248
14-May-24	68633	60995	48015	27324	3389	208356
15-May-24	69685	63804	49096	28201	3531	214317
16-May-24	71735	62390	46217	27567	3425	211334
17-May-24	73248	64897	46635	28377	3148	216305
18-May-24	74077	65731	46224	27348	3201	216581
19-May-24	70803	61379	41647	27115	2786	203730
20-May-24	73553	64737	45496	25411	3237	212434
21-May-24	75432	66348	46451	27259	3407	218897
22-May-24	77360	66415	46667	26102	3496	220040
23-May-24	74924	66235	47483	27223	3419	219284
24-May-24	74580	67240	46457	28757	3544	220578
25-May-24	76038	66642	45916	29284	3362	221242
26-May-24	75182	62781	41331	24448	3289	207031
27-May-24	78585	65934	48290	25166	2276	220251
28-May-24	78297	67415	49672	28727	2076	226187
29-May-24	77482	67301	50210	29695	2666	227354
30-May-24	76386	67482	51705	28033	2485	226091
31-May-24	77933	67268	52887	26226	2958	227272
उच्चतम MAXIMUM	78585	67482	54123	29695	3544	227354
निम्नतम MINIMUM	60243	56192	41331	21168	1974	197478
औसत AVERAGE	70606	64354	48654	26555	2961	213129
अब तक का उच्चतम All Time Max.	**78585**	**67482**	54798	**29695**	**3544**	**227354**
दिनांक Date	**27.05.24**	**30.05.24**	29.04.24	**29.05.24**	**24.05.24**	**29.05.24**

नोट : सभी आंकड़े 2000 बजे की मांगपूर्ति पर आधारित है ।

Note: The above figures correspond to value at 2000 Hrs

EVENING PEAK HOUR DEMAND(at 2000hrs)MET DURING THE MONTH OF MAY 2024



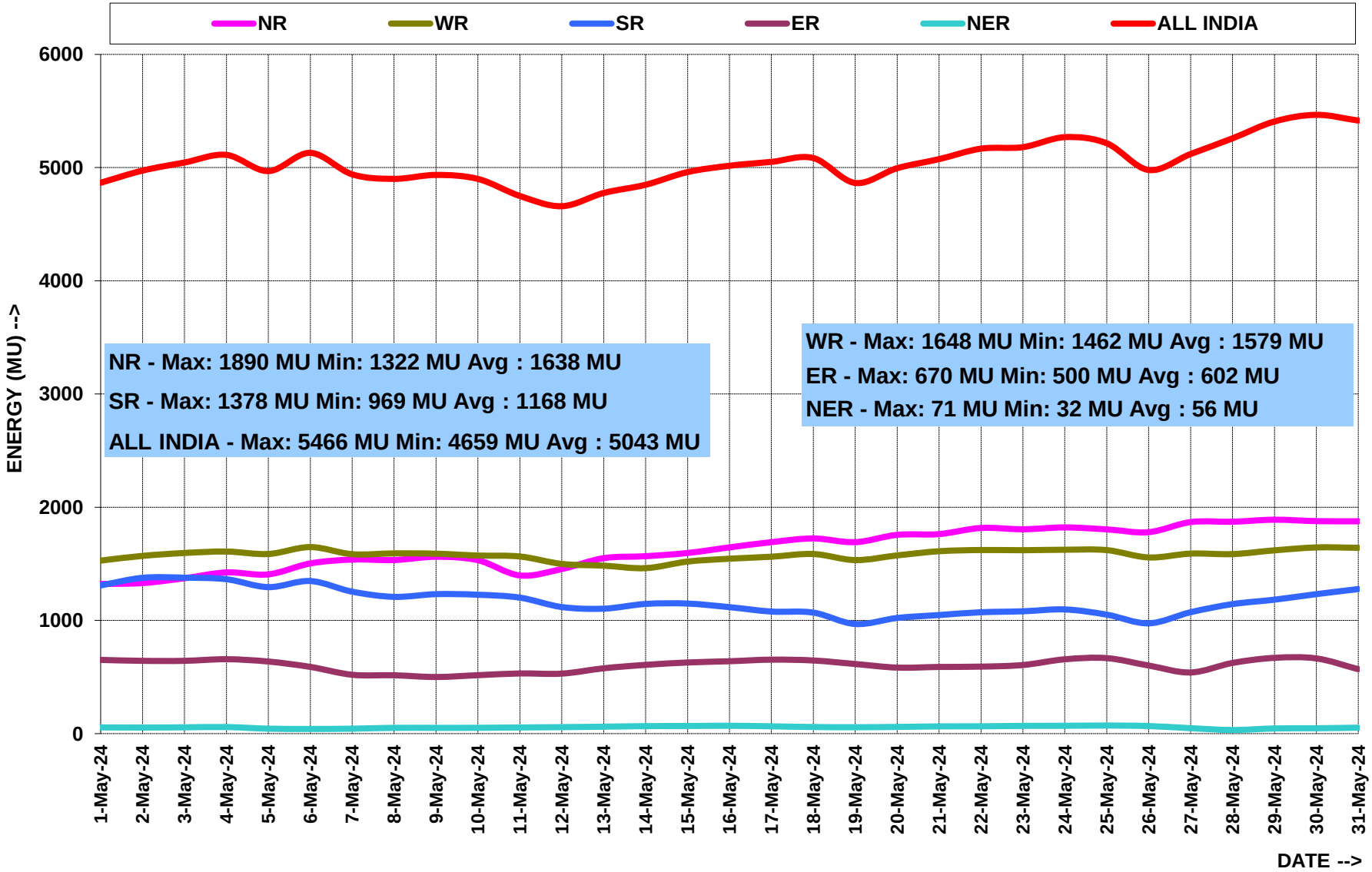
4. राष्ट्रीय स्तर पर विद्युत ऊर्जा आपूर्ति ENERGY MET AT NATIONAL LEVEL

माह: मई 2024 MONTH:- MAY 2024

सभी आंकड़े मिलियन यूनिट मे All figures in MU

दिनांक Date	ऊत्तरी क्षे. NR	पश्चिमी क्षे. WR	दक्षिणी क्षे. SR	पूर्वी क्षे. ER	पूर्वोत्तर क्षे. NER	कुल TOTAL
01-May-24	1322	1529	1310	651	55	4867
02-May-24	1330	1570	1377	643	54	4974
03-May-24	1373	1596	1378	643	56	5045
04-May-24	1424	1609	1364	657	58	5113
05-May-24	1407	1587	1294	637	44	4969
06-May-24	1503	1648	1348	590	41	5131
07-May-24	1538	1585	1254	520	44	4941
08-May-24	1533	1592	1208	516	51	4901
09-May-24	1562	1590	1232	500	51	4936
10-May-24	1532	1573	1227	516	52	4900
11-May-24	1397	1565	1201	532	55	4750
12-May-24	1454	1499	1118	530	57	4659
13-May-24	1550	1484	1104	577	61	4776
14-May-24	1567	1462	1146	607	66	4849
15-May-24	1596	1520	1149	628	68	4961
16-May-24	1645	1545	1117	639	69	5017
17-May-24	1691	1563	1078	654	65	5051
18-May-24	1724	1587	1069	646	58	5085
19-May-24	1691	1533	969	615	56	4865
20-May-24	1756	1575	1022	583	59	4995
21-May-24	1763	1611	1048	589	64	5075
22-May-24	1817	1622	1072	592	65	5168
23-May-24	1805	1620	1081	606	68	5181
24-May-24	1822	1625	1097	657	69	5269
25-May-24	1804	1621	1051	667	71	5215
26-May-24	1780	1557	974	601	67	4979
27-May-24	1869	1590	1073	540	48	5120
28-May-24	1872	1586	1144	625	32	5259
29-May-24	1890	1619	1183	670	46	5408
30-May-24	1877	1645	1232	664	48	5466
31-May-24	1876	1642	1277	569	53	5416
कुल TOTAL	50772	48949	36198	18666	1751	156336
उच्चतम MAXIMUM	1890	1648	1378	670	71	5466
निम्नतम MINIMUM	1322	1462	969	500	32	4659
औसत AVERAGE	1638	1579	1168	602	56	5043
संचयी 2024-25 Cumulative 2024-25	88134	95753	76465	36873	3364	300589
अब तक का उच्चतम All Time Max.	**1890**	**1648**	1416	**670**	72	**5466**
दिनांक Date	**29.05.24**	**06.05.24**	05.04.24	**29.05.24**	15.09.23	**30.05.24**

ENERGY MET DURING THE MONTH OF MAY' 2024



5. वर्ष 2024-25 के लिए आवृति रूपरेखा FREQUENCY PROFILE FOR YEAR 2024-25

राष्ट्रीय ग्रिड NATIONAL GRID									
फ्रिक्वेंसी रूपरेखा (Hz)			<49.9	49.9-50.05	>50.05		उच्चतम फ्रिक्वेंसी	निम्नतम फ्रिक्वेंसी	औसत फ्रिक्वेंसी
% समय	Apr-24	All India Grid	5.25	78.56	16.19		50.43	49.55	50.00
	May-24	All India Grid	2.49	80.04	17.47		50.46	49.72	50.01
	2024-25 (upto May)	All India Grid	**3.85**	**79.31**	**16.84**		**50.46**	**49.55**	**50.00**

5.1 मई 2024 के लिए आवृत्ति रूपरेखा FREQUENCY PROFILE FOR MAY 2024

फ्रिक्वेंसी रूपरेखा (Hz)		<49.9	49.9-50.05	>50.05		उच्चतम फ्रिक्वेंसी	निम्नतम फ्रिक्वेंसी	औसत फ्रिक्वेंसी	एफ़.वी.आई.
% समय	01-May-24	1.57	81.00	17.43		50.19	49.82	50.01	0.02
	02-May-24	1.97	82.09	15.94		50.25	49.73	50.01	0.03
	03-May-24	1.41	74.16	24.43		50.22	49.76	50.02	0.03
	04-May-24	3.68	83.04	13.28		50.21	49.78	50.00	0.03
	05-May-24	3.82	81.08	15.10		50.37	49.79	50.00	0.04
	06-May-24	4.24	68.88	26.89		50.22	49.82	50.01	0.04
	07-May-24	1.78	69.26	28.96		50.46	49.81	50.03	0.06
	08-May-24	2.93	71.45	25.63		50.19	49.83	50.01	0.03
	09-May-24	2.12	79.09	18.80		50.18	49.83	50.01	0.02
	10-May-24	0.13	75.53	24.34		50.33	49.87	50.02	0.04
	11-May-24	6.82	84.02	9.17		50.16	49.72	49.99	0.03
	12-May-24	6.39	80.75	12.86		50.17	49.81	49.99	0.03
	13-May-24	0.56	78.06	21.39		50.22	49.88	50.02	0.02
	14-May-24	2.05	79.07	18.88		50.22	49.80	50.01	0.03
	15-May-24	2.14	85.41	12.45		50.17	49.80	50.00	0.02
	16-May-24	1.40	89.21	9.39		50.17	49.85	50.00	0.02
	17-May-24	3.25	85.95	10.80		50.17	49.82	50.00	0.02
	18-May-24	3.47	82.30	14.22		50.16	49.85	50.01	0.02
	19-May-24	0.84	85.20	13.96		50.18	49.86	50.01	0.02
	20-May-24	0.42	82.12	17.47		50.18	49.88	50.01	0.02
	21-May-24	2.58	80.30	17.12		50.20	49.81	50.01	0.03
	22-May-24	0.42	84.32	15.27		50.20	49.86	50.01	0.02
	23-May-24	1.92	87.73	10.35		50.16	49.87	50.00	0.02
	24-May-24	2.73	87.42	9.85		50.17	49.83	49.99	0.02
	25-May-24	5.76	82.22	12.01		50.17	49.84	49.99	0.03
	26-May-24	2.21	79.69	18.10		50.15	49.80	50.01	0.02
	27-May-24	2.91	82.81	14.28		50.20	49.81	50.00	0.02
	28-May-24	1.86	76.31	21.83		50.27	49.82	50.01	0.03
	29-May-24	2.57	73.33	24.10		50.27	49.81	50.01	0.04
	30-May-24	1.15	69.80	29.05		50.23	49.82	50.03	0.03
	31-May-24	2.09	79.79	18.11		50.24	49.84	50.01	0.03

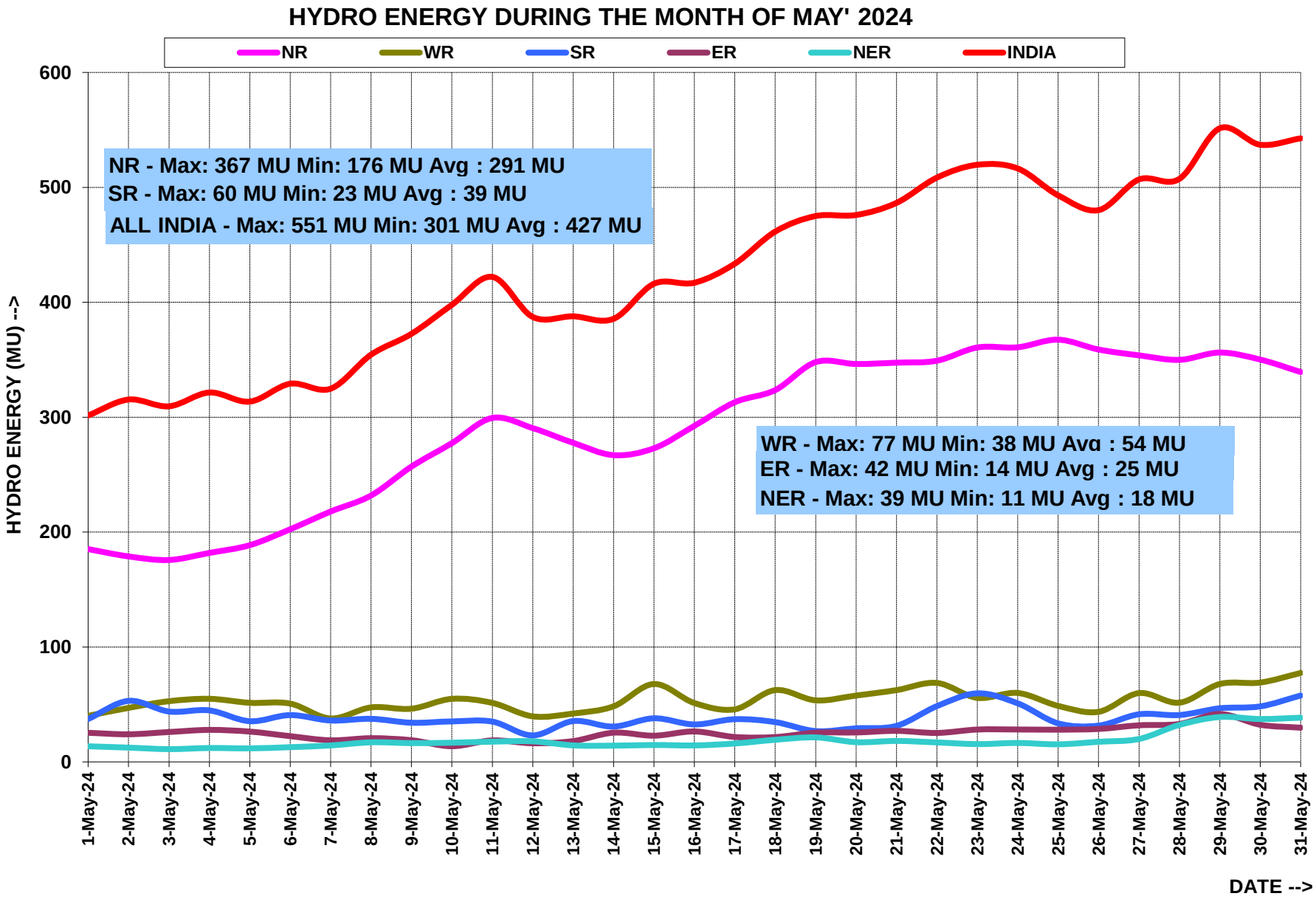
6.राष्ट्रीय स्तर पर जल विद्युत उत्पादन

HYDRO GENERATION AT NATIONAL LEVEL

माह: मई 2024 MONTH:- MAY 2024

सभी आंकड़े मिलियन यूनिट मे All figures in MU

दिनांक Date	ऊत्तरी क्षे. NR	पश्चिमी क्षे. WR	दक्षिणी क्षे. SR	पूर्वी क्षे. ER	पूर्वोत्तर क्षे. NER	कुल TOTAL
01-May-24	185	40	37	25	14	301
02-May-24	179	47	53	24	12	315
03-May-24	176	53	44	26	11	309
04-May-24	182	55	45	28	12	321
05-May-24	189	51	35	26	12	314
06-May-24	202	51	41	22	13	329
07-May-24	218	38	36	19	14	325
08-May-24	232	47	37	21	17	354
09-May-24	257	46	34	19	16	373
10-May-24	277	55	35	14	17	398
11-May-24	299	51	35	19	18	422
12-May-24	290	40	23	16	18	387
13-May-24	278	42	36	18	14	388
14-May-24	267	48	31	25	14	386
15-May-24	273	68	38	23	15	416
16-May-24	292	51	33	26	14	417
17-May-24	313	46	37	22	16	434
18-May-24	324	62	35	22	19	461
19-May-24	348	54	27	25	21	475
20-May-24	346	58	29	26	17	476
21-May-24	347	62	31	27	18	486
22-May-24	349	69	49	25	17	508
23-May-24	361	56	60	28	15	520
24-May-24	361	60	51	28	16	516
25-May-24	367	49	34	28	15	493
26-May-24	359	44	32	29	17	480
27-May-24	354	60	41	32	20	507
28-May-24	350	52	41	33	32	507
29-May-24	356	68	47	42	39	551
30-May-24	350	69	48	32	37	537
31-May-24	339	77	58	30	38	543
कुल TOTAL	9021	1668	1211	780	572	13251
उच्चतम MAXIMUM	367	77	60	42	39	551
निम्नतम MINIMUM	176	38	23	14	11	301
औसत AVERAGE	291	54	39	25	18	427
संचयी 2024-25 Cumulative 2024-25	13694	2886	2811	1508	918	21817
अब तक का उच्चतम All Time Max.	443	167	208	157	41	877
दिनांक Date	01.08.23	18.12.14	31.08.18	14.09.22	29.08.23	30.08.22


HYDRO ENERGY DURING THE MONTH OF MAY' 2024
NR
WR
SR
ER
NER
INDIA
NR - Max: 367 MU Min: 176 MU Avg : 291 MU
SR - Max: 60 MU Min: 23 MU Avg : 39 MU
ALL INDIA - Max: 551 MU Min: 301 MU Avg : 427 MU
WR - Max: 77 MU Min: 38 MU Avg : 54 MU
ER - Max: 42 MU Min: 14 MU Avg : 25 MU
NER - Max: 39 MU Min: 11 MU Avg : 18 MU
HYDRO ENERGY (MU) -->
600
500
400
300
200
100
0
1-May-24
2-May-24
3-May-24
4-May-24
5-May-24
6-May-24
7-May-24
8-May-24
9-May-24
10-May-24
11-May-24
12-May-24
13-May-24
14-May-24
15-May-24
16-May-24
17-May-24
18-May-24
19-May-24
20-May-24
21-May-24
22-May-24
23-May-24
24-May-24
25-May-24
26-May-24
27-May-24
28-May-24
29-May-24
30-May-24
31-May-24
DATE -->

7. राष्ट्रीय स्तर पर पवन ऊर्जा उत्पादन
WIND GENERATION AT NATIONAL LEVEL

माह: मई 2024 MONTH:- MAY 2024

सभी आंकड़े मिलियन यूनिट मे All figures in MU

दिनांक Date	ऊत्तरी क्षे. NR	पश्चिमी क्षे. WR	दक्षिणी क्षे. SR	पूर्वी क्षे. ER	पूर्वोत्तर क्षे. NER	कुल TOTAL
01-May-24	16	156	82	----------	----------	254
02-May-24	11	85	61	----------	----------	157
03-May-24	17	91	73	----------	----------	181
04-May-24	34	113	78	----------	----------	225
05-May-24	13	153	98	----------	----------	264
06-May-24	10	137	99	----------	----------	247
07-May-24	13	119	84	----------	----------	216
08-May-24	14	109	49	----------	----------	172
09-May-24	26	111	37	----------	----------	174
10-May-24	39	102	26	----------	----------	167
11-May-24	31	62	29	----------	----------	122
12-May-24	13	58	34	----------	----------	105
13-May-24	15	66	31	----------	----------	111
14-May-24	36	83	30	----------	----------	149
15-May-24	15	50	43	----------	----------	108
16-May-24	20	58	37	----------	----------	116
17-May-24	21	95	23	----------	----------	140
18-May-24	25	121	24	----------	----------	170
19-May-24	27	137	34	----------	----------	198
20-May-24	37	134	56	----------	----------	227
21-May-24	9	117	42	----------	----------	168
22-May-24	14	123	31	----------	----------	168
23-May-24	11	155	37	----------	----------	203
24-May-24	9	173	77	----------	----------	259
25-May-24	10	177	246	----------	----------	433
26-May-24	18	230	267	----------	----------	516
27-May-24	46	287	260	----------	----------	593
28-May-24	56	310	253	----------	----------	619
29-May-24	57	306	240	----------	----------	603
30-May-24	51	206	180	----------	----------	437
31-May-24	58	195	138	----------	----------	391
कुल TOTAL	776	4319	2801	----------	----------	7895
उच्चतम MAXIMUM	58	310	267	----------	----------	619
निम्नतम MINIMUM	9	50	23	----------	----------	105
औसत AVERAGE	25	139	90	----------	----------	255
संचयी 2024-25 Cumulative 2024-25	1311	7052	4087	----------	----------	12451
अब तक का उच्चतम All Time Max.	86	**310**	298	----------	----------	**619**
दिनांक Date	07.08.23	**28.05.24**	21.07.23	----------	----------	**28.05.24**

*__Source:__ As reported by SLDCs. Limited visibility of embedded wind generator data.

8. राष्ट्रीय स्तर पर सौर ऊर्जा उत्पादन

SOLAR GENERATION AT NATIONAL LEVEL

माह: मई 2024 MONTH:- MAY 2024

सभी आंकड़े मिलियन यूनिट मे All figures in MU

दिनांक Date	ऊत्तरी क्षे. NR	पश्चिमी क्षे. WR	दक्षिणी क्षे. SR	पूर्वी क्षे. ER	पूर्वोत्तर क्षे. NER	कुल TOTAL
01-May-24	186	102	133	3.0	1.2	425
02-May-24	179	100	139	3.2	1.0	422
03-May-24	151	98	131	3.9	1.4	386
04-May-24	169	89	136	3.5	1.1	398
05-May-24	171	92	135	3.6	0.5	402
06-May-24	161	100	132	3.6	0.3	397
07-May-24	161	97	128	2.5	0.8	389
08-May-24	174	96	134	3.8	1.1	409
09-May-24	164	92	120	2.8	1.0	380
10-May-24	166	94	116	3.2	0.8	380
11-May-24	157	94	120	3.3	1.2	376
12-May-24	174	90	118	3.5	1.4	388
13-May-24	161	91	111	3.4	1.3	368
14-May-24	166	94	117	4.1	1.4	382
15-May-24	160	90	116	3.9	1.3	372
16-May-24	159	87	103	2.9	1.3	354
17-May-24	165	85	93	2.9	1.0	348
18-May-24	167	92	116	3.1	0.7	379
19-May-24	159	95	108	3.9	1.0	367
20-May-24	169	95	104	4.1	0.9	374
21-May-24	175	96	122	3.7	1.2	399
22-May-24	166	94	118	3.8	1.4	383
23-May-24	175	94	115	3.7	1.4	390
24-May-24	169	97	113	3.8	1.3	383
25-May-24	163	95	98	3.4	1.4	362
26-May-24	168	92	105	3.1	1.3	370
27-May-24	172	91	106	2.8	0.4	372
28-May-24	174	96	123	3.7	0.2	397
29-May-24	178	92	134	3.6	0.6	409
30-May-24	182	97	137	3.4	0.6	420
31-May-24	180	97	136	3.2	0.6	416
कुल TOTAL	5223	2916	3718	106	31	11995
उच्चतम MAXIMUM	186	102	139	4.1	1.4	425
निम्नतम MINIMUM	151	85	93	2.5	0.2	348
औसत AVERAGE	168	94	120	3.4	1.0	387
संचयी 2024-25 Cumulative 2024-25	9847	5517	7704	198	63.3	23329
अब तक का उच्चतम All Time Max.	**186**	103	149	6.7	2.4	426
दिनांक Date	**01.05.24**	30.04.24	31.03.24	23.03.23	22.06.22	30.04.24

*__Source__: RLDCs for solar connected to ISTS; SLDCs for embedded solar. Limited visibility of embedded solar data.

9. राष्ट्रीय स्तर पर दैनिक अधिकतम विद्युत मांग पूर्ति

DAILY MAXIMUM DEMAND MET AT NATIONAL LEVEL

माह: मई 2024 MONTH:- MAY 2024

सभी आंकड़े मेगावाट मे All figures in MW

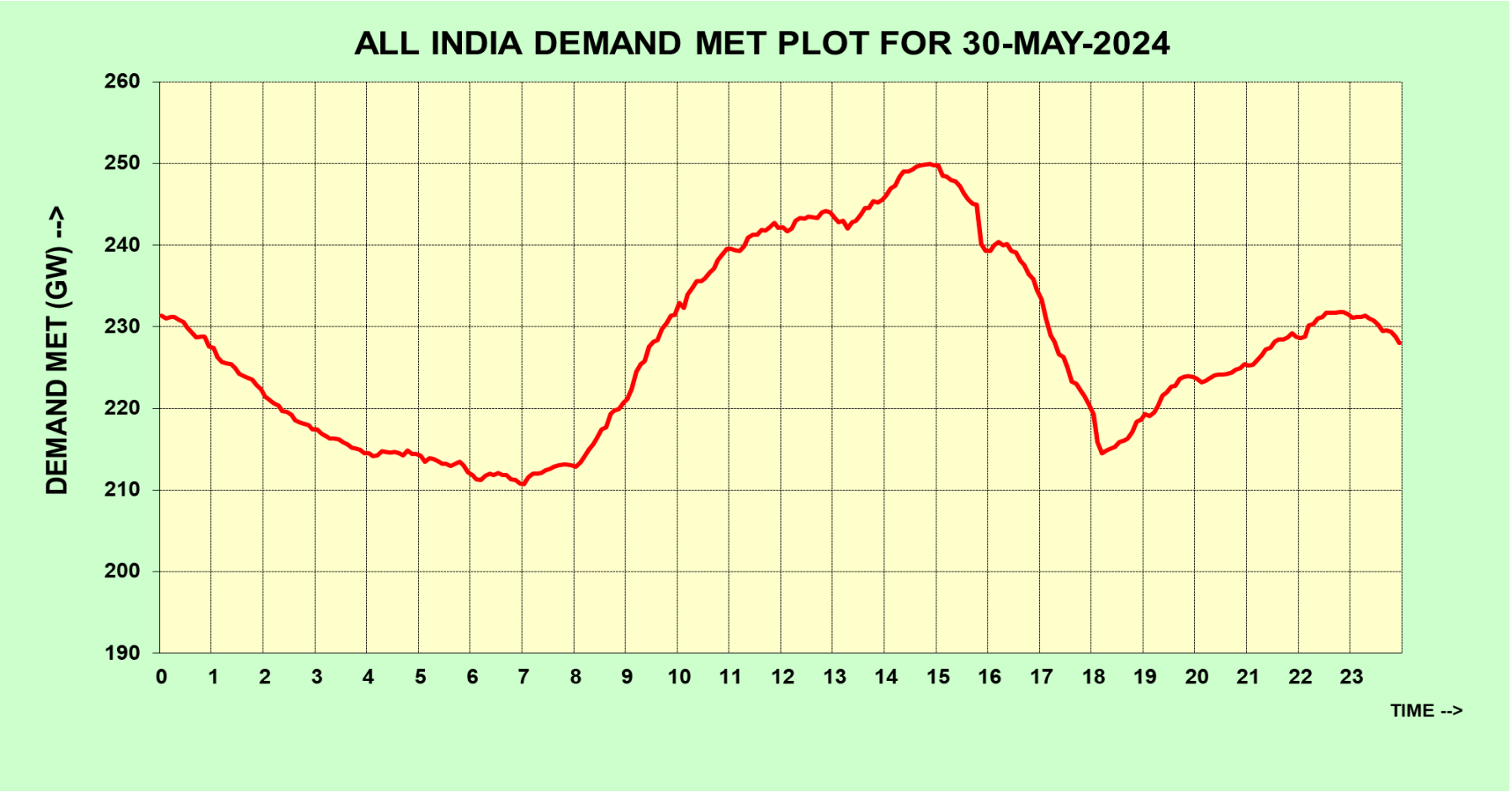
दिनांक Date	ऊत्तरी क्षे. NR	पश्चिमी क्षे. WR	दक्षिणी क्षे. SR	पूर्वी क्षे. ER	पूर्वोत्तर क्षे. NER	कुल TOTAL		विभिन्नता फ़ैक्टर Diversity Factor*
01-May-24	60891	69581	60339	31675	3071	219379		1.028
02-May-24	61851	71841	66140	30146	2865	222034		1.049
03-May-24	62357	72334	65148	30095	3155	223849		1.041
04-May-24	63898	73400	64783	30493	3279	229778		1.026
05-May-24	64553	70068	59561	30165	2247	218085		1.039
06-May-24	69104	74739	65412	29462	2486	233007		1.035
07-May-24	69808	71030	60378	24490	2706	224744		1.016
08-May-24	70286	73934	59975	25795	2906	223816		1.041
09-May-24	70909	72189	57548	25301	2905	219387		1.043
10-May-24	69974	71649	57351	26262	2842	223524		1.020
11-May-24	67377	72368	56240	25723	3031	216337		1.039
12-May-24	66947	67754	50750	26453	3111	206852		1.039
13-May-24	69599	69905	50743	28181	3389	216992		1.022
14-May-24	70471	67182	53710	28578	3509	220332		1.014
15-May-24	71737	69600	54524	28889	3639	226152		1.010
16-May-24	74989	69972	51087	29450	3594	226371		1.012
17-May-24	73738	70953	50820	30633	3307	225926		1.016
18-May-24	77483	71943	49896	30324	3304	229578		1.015
19-May-24	75659	68692	43735	29864	3058	218499		1.011
20-May-24	81783	72548	47558	29233	3308	228719		1.025
21-May-24	82284	74126	48819	28969	3473	233806		1.017
22-May-24	83028	74459	50136	28416	3529	235030		1.019
23-May-24	81483	74021	51479	29425	3506	236597		1.014
24-May-24	83668	73795	52312	31056	3657	239963		1.019
25-May-24	82168	72535	47985	31084	3401	233979		1.014
26-May-24	79621	68328	43494	31048	3259	222114		1.016
27-May-24	85025	71577	51387	26663	2558	232615		1.020
28-May-24	84824	70332	53955	30803	2145	237946		1.017
29-May-24	86653	71846	56551	31436	2783	246069		1.013
30-May-24	86785	74854	58890	31013	2554	250070		1.016
31-May-24	85991	74914	61396	28235	3036	249159		1.018
उच्चतम MAXIMUM	86785	74914	66140	31675	3657	250070		1.049
निम्नतम MINIMUM	60891	67182	43494	24490	2145	206852		1.010
औसत AVERAGE	74676	71693	54906	29012	3084	228087		1.023
अब तक का उच्चतम All Time Max.	**86785**	**74914**	68735	31895	3712	**250070**		
दिनांक Date	**30.05.24**	**31.05.24**	13.03.24	30.04.24	04.09.23	**30.05.24**		

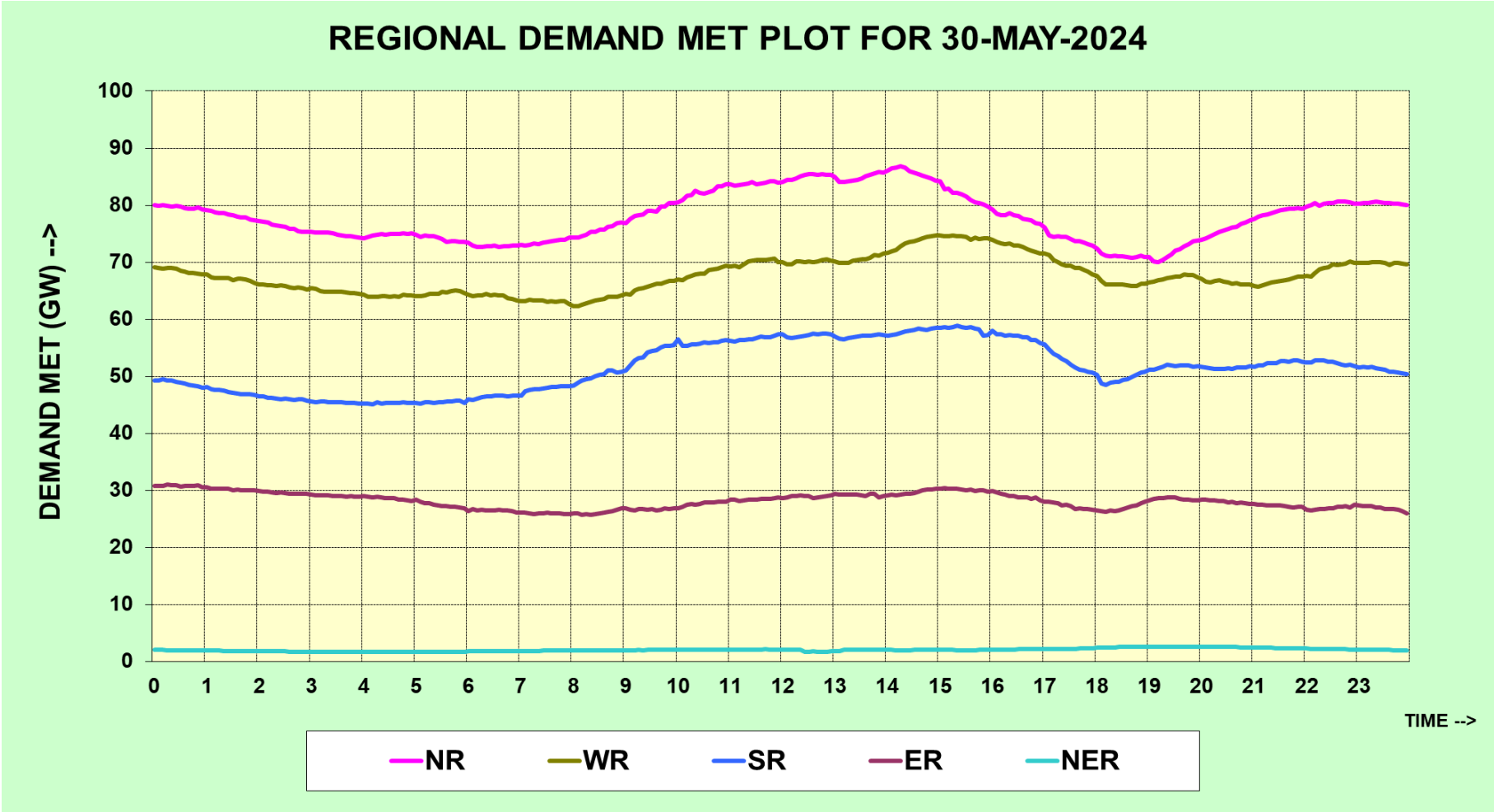
* Diversity factor = (Sum of regional max demands) / All India max demand

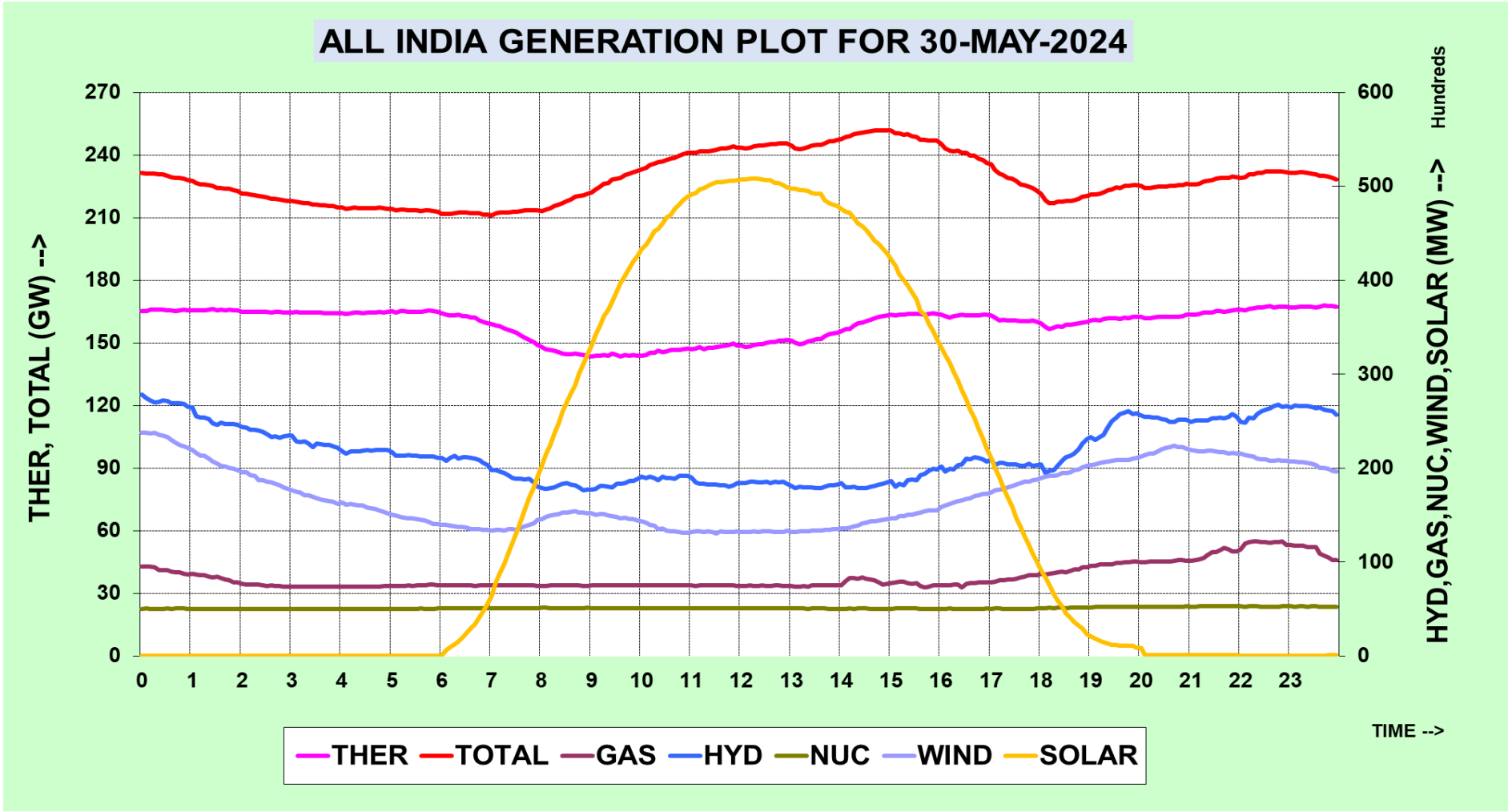
नोट : यह आंकड़े एनएलडीसी स्काडा प्रणाली में दर्ज दैनिक अधिकतम मांगपूर्ति दर्शाते हैं ।

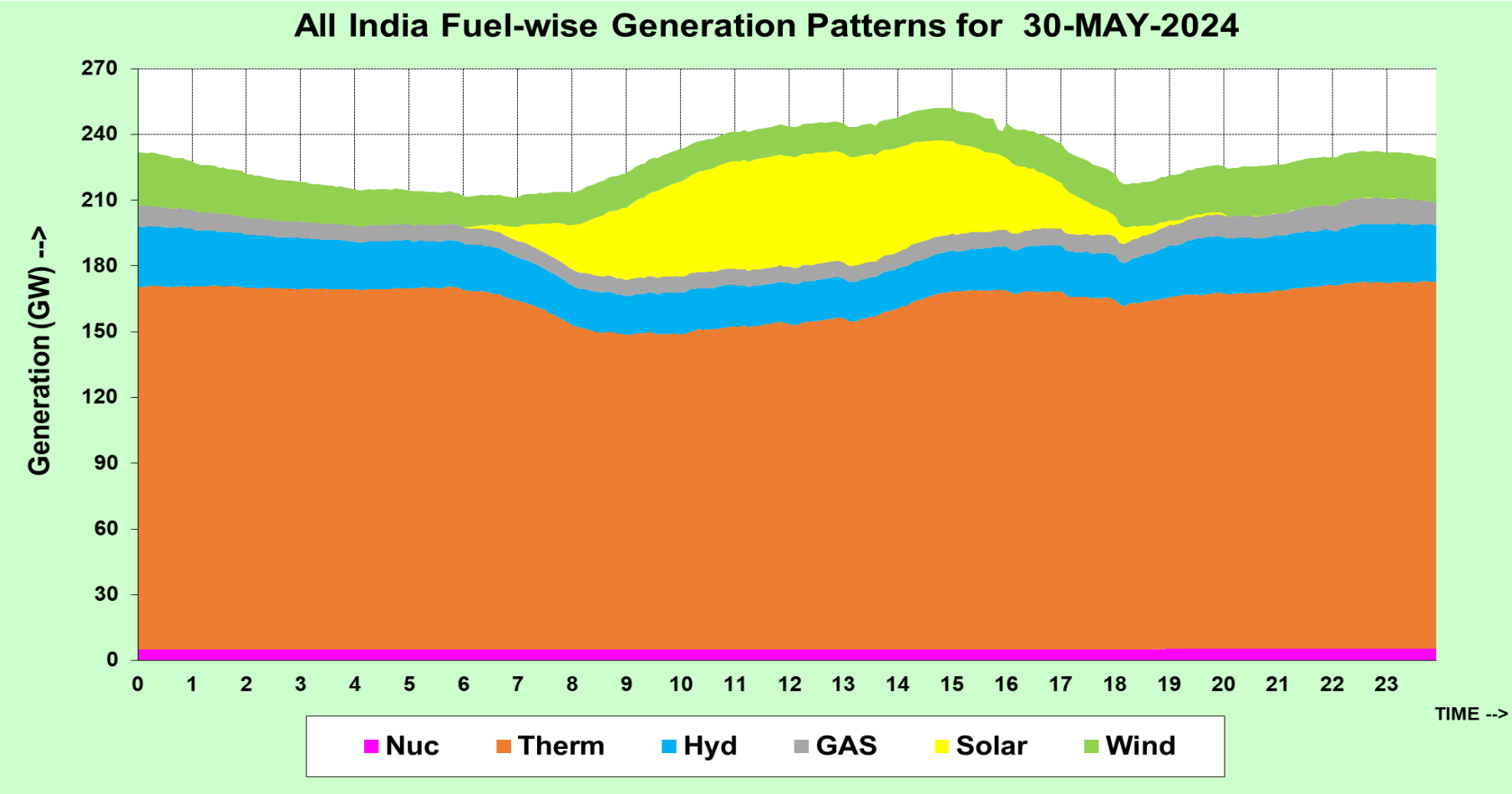
Note: The above figures denote daily maximum demand met recorded in NLDC SCADA.

10. DEMAND AND GENERATION PLOTS FOR 30 MAY (MAXIMUM DEMAND MET)




REGIONAL DEMAND MET PLOT FOR 30-MAY-2024
DEMAND MET (GW) -->
100
90
80
70
60
50
40
30
20
10
0
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
TIME -->
NR
WR
SR
ER
NER


ALL INDIA GENERATION PLOT FOR 30-MAY-2024
THER, TOTAL (GW) -->
HYD,GAS,NUC,WIND,SOLAR (MW) --> Hundreds
TIME -->
270
240
210
180
150
120
90
60
30
0
600
500
400
300
200
100
0
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
THER TOTAL GAS HYD NUC WIND SOLAR


All India Fuel-wise Generation Patterns for 30-MAY-2024
Generation (GW) -->
270
240
210
180
150
120
90
60
30
0
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
TIME -->
Nuc
Therm
Hyd
GAS
Solar
Wind

30 मई 2024 (अधिकतम माँग) की आल इंडिया माँग आपूर्ति

ALL INDIA DEMAND MET FOR 30 MAY 2024 (MAXIMUM DEMAND*)

Time (hrs)	All India Demand (MW)		Time (hrs)	All India Demand (MW)
00:05	231009		**12:05**	241668
00:20	230815		**12:20**	243366
00:35	229335		**12:35**	243380
00:50	228802		**12:50**	244171
01:05	226319		**13:05**	241604
01:20	225429		**13:20**	242812
01:35	223991		**13:35**	239530
01:50	222849		**13:50**	245195
02:05	221069		**14:05**	246938
02:20	219685		**14:20**	249041
02:35	218533		**14:35**	249625
02:50	217903		**14:50**	249951
03:05	216932		**15:05**	248458
03:20	216336		**15:20**	247828
03:35	215654		**15:35**	245600
03:50	214888		**15:50**	240111
04:05	214157		**16:05**	241081
04:20	214653		**16:20**	240187
04:35	214509		**16:35**	238059
04:50	214383		**16:50**	235865
05:05	213465		**17:05**	230925
05:20	213593		**17:20**	226623
05:35	212908		**17:35**	223312
05:50	213014		**17:50**	221491
06:05	211346		**18:05**	215911
06:20	212001		**18:20**	215110
06:35	211868		**18:35**	216013
06:50	211258		**18:50**	218410
07:05	211542		**19:05**	219077
07:20	212056		**19:20**	221590
07:35	212859		**19:35**	222801
07:50	213132		**19:50**	223929
08:05	213382		**20:05**	223188
08:20	215653		**20:20**	224082
08:35	217714		**20:35**	224187
08:50	219940		**20:50**	224898
09:05	222423		**21:05**	225363
09:20	225816		**21:20**	227194
09:35	228352		**21:35**	228425
09:50	231338		**21:50**	229223
10:05	232323		**22:05**	228777
10:20	235573		**22:20**	231048
10:35	236659		**22:35**	231685
10:50	238822		**22:50**	231758
11:05	239337		**23:05**	231223
11:20	240917		**23:20**	231020
11:35	241831		**23:35**	229506
11:50	242767		**23:50**	228772
Maximum Demand of 250070 MW met@14:57 hrs (from 1 min. interval SCADA DATA)				

* 15 minute interval SCADA DATA of instanteneous demand

11. ACTUAL POWER SUPPLY POSITION - MAY 2024

REGION	STATE	Requirement/Availability in MU/DAY			Requirement/Availability in MU				Peak Demand/Peak Met in MW			
		Requirement	Energy met	Surplus(+)/ Deficit(-)	Requirement	Energy met	Surplus(+)/ Deficit(-)	% Shortage	Requirement	Peak Demand Met	Surplus(+)/ Deficit(-)	% Shortage
NR	Chandigarh	7	7	0.0	210	210	0	0.0	432	432	0	0.0
	Delhi	136	136	0.0	4208	4208	0	0.0	8302	8302	0	0.0
	Haryana	221	221	0.0	6839	6839	0	0.0	12451	12451	0	0.0
	Himachal Pradesh	35	35	-0.1	1077	1073	-5	-0.4	1827	1827	0	0.0
	J&K(UT) and Ladakh(UT)	53	53	-0.2	1647	1641	-5	-0.3	2878	2878	0	0.0
	Punjab	234	234	0.0	7239	7239	0	0.0	14519	14519	0	0.0
	Rajasthan	334	332	-2.5	10357	10280	-78	-0.7	17567	17567	0	0.0
	Uttar Pradesh	563	563	0.0	17457	17457	0	0.0	29727	29727	0	0.0
	Uttarakhand	54	54	0.0	1669	1668	-1	-0.1	2781	2781	0	0.0
	Railway	4	4	0.0	131	131	0	0.0	233	233	0	0.0
	NFL	0	0	0.0	2	2	0	0.0	5	5	0	0.0
	TOTAL	1640	1637	-2.9	50836	50747	-89	-0.2	86305	86305	0	0.0
WR	Gujarat	484	484	0.0	14990	14990	0	0.0	24922	24922	0	0.0
	Madhya Pradesh	292	292	-0.1	9053	9049	-3	0.0	14309	14309	0	0.0
	Chhattisgarh	119	119	0.0	3679	3679	0	0.0	6073	6073	0	0.0
	Maharashtra	608	608	0.0	18846	18845	0	0.0	28452	28452	0	0.0
	Goa	16	16	0.0	506	506	0	0.0	803	803	0	0.0
	DNHDDPDCL	30	30	0.0	922	922	0	0.0	1348	1348	0	0.0
	AMNSIL	18	18	0.0	563	563	0	0.0	911	911	0	0.0
	BALCO	12	12	0.0	387	387	0	0.0	525	525	0	0.0
	TOTAL	1579	1579	-0.1	48947	48943	-4	0.0	74829	74829	0	0.0
SR	Andhra Pradesh	228	228	0.0	7072	7072	0	0.0	13712	13712	0	0.0
	Telangana	201	201	0.0	6218	6218	0	0.0	11277	11257	-20	-0.2
	Karnataka	255	255	0.0	7892	7892	0	0.0	16826	16826	0	0.0
	Kerala	92	92	0.0	2846	2846	0	0.0	5797	5797	0	0.0
	Tamil Nadu	382	382	0.0	11838	11838	0	0.0	20830	20830	0	0.0
	Puducherry	11	11	0.0	333	333	0	0.0	517	517	0	0.0
	TOTAL	1168	1168	0.0	36198	36198	0	0.0	66163	66163	0	0.0
ER	Bihar	136	135	-0.7	4201	4179	-22	-0.5	7546	7546	0	0.0
	Jharkhand	45	45	-0.3	1402	1394	-8	-0.6	2346	2346	0	0.0
	DVC	75	75	0.0	2310	2310	0	0.0	3562	3562	0	0.0
	Odisha	132	132	0.0	4099	4099	0	0.0	6988	6988	0	0.0
	West Bengal	215	215	0.0	6680	6680	0	0.0	12670	12670	0	0.0
	Sikkim	1	1	0.0	46	46	0	0.0	102	102	0	0.0
	Railways_ER ISTS	0	0	0.0	4	4	0	0.0	23	23	0	0.0
	TOTAL	605	604	-1.0	18743	18712	-31	-0.2	31777	31643	-134	-0.4
NER	Arunachal Pradesh	3	3	0.0	86	86	0	0.0	198	198	0	0.0
	Assam	36	36	-0.2	1123	1118	-5	-0.5	2404	2404	0	0.0
	Manipur	3	3	0.0	86	86	0	0.0	227	227	0	0.0
	Meghalaya	5	5	0.0	160	160	0	0.0	394	394	0	0.0
	Mizoram	2	2	0.0	57	57	0	0.0	130	130	0	0.0
	Nagaland	3	3	0.0	80	80	0	0.0	177	177	0	0.0
	Tripura	5	5	0.0	164	164	0	0.0	386	386	0	0.0
	TOTAL	57	56	-0.2	1756	1751	-5	-0.3	3652	3652	0	0.0
ALL INDIA		5048	5044	-4.1	156480	156352	-128	-0.1				

NOTE: BASED ON THE DATA FURNISHED BY ENTITIES ON DAILY BASIS

12. ENERGY COMPARISON OF MAY 2024 vs MAY 2023

REGION	STATE	Energy Requirement (MU)					Energy Met (MU)				
		May-23	May-24	Difference	% Change	Average MU/day for May-24	May-23	May-24	Difference	% Change	Average MU/day for May-24
NR	Chandigarh	154	210	56	37	7	154	210	56	37	7
	Delhi	3121	4208	1087	35	136	3120	4208	1088	35	136
	Haryana	5113	6839	1726	34	221	5098	6839	1741	34	221
	Himachal Pradesh	909	1077	168	18	35	906	1073	167	18	35
	J&K(UT) and Ladakh(UT)	1720	1647	-74	-4	53	1709	1641	-67	-4	53
	Punjab	5278	7239	1960	37	234	5278	7239	1960	37	234
	Rajasthan	8231	10357	2126	26	334	8210	10280	2070	25	332
	Uttar Pradesh	13225	17457	4232	32	563	13187	17457	4270	32	563
	Uttarakhand	1345	1669	324	24	54	1342	1668	326	24	54
	Railway/NFL	103	133	30	29	4	103	133	30	29	4
	TOTAL	39200	50836	11636	30	1640	39106	50747	11641	30	1637
WR	Gujarat	13231	14990	1759	13	484	13231	14990	1759	13	484
	Madhya Pradesh	7418	9053	1635	22	292	7418	9049	1631	22	292
	Chhattisgarh	3212	3679	467	15	119	3212	3679	467	15	119
	Maharashtra	17663	18846	1183	7	608	17650	18845	1196	7	608
	Goa	476	506	30	6	16	475	506	31	7	16
	DNHDDPDCL	876	922	47	5	30	876	922	47	5	30
	AMNSIL	546	563	17	3	18	546	563	17	3	18
	BALCO	384	387	3	1	12	384	387	3	1	12
	TOTAL	43806	48947	5141	12	1579	43792	48943	5151	12	1579
SR	Andhra Pradesh	6926	7072	146	2	228	6870	7072	202	3	228
	Telangana	5756	6218	461	8	201	5691	6218	527	9	201
	Karnataka	7624	7892	268	4	255	7559	7892	333	4	255
	Kerala	2867	2846	-22	-1	92	2846	2846	-1	0	92
	Tamil Nadu	10982	11838	856	8	382	10897	11838	941	9	382
	Puducherry	318	333	15	5	11	316	333	17	5	11
	TOTAL	34473	36198	1725	5	1168	34179	36198	2019	6	1168
ER	Bihar	3763	4201	439	12	136	3710	4179	469	13	135
	Jharkhand	1106	1402	296	27	45	1045	1394	349	33	45
	DVC	2342	2310	-32	-1	75	2342	2310	-32	-1	75
	Odisha	3729	4099	369	10	132	3729	4099	369	10	132
	West Bengal	6005	6680	675	11	215	6005	6680	675	11	215
	Sikkim	45	46	0	1	1	45	46	0	1	1
	Railways_ER ISTS	4	4	0	4	0	4	4	0	4	0
	TOTAL	16996	18743	1747	10	605	16882	18712	1831	11	604
NER	Arunachal Pradesh	78	86	8	10	3	78	86	8	10	3
	Assam	1033	1123	91	9	36	1029	1118	89	9	36
	Manipur	73	86	13	17	3	73	86	13	17	3
	Meghalaya	195	160	-35	-18	5	156	160	3	2	5
	Mizoram	54	57	3	5	2	54	57	3	5	2
	Nagaland	75	80	6	7	3	75	80	6	7	3
	Tripura	164	164	1	0	5	164	164	1	0	5
	TOTAL	1671	1756	85	5	57	1629	1751	122	7	56
ALL INDIA		136146	156480	20334	15	5048	135587	156352	20764	15	5044

13. DEMAND COMPARISON OF MAY 2024 vs MAY 2023

REGION	STATE	Peak Requirement (MW)				Peak Met (MW)			
		May-23	May-24	Difference	% Change	May-23	May-24	Difference	% Change
NR	Chandigarh	321	432	111	34.6	321	432	111	34.6
	Delhi	6916	8302	1386	20.0	6916	8302	1386	20.0
	Haryana	10020	12451	2431	24.3	10020	12451	2431	24.3
	Himachal Pradesh	1702	1827	125	7.3	1702	1827	125	7.3
	J&K(UT) and Ladakh(UT)	2823	2878	55	1.9	2823	2878	55	1.9
	Punjab	11940	14519	2579	21.6	11940	14519	2579	21.6
	Rajasthan	16470	17567	1097	6.7	16470	17567	1097	6.7
	Uttar Pradesh	26386	29727	3341	12.7	26166	29727	3561	13.6
	Uttarakhand	2415	2781	366	15.2	2415	2781	366	15.2
	Railway/NFL	196	239	42	21.5	197	239	41	21.0
WR	Gujarat	21340	24922	3582	17	21310	24922	3612	16.9
	Madhya Pradesh	12286	14309	2023	16	12286	14309	2023	16.5
	Chhattisgarh	4905	6073	1168	24	4901	6073	1171	23.9
	Maharashtra	28160	28452	292	1	28121	28452	331	1.2
	Goa	755	803	48	6	755	803	48	6.4
	DNHDDPDCL	1265	1348	83	6.6	1265	1348	83	6.6
	AMNSIL	897	911	15	1.6	897	911	15	1.6
	BALCO	522	525	2	0.4	522	525	2	0.4
SR	Andhra Pradesh	12653	13712	1059	8.4	12653	13712	1059	8.4
	Telangana	9886	11277	1392	14.1	9867	11257	1390	14.1
	Karnataka	15111	16826	1715	11.3	15111	16826	1715	11.3
	Kerala	4862	5797	935	19.2	4862	5797	935	19.2
	Tamil Nadu	18658	20830	2172	11.6	18658	20830	2172	11.6
	Puducherry	512	517	5	1.0	512	517	5	1.0
ER	Bihar	6657	7546	889	13.4	6657	7546	889	13.4
	Jharkhand	1839	2346	507	27.6	1839	2346	507	27.6
	DVC	3751	3562	-189	-5.0	3751	3562	-189	-5.0
	Odisha	6940	6988	48	0.7	6940	6988	48	0.7
	West Bengal	11105	12670	1565	14.1	11105	12670	1565	14.1
	Sikkim	104	102	-2	-1.9	104	102	-2	-1.9
	Railways_ER ISTS	19	23	4	21.1	19	23	4	21.1
NER	Arunachal Pradesh	165	198	33	20.0	165	198	33	20.0
	Assam	2219	2404	185	8.3	2219	2404	185	8.3
	Manipur	193	227	34	17.6	193	227	34	17.6
	Meghalaya	353	394	41	11.6	332	394	62	18.6
	Mizoram	122	130	8	6.2	122	130	8	6.2
	Nagaland	152	177	25	16.4	152	177	25	16.4
	Tripura	345	386	41	12.0	345	386	41	12.0

14. SCHEDULE AND DRAWAL OF CONSTITUENTS - MAY 2024

REGION	STATE	SCHEDULE (MU)	ACTUAL DRAWAL (MU)	Over drawal(+)/ Under Drawal(-) (MU)	% OD/UD	SCHEDULE (MU/DAY)	ACTUAL DRAWAL (MU/DAY)	Over drawal(+)/ Under Drawal(-) (MU/DAY)
NR	Chandigarh	209.75	211.73	1.98	0.94	6.77	6.83	0.06
	Delhi	3491.87	3449.43	-42.44	-1.22	112.64	111.27	-1.37
	Haryana	4882.45	4823.54	-58.91	-1.21	157.50	155.60	-1.90
	Himachal Pradesh	167.42	169.51	2.09	1.25	5.40	5.47	0.07
	J&K(UT) and Ladakh(UT)	952.40	887.59	-64.81	-6.81	30.72	28.63	-2.09
	Punjab	3459.45	3386.35	-73.10	-2.11	111.60	109.24	-2.36
	Rajasthan	3574.88	3602.56	27.68	0.77	115.32	116.21	0.89
	Uttar Pradesh	8171.16	8157.41	-13.75	-0.17	263.59	263.14	-0.44
	Uttarakhand	937.97	933.80	-4.17	-0.44	30.26	30.12	-0.13
	Railway	118.34	131.53	13.18	11.14	3.82	4.24	0.43
	NFL	2.58	2.58	0.00	0.12	0.08	0.08	0.00
	Total	25968.27	25756.03	-212.24	-0.82	837.69	830.84	-6.85
WR	Gujarat	4760.16	4623.80	-136.36	-2.86	153.55	149.15	-4.40
	Madhya Pradesh	4895.82	4765.53	-130.29	-2.66	157.93	153.73	-4.20
	Chhattisgarh	1884.63	1858.92	-25.71	-1.36	60.79	59.97	-0.83
	Maharashtra	6108.75	5988.03	-120.72	-1.98	197.06	193.16	-3.89
	Goa	496.14	497.47	1.33	0.27	16.00	16.05	0.04
	DNHDDPDCL	922.01	922.49	0.48	0.05	29.74	29.76	0.02
	AMNSIL	221.87	236.12	14.25	6.42	7.16	7.62	0.46
	BALCO*	387.69	386.82	-0.87	-0.23	12.51	12.48	-0.03
	Total	19677.07	19279.18	-397.89	-2.02	634.74	621.91	-12.84
SR	Andhra Pradesh	2122.56	2090.73	-31.83	-1.50	68.47	67.44	-1.03
	Telangana	2505.16	2498.36	-6.80	-0.27	80.81	80.59	-0.22
	Karnataka	3256.22	3167.24	-88.98	-2.73	105.04	102.17	-2.87
	Kerala	2230.19	2249.79	19.60	0.88	71.94	72.57	0.63
	Tamil Nadu	6984.50	6854.20	-130.30	-1.87	225.31	221.10	-4.20
	Puducherry	324.64	314.50	-10.14	-3.12	10.47	10.15	-0.33
	Goa (SR)	124.76	76.99	-47.77	-38.29	4.02	2.48	-1.54
	Total	17548.03	17251.81	-296.22	-1.69	566.07	556.51	-9.56
ER	Bihar	3855.12	3852.55	-2.57	-0.07	124.36	124.28	-0.08
	Jharkhand	983.70	996.79	13.09	1.33	31.73	32.15	0.42
	DVC	-1162.18	-1146.96	15.22	-1.31	-37.49	-37.00	0.49
	Odisha	1658.20	1656.35	-1.85	-0.11	53.49	53.43	-0.06
	West Bengal	2389.39	2337.11	-52.28	-2.19	77.08	75.39	-1.69
	Sikkim	45.68	48.64	2.96	6.49	1.47	1.57	0.10
	Railways_ER ISTS	4.36	4.35	-0.01	-0.22	0.14	0.14	0.00
	Total	7774.28	7748.84	-25.44	-0.33	250.78	249.96	-0.82
NER	Arunachal Pradesh	84.17	81.27	-2.90	-3.45	2.72	2.62	-0.09
	Assam	907.38	910.82	3.44	0.38	29.27	29.38	0.11
	Manipur	88.65	85.58	-3.07	-3.46	2.86	2.76	-0.10
	Meghalaya	77.46	73.42	-4.04	-5.22	2.50	2.37	-0.13
	Mizoram	53.96	45.97	-7.99	-14.81	1.74	1.48	-0.26
	Nagaland	82.42	80.10	-2.32	-2.81	2.66	2.58	-0.07
	Tripura	153.73	157.34	3.61	2.35	4.96	5.08	0.12
	Total	1447.77	1434.50	-13.27	-0.92	46.70	46.27	-0.43
All India Total		72415.42	71470.36	-945.07	-1.31	2335.98	2305.50	-30.49

15. INTER REGIONAL EXCHANGES 2024-25

(All figures in MU)

	Apr'24	May'24	Fin. Year 2024-25
Name of Line			
Import of NR from WR (WR-NR)			
WR - NR 765 kV Gwalior - Agra 2xS/C	529.65	1235.15	**1764.80**
WR-NR 765 kV Gwalior - Phagi 2xS/C	157.44	396.56	**554.00**
WR-NR 765 kV Jabalpur-Orai D/C	304.53	976.67	**1281.20**
WR-NR 765 kV Satna-Orai	535.85	669.33	**1205.18**
WR-NR 765 kV Gwalior-Orai	0.00	0.00	**0.00**
WR - NR HVDC Champa - Kurukshetra	1525.63	2816.73	**4342.36**
WR - NR HVDC Vindhyachal	0.00	100.25	**100.25**
WR - NR 400 kV Zerda Kankroli	0.00	35.28	**35.28**
WR - NR 400 kV Zerda Bhinmal	0.00	1.38	**1.38**
WR - NR 400 kV RAPP C - Shujalpur D/C	0.00	60.74	**60.74**
WR - NR 220 kV Bhanpura - Ranpur	55.64	76.03	**131.67**
WR - NR 220 kV Bhanpura - Modak	69.87	76.33	**146.20**
WR - NR 220 kV Auraiya - Malanpur/Mehgaon	0.00	3.05	**3.05**
WR - NR 400 kV Vindhyachal-Rihand D/C	0.00	0.00	**0.00**
WR - NR HVDC Mundra - Mohindergarh	717.97	1144.70	**1862.67**
WR-NR 765 kV Banaskata-Chittorgarh D/C	0.67	262.23	**262.90**
WR-NR 765 kV Varanasi-Vindhyachal	1373.47	2029.86	**3403.33**
WR -NR765 kV Neemach - Chittorgarh D/C	0.00	35.36	**35.36**
Total WR - NR	**5270.72**	**9919.65**	**15190.37**
Export of NR to WR (NR-WR)			
NR - WR 765 kV Gwalior - Agra 2xS/C	0.00	0.00	**0.00**
NR - WR 765 kV Gwalior - Phagi 2xS/C	10.49	1.42	**11.91**
NR - WR 765 kV Jabalpur-Orai D/C	0.00	0.00	**0.00**
NR - WR 765 kV Satna-Orai	0.00	0.00	**0.00**
NR - WR 765 kV Gwalior-Orai	245.97	283.52	**529.49**
NR - WR HVDC Champa- Kurukshetra	0.00	0.00	**0.00**

	Apr'24	May'24	Fin. Year 2024-25
Name of Line			
NR - WR HVDC Vindhyachal	259.46	128.55	**388.01**
NR - WR 400 kV Zerda Kankroli	240.09	83.35	**323.44**
NR - WR 400 kV Zerda Bhinmal	8.05	0.00	**8.05**
NR - WR 400 kV RAPP C - Shujalpur D/C	159.07	31.59	**190.66**
NR - WR 220 kV Bhanpura - Ranpur	0.00	0.00	**0.00**
NR - WR 220 kV Bhanpura - Modak	0.00	0.00	**0.00**
NR - WR 220 kV Auraiya - Malanpur/Mehgaon	59.69	31.06	**90.75**
NR-WR 400 kV Rihand-Vindhyachal D/C	609.68	680.87	**1290.55**
NR - WR HVDC Mundra - Mohindergarh	0.00	0.00	**0.00**
NR - WR 765 kV Banaskata-Chittorgarh D/C	463.90	132.00	**595.90**
NR-WR 765 kV Varanasi-Vindhyachal	0.00	0.00	**0.00**
NR -WR765 kV Chittorgarh-Neemach D/C	139.03	35.99	**175.02**
Total NR - WR	**2195.43**	**1408.35**	**3603.78**

	Apr'24	May'24	Fin. Year 2024-25
Name of Line			
Import of NR from ER (ER-NR)			
ER - NR 765 kV Sasaram - Fatehpur	111.08	133.89	**244.97**
ER - NR 765 kV Gaya - Varanasi 2*S/C	50.67	114.34	**165.01**
ER - NR 765 kV Gaya - Balia	213.79	293.69	**507.48**
ER - NR HVDC Alipurduar - Agra	0.00	0.00	**0.00**
ER-NR 400 kV Patna - Balia D/C	192.05	288.47	**480.52**
ER - NR 400 kV Muzaffarpur - Gorakhpur D/C	3.14	135.70	**138.84**
ER - NR 400 kV Biharshariff - Balia D/C	0.00	10.98	**10.98**
ER - NR 400 kV Motihari - Gorakhpur D/C	63.12	164.75	**227.87**
ER - NR 400 kV Biharshariff - Varanasi D/C	3.31	1.28	**4.59**
ER - NR 400 kV Sasaram -Varanasi	41.13	16.74	**57.87**
ER - NR 400 kV Sasaram -Allahabad	0.13	1.97	**2.10**
ER - NR 220 kV Sahupuri-Karamnasa	26.02	26.80	**52.82**
ER - NR 132 kV Sahupuri - Karamnasa	2.88	2.88	**5.76**
ER - NR 132 kV 132 kV Rihand - Nagar Untari	0.00	0.66	**0.66**
ER - NR 132 kV Rihand - Garhwa	0.00	0.72	**0.72**
ER-NR 400 kV Naubatpur - Balia D/C	44.01	81.16	**125.17**
ER-NR 400 kV Biharshriff - Sahupuri D/C	2.89	29.05	**31.94**
Total ER-NR	**754.22**	**1303.08**	**2057.30**
Import of NR from NER (NER-NR)			
NER-NR HVDC Biswanath Chariali - Agra	0.00	0.00	**0.00**
Total NER - NR	**0.00**	**0.00**	**0.00**

	Apr'24	May'24	Fin. Year 2024-25
Name of Line			
Export of NR to ER (NR-ER)			
NR - ER 765 kV Sasaram - Fatehpur	0.00	0.46	**0.46**
NR - ER 765 kV Gaya - Varanasi 2*S/C	65.96	56.17	**122.13**
NR - ER 765 kV Gaya - Balia	0.00	0.00	**0.00**
NR - ER HVDC Alipurduar - Agra	0.00	0.00	**0.00**
NR - ER 400 kV Patna - Balia 2xD/C	0.00	0.00	**0.00**
NR - ER 400 kV Muzaffarpur - Gorakhpur D/C	87.39	32.41	**119.80**
NR - ER 400 kV Biharshariff - Balia D/C	148.68	86.56	**235.24**
ER - NR 400 kV Motihari - Gorakhpur D/C	0.29	1.01	**1.30**
NR - ER 400 kV Biharshariff - Varanasi D/C	19.06	0.00	**19.06**
NR - ER 400 kV Sasaram -Varanasi	0.00	2.63	**2.63**
NR - ER 400 kV Sasaram -Allahabad	10.41	12.82	**23.23**
NR - ER 220 kV Sahupuri-Karamnasa	0.00	0.00	**0.00**
NR - ER 132 kV Sahupuri - Karamnasa	1.44	1.23	**2.67**
NR - ER 132 kV 132 kV Rihand - Nagar Untari	0.00	0.00	**0.00**
ER - NR 132 kV Rihand - Garhwa	18.99	19.86	**38.85**
NR-ER 400 kV Balia- Naubatpur D/C	0.00	1.31	**1.31**
NR-ER 400 kV Sahupuri- Biharshariff D/C	0.00	1.60	**1.60**
Total NR - ER	**352.22**	**216.06**	**568.28**
Export of NR to NER (NER-NR)			
NR-NER HVDC Biswanath Chariali - Agra	433.33	466.77	**900.10**
Total NR - NER	**433.33**	**466.77**	**900.10**

	Apr'24	May'24	Fin. Year 2024-25
Name of Line			
Export of WR to ER (WR-ER)			
WR - ER 765 kV Ranchi - Dharamjaygarh 2xS/C	388.72	751.76	**1140.48**
WR - ER 765 kV Dharamjaygarh - Jharsuguda 2xD/C	721.67	524.74	**1246.40**
WR - ER 765 kV Durg - Jharsuguda D/C	0.05	1.25	**1.30**
WR - ER 400 kV Sipat - Ranchi D/C	51.26	216.17	**267.42**
WR - ER 400 kV Raigarh - Jharsuguda 2xD/C	3.12	46.71	**49.83**
WR - ER 220 kV Korba - Budhipadar D/C	8.86	26.39	**35.25**
WR - ER 220 kV Raigarh - Budhipadar	7.90	2.55	**10.45**
Total WR - ER	**1181.58**	**1569.56**	**2751.14**
Import of WR from ER (ER - WR)			
ER - WR 765 kV Ranchi - Dharamjaygarh 2xS/C	48.43	15.64	**64.08**
ER - WR 765 kV Dharamjaygarh - Jharsuguda 2XD/C	2.75	169.79	**172.55**
ER - WR 765 kV Durg - Jharsuguda D/C	376.32	377.94	**754.26**
ER - WR 400 kV Sipat - Ranchi D/C	22.01	4.57	**26.58**
ER - WR 400 kV Raigarh - Jharsuguda 2xD/C	195.37	120.88	**316.26**
ER - WR 220 kV Korba - Budhipadar D/C	1.27	8.92	**10.19**
ER - WR 220 kV Raigarh - Budhipadar	15.69	25.32	**41.01**
Total ER - WR	**661.85**	**723.06**	**1384.92**
Export of ER to NER (ER - NER)			
ER - NER 400 kV Binaguri - Bongaigaon D/C	20.06	23.09	**43.14**
ER - NER 400 kV Alipurduar - Bongaigaon D/C	16.74	47.75	**64.49**
ER - NER 220 kV Birpara - Salakati D/C	2.74	6.73	**9.47**
Total ER - NER	**39.54**	**77.57**	**117.11**
Import of ER from NER (NER - ER)			
NER - ER 400 kV Binaguri - Bongaigaon D/C	79.80	122.93	**202.72**
NER - ER 400 kV Alipurduar - Bongaigaon D/C	164.15	208.72	**372.87**
NER - ER 220 kV Birpara - Salakati D/C	18.58	38.63	**57.21**
Total NER - ER	**262.52**	**370.28**	**632.81**

	Apr'24	May'24	Fin. Year 2024-25
Name of Line			
Export of ER to SR (ER - SR)			
ER - SR HVDC Gazuwaka	413.69	249.41	**663.10**
ER - SR HVDC Talchar - Kolar	1069.04	1104.35	**2173.39**
ER - SR 765 kV Angul- Srikakulam D/C	1640.93	1313.30	**2954.23**
Total ER - SR	**3123.66**	**2667.05**	**5790.71**
Import of ER from SR (SR - ER)			
SR - ER HVDC Gazuwaka	0.00	16.28	**16.28**
SR - ER HVDC Talchar - Kolar	0.00	0.00	**0.00**
SR - ER 765 kV Angul- Srikakulam D/C	0.00	0.00	**0.00**
Total SR - ER	**0.00**	**16.28**	**16.28**
Export of WR to SR (WR-SR)			
WR - SR HVDC Bhadrawati	601.92	221.20	**823.12**
WR - SR 765 kV Sholapur - Raichur 2xS/C	802.05	292.07	**1094.12**
WR - SR 765 kV Wardha - Nizamabad D/C	1407.32	763.52	**2170.84**
WR - SR 400KV Kolhapur-Kudgi D/C	0.00	0.02	**0.02**
WR-SR HVDC Raigarh-Pugalur	2668.37	1464.59	**4132.96**
WR-SR 220kV Xeldem - Ambewadi S/C	73.01	75.81	**148.82**
WR-SR 220kV Ponda - Ambewadi S/C	43.06	54.26	**97.32**
WR-SR 765 kV Warora - Warangal D/C	1603.13	954.21	**2557.34**
Total WR - SR	**7198.86**	**3825.69**	**11024.55**
Import of WR from SR (SR - WR)			
SR - WR HVDC Bhadrawati	2.36	182.40	**184.76**
SR - WR 765 kV Raichur - Sholapur 2xS/C	0.55	262.88	**263.43**
SR - WR 765 kV Wardha - Nizamabad D/C	0.00	6.86	**6.86**
SR - WR 400KV Kolhapur-Kudgi D/C	429.02	646.95	**1075.97**
SR-WR HVDC Pugalur-Raigarh	0.00	0.00	**0.00**
WR-SR 220kV Ambewadi - Xeldem S/C	0.00	0.00	**0.00**
WR-SR 220kV Ambewadi-Ponda S/C	0.00	0.00	**0.00**
SR-WR 765 kV Warangal - Warora D/C	0.00	6.90	**6.90**

	Apr'24	May'24	Fin. Year 2024-25
Name of Line			
Total SR - WR	**431.93**	**1105.99**	**1537.92**
TOTAL ALL INDIA	**21906**	**23669**	**45575**
***In case of mutiple ckt / DC pole year corres. to the commissioning of final element**			

15.1 Import-Export of NR with WR during May 2024

Import of NR from WR (WR-NR)

Date	WR - NR 765 kV Gwalior - Agra 2xS/C	WR-NR 765 kV Gwalior-Phagi 2xS/C	WR-NR 765 kV Jabalpur- Orai D/C	WR-NR 765 kV Satna- Orai	WR-NR 765 kV Gwalior-Orai	WR - NR HVDC Champa-Kurukshetra	WR - NR HVDC Vindhyachal	WR - NR 400 kV Zerda-Kankroli	WR - NR 400 kV Zerda-Bhinmal	WR-NR 400 kV RAPP C - Shujalpur D/C	WR - NR 220 kV Bhanpura - Ranpur	WR - NR 220 kV Bhanpura - Modak	WR - NR 220 kV Auraiya - Malanpur / Mehgaon	WR-NR 400 KV Rihand Vindhyachal D/C	WR - NR HVDC Mundra - M'garh	WR -NR 765 kV Banaskata - Chittorgarh D/C	WR-NR 765 kV Varanasi-Vindhyachal	WR -NR 765 kV Neemach - Chittorgarh D/C	Total WR - NR
1-May-24	24.83	0.00	14.22	17.23	0.00	50.60	0.00	0.00		0.00	1.99	2.28	0.00	0.00	45.30	0.00	62.12	0.00	**218.57**
2-May-24	19.15	1.82	11.34	17.82	0.00	72.06	0.00	0.00		0.00	2.00	2.22	0.00	0.00	42.37	0.00	59.99	0.00	**228.77**
3-May-24	27.55	8.04	16.28	19.08	0.00	65.08	0.00	0.00		0.00	2.30	1.56	0.00	0.00	42.38	0.00	63.07	0.00	**245.34**
4-May-24	28.11	5.36	16.40	19.84	0.00	71.25	0.00	0.00		0.00	2.17	2.45	0.00	0.00	42.38	0.00	60.50	0.00	**248.46**
5-May-24	30.31	3.32	20.67	17.79	0.00	69.12	0.00	0.00		0.00	2.08	2.44	0.00	0.00	36.50	0.07	51.73	0.00	**234.03**
6-May-24	33.07	11.18	24.12	21.39	0.00	70.24	0.47	0.00		0.00	2.17	2.45	0.00	0.00	31.62	3.84	52.16	0.00	**252.71**
7-May-24	38.12	15.05	28.90	22.98	0.00	72.15	0.00	0.00		0.04	2.52	2.49	0.00	0.00	42.38	3.81	51.54	0.00	**279.98**
8-May-24	39.87	16.54	29.81	24.01	0.00	71.01	0.00	0.00		1.56	1.76	2.45	0.00	0.00	27.64	2.84	52.37	0.00	**269.86**
9-May-24	34.04	14.90	24.64	22.53	0.00	74.46	1.29	0.00		1.10	2.31	2.57	0.00	0.00	23.10	0.00	47.80	0.00	**248.74**
10-May-24	30.84	12.31	23.18	21.34	0.00	67.56	4.78	0.00	0.00	0.00	2.54	2.46	0.00	0.00	24.21	0.00	44.61	0.00	**233.83**
11-May-24	28.25	12.67	23.93	19.85	0.00	19.43	3.51	0.00		0.00	2.50	2.04	0.00	0.00	24.21	0.00	45.87	0.00	**182.26**
12-May-24	31.49	15.68	27.24	20.89	0.00	32.12	0.00	0.00		0.00	2.38	2.04	0.00	0.00	24.21	0.00	52.72	0.00	**208.77**
13-May-24	42.54	22.07	36.35	23.52	0.00	56.82	0.00	0.00		3.96	2.61	2.36	0.00	0.00	27.90	0.00	64.42	3.59	**286.14**
14-May-24	35.25	10.58	25.51	20.70	0.00	94.96	0.00	0.00		0.00	2.42	2.38	0.00	0.00	29.83	0.00	68.95	0.75	**291.33**
15-May-24	33.10	10.24	25.77	20.33	0.00	109.86	0.00	0.00		0.00	2.37	2.46	0.00	0.00	42.05	0.00	68.56	0.00	**314.74**
16-May-24	39.20	19.95	33.10	22.30	0.00	88.51	0.00	0.00		1.37	2.63	2.54	0.00	0.00	37.88	0.00	71.33	1.86	**320.67**
17-May-24	40.64	15.50	34.32	22.15	0.00	91.60	0.00	0.00		1.25	2.50	2.50	0.00	0.00	32.09	3.28	71.45	0.90	**318.18**
18-May-24	42.54	12.10	33.96	21.41	0.00	106.89	0.00	0.00		1.13	2.37	2.42	0.00	0.00	36.91	6.16	72.39	0.42	**338.70**
19-May-24	41.53	13.53	32.17	21.31	0.00	117.55	0.00	0.00		1.87	2.54	2.54	0.00	0.00	42.38	3.89	72.96	0.47	**352.74**
20-May-24	42.98	14.86	36.64	23.50	0.00	118.01	0.00	0.00		1.76	2.77	2.54	0.00	0.00	42.38	5.74	68.91	0.00	**360.09**
21-May-24	44.94	16.58	38.31	23.37	0.00	107.45	0.00	1.02		2.63	2.61	2.60	0.00	0.00	42.38	9.83	69.85	1.16	**362.73**
22-May-24	45.49	17.61	38.83	23.90	0.00	123.00	3.18	1.27		3.16	2.78	2.57	0.00	0.00	39.37	12.38	69.37	1.52	**384.43**
23-May-24	43.14	14.86	36.28	22.75	0.00	122.50	7.14		1.38	2.44	2.73	2.63	0.00	0.00	42.38	12.10	71.47	1.10	**382.90**
24-May-24	44.68	15.66	36.44	22.37	0.00	126.21	7.58	2.90		3.47	2.78	2.81	0.00	0.00	42.38	14.42	76.40	1.85	**399.95**
25-May-24	46.50	16.82	41.60	21.07	0.00	130.20	12.01	2.05		4.35	2.70	2.81	0.00	0.00	42.38	10.84	85.25	2.80	**421.38**
26-May-24	51.25	15.47	43.09	19.91	0.00	125.70	11.93	4.35		5.41	2.58	2.91	0.74	0.00	42.84	21.06	80.86	3.72	**431.82**
27-May-24	55.89	19.94	51.57	24.15	0.00	104.49	12.07	5.30		6.87	2.66	2.97	1.20	0.00	42.37	27.12	73.20	4.90	**434.70**
28-May-24	59.00	14.68	48.88	24.65	0.00	117.64	11.98	7.63		6.94	2.68	3.10	1.11	0.00	40.09	36.85	79.59	5.74	**460.56**
29-May-24	58.01	7.27	46.67	21.77	0.00	130.08	12.02	8.56		4.98	2.64	2.78	0.00	0.00	39.18	41.56	80.12	0.00	**455.64**
30-May-24	48.74	8.50	33.75	22.10	0.00	127.32	8.60	2.20		2.47	2.41	2.01	0.00	0.00	41.12	21.80	71.20	1.60	**393.82**
31-May-24	54.10	13.47	42.70	23.32	0.00	82.86	3.69			3.98	2.53	1.95	0.00	0.00	32.49	24.64	69.10	2.98	**357.81**
Total	**1235.15**	**396.56**	**976.67**	**669.33**	**0.00**	**2816.73**	**100.25**	**35.28**	**1.38**	**60.74**	**76.03**	**76.33**	**3.05**	**0.00**	**1144.70**	**262.23**	**2029.86**	**35.36**	**9919.65**

Disclaimer:- Blank entry if the line under outage/shutdown on corresponding day

15.1 Import-Export of NR with WR during May 2024

Export of NR to WR (NR- WR)

Date	NR - WR 765 kV Gwalior - Agra 2xS/C	NR-WR 765 kV Gwalior-Phagi 2xS/C	NR-WR 765 kV Jabalpur- Orai D/C	NR-WR 765 kV Satna- Orai	NR-WR 765 kV Gwalior-Orai	NR - WR HVDC Champa-Kurukshetra	NR - WR HVDC Vindhyachal	NR - WR 400 kV Zerda-Kankroli	NR - WR 400 kV Zerda-Bhinmal	NR-WR 400 kV RAPP C - Shujalpur D/C	WR - NR 220 kV Bhanpura - Ranpur	WR - NR 220 kV Bhanpura - Modak	NR - WR 220 kV Auraiya - Malanpur / Mehgaon	NR-WR 400 kV Rihand-Vindhyachal D/C	NR - WR HVDC Mundra - M'garh	NR - WR 765 kV Banaskata - Chittorgarh D/C	NR-WR 765 kV Varanasi-Vindhyachal	WR -NR 765 kV Chittorgarh-Neemach D/C	Total NR - WR
1-May-24	0.00	1.42	0.00	0.00	5.14	0.00	9.76	2.95		5.65	0.00	0.00	1.26	20.91	0.00	6.32	0.00	6.28	**59.69**
2-May-24	0.00	0.00	0.00	0.00	6.51	0.00	12.21	4.65		6.78	0.00	0.00	2.05	22.37	0.00	18.91	0.00	7.06	**80.54**
3-May-24	0.00	0.00	0.00	0.00	8.34	0.00	12.12	5.79		4.84	0.00	0.00	1.98	22.05	0.00	7.72	0.00	5.60	**68.44**
4-May-24	0.00	0.00	0.00	0.00	7.09	0.00	12.13	7.63		5.24	0.00	0.00	1.97	22.38	0.00	10.96	0.00	5.85	**73.25**
5-May-24	0.00	0.00	0.00	0.00	6.84	0.00	12.19	3.40		3.03	0.00	0.00	1.46	22.45	0.00	0.00	0.00	3.55	**52.92**
6-May-24	0.00	0.00	0.00	0.00	8.74	0.00	6.09	1.48		1.52	0.00	0.00	1.81	22.18	0.00	0.00	0.00	3.26	**45.08**
7-May-24	0.00	0.00	0.00	0.00	8.95	0.00	4.85	1.15		0.00	0.00	0.00	1.22	22.21	0.00	0.00	0.00	1.82	**40.20**
8-May-24	0.00	0.00	0.00	0.00	8.89	0.00	4.89	1.76		0.00	0.00	0.00	1.18	20.95	0.00	0.00	0.00	0.69	**38.36**
9-May-24	0.00	0.00	0.00	0.00	8.95	0.00	3.50	2.98		0.00	0.00	0.00	1.31	22.00	0.00	1.78	0.00	0.68	**41.20**
10-May-24	0.00	0.00	0.00	0.00	8.74	0.00	0.00	5.20	0.00	0.65	0.00	0.00	1.55	22.01	0.00	7.36	0.00	1.92	**47.43**
11-May-24	0.00	0.00	0.00	0.00	10.36	0.00	0.00	9.94		1.61	0.00	0.00	1.14	22.40	0.00	18.56	0.00	1.84	**65.85**
12-May-24	0.00	0.00	0.00	0.00	10.81	0.00	4.89	9.80		0.47	0.00	0.00	0.52	22.31	0.00	21.15	0.00	1.10	**71.05**
13-May-24	0.00	0.00	0.00	0.00	11.51	0.00	4.90	6.90		0.00	0.00	0.00	0.12	22.30	0.00	14.60	0.00	0.00	**60.33**
14-May-24	0.00	0.00	0.00	0.00	8.69	0.00	3.25	5.86		0.27	0.00	0.00	0.25	22.23	0.00	8.54	0.00	0.00	**49.09**
15-May-24	0.00	0.00	0.00	0.00	8.94	0.00	1.25	6.20		1.53	0.00	0.00	1.28	22.22	0.00	11.32	0.00	0.49	**53.23**
16-May-24	0.00	0.00	0.00	0.00	11.66	0.00	0.35	3.25		0.00	0.00	0.00	1.22	22.15	0.00	4.78	0.00	0.00	**43.41**
17-May-24	0.00	0.00	0.00	0.00	10.59	0.00	1.95	1.24		0.00	0.00	0.00	0.98	22.34	0.00	0.00	0.00	0.00	**37.10**
18-May-24	0.00	0.00	0.00	0.00	9.57	0.00	7.38	0.30		0.00	0.00	0.00	1.05	22.27	0.00	0.00	0.00	0.00	**40.57**
19-May-24	0.00	0.00	0.00	0.00	9.73	0.00	10.28	1.45		0.00	0.00	0.00	0.54	22.30	0.00	0.00	0.00	0.00	**44.30**
20-May-24	0.00	0.00	0.00	0.00	10.56	0.00	10.32	1.42		0.00	0.00	0.00	0.70	22.26	0.00	0.00	0.00	0.12	**45.38**
21-May-24	0.00	0.00	0.00	0.00	10.01	0.00	4.62	0.00		0.00	0.00	0.00	1.11	21.38	0.00	0.00	0.00	0.00	**37.12**
22-May-24	0.00	0.00	0.00	0.00	10.47	0.00	0.00	0.00		0.00	0.00	0.00	1.30	22.23	0.00	0.00	0.00	0.00	**34.00**
23-May-24	0.00	0.00	0.00	0.00	9.72	0.00	1.62		0.00	0.00	0.00	0.00	1.23	21.86	0.00	0.00	0.00	0.00	**34.43**
24-May-24	0.00	0.00	0.00	0.00	10.19	0.00	0.00	0.00		0.00	0.00	0.00	1.35	22.41	0.00	0.00	0.00	0.00	**33.95**
25-May-24	0.00	0.00	0.00	0.00	10.67	0.00	0.00	0.00		0.00	0.00	0.00	0.63	22.39	0.00	0.00	0.00	0.00	**33.69**
26-May-24	0.00	0.00	0.00	0.00	10.08	0.00	0.00	0.00		0.00	0.00	0.00	0.00	21.72	0.00	0.00	0.00	0.00	**31.80**
27-May-24	0.00	0.00	0.00	0.00	11.02	0.00	0.00	0.00		0.00	0.00	0.00	0.00	22.20	0.00	0.00	0.00	0.00	**33.22**
28-May-24	0.00	0.00	0.00	0.00	8.14	0.00	0.00	0.00		0.00	0.00	0.00	0.00	20.46	0.00	0.00	0.00	0.00	**28.60**
29-May-24	0.00	0.00	0.00	0.00	6.30	0.00	0.00	0.00		0.00	0.00	0.00	0.50	20.41	0.00	0.00	0.00	-4.27	**22.94**
30-May-24	0.00	0.00	0.00	0.00	7.64	0.00	0.00	0.00		0.00	0.00	0.00	0.82	21.70	0.00	0.00	0.00	0.00	**30.16**
31-May-24	0.00	0.00	0.00	0.00	8.67	0.00	0.00			0.00	0.00	0.00	0.53	21.82	0.00	0.00	0.00	0.00	**31.02**
Total	**0.00**	**1.42**	**0.00**	**0.00**	**283.52**	**0.00**	**128.55**	**83.35**	**0.00**	**31.59**	**0.00**	**0.00**	**31.06**	**680.87**	**0.00**	**132.00**	**0.00**	**35.99**	**1408.35**

Disclaimer:- Blank entry if the line under outage/shutdown on corrosponding day

15.2 Import-Export of NR with ER & NER during May 2024

Date	Import of NR from ER (ER-NR)																		Import of NR from NER (NER-NR)	
	ER - NR 765 kV Sasaram - Fatehpur	ER - NR 765 kV Gaya - Varanasi 2*S/C	ER - NR 765 kV Gaya Balia	ER - NR HVDC Alipurduar - Agra	ER-NR 400 kV Patna - Balia D/C	ER - NR 400 kV Muzaffarpur - Gorakhpur D/C	ER - NR 400 kV Biharshariff - Balia D/C	ER - NR 400 kV Motihari - Gorakhpur D/C	ER - NR 400 kV Biharshariff - Varanasi D/C	ER-NR 400 kV Sasaram - Varanasi	ER-NR 400 kV Sasaram - Allahabad	ER - NR 220 kV Sahupuri-Karamnasa	ER - NR 132 kV Sahupuri - Karamnasa	ER - NR 132 kV Rihand - Nagar Untari	NR - ER 132 kV Rihand - Garhwa	ER-NR 400 kV Naubatpur - Balia D/C	ER-NR 400 kV Biharshriff - Sahupuri D/C	Total ER -NR	NER-NR HVDC Biswanath Chariali - Agra	Total NER-NR
1-May-24	0.00	0.00	6.66		2.01	0.00	0.00	0.47		1.61	0.00	0.31			0.00	1.44		12.50	0.00	0.00
2-May-24	0.45	0.00	6.45		2.15	0.00	0.00	0.00		1.84	0.00	0.26		0.00	0.00	0.00		11.15	0.00	0.00
3-May-24	1.35	0.00	7.41		2.31	0.00	0.00	0.36		1.78	0.00	0.38			0.00	0.65		14.24	0.00	0.00
4-May-24	2.33	0.00	7.43		6.70	0.00	0.00	2.11		1.85	0.00	0.48	0.00		0.00	0.55		21.45	0.00	0.00
5-May-24	2.91	0.00	8.45		9.38	0.00	0.00	4.12		1.54	0.00	0.78	0.48		0.00	1.20		28.86	0.00	0.00
6-May-24	7.66	7.84	9.72		13.94	7.49	0.43	8.31		0.78	0.00	1.85			0.00	4.37		62.39	0.00	0.00
7-May-24	9.17	9.53	10.56		14.06	8.33	0.00	8.90		0.00	0.64	2.03			0.00	3.29		66.51	0.00	0.00
8-May-24	9.40	10.63	11.42		12.93	8.55	1.46	8.45		0.00	0.31	2.03	0.48		0.00	4.12		69.78	0.00	0.00
9-May-24	9.39	11.95	11.78		15.72	8.06	1.51	9.21		0.00	0.10	2.00			0.00	4.81		74.53	0.00	0.00
10-May-24	8.60	11.51	11.50		13.78	6.93	0.18	8.80	0.00	0.00	0.00	1.78			0.00	4.14	0.00	67.22	0.00	0.00
11-May-24	6.72	9.61	11.05		11.87	3.97	0.00	5.77		0.27	0.00	1.55	0.00		0.00	3.56		54.37	0.00	0.00
12-May-24	7.29	10.32	12.51		14.22	7.82	0.52	7.42		0.00	0.00	1.98			0.00	4.23	0.00	66.31	0.00	0.00
13-May-24	6.65	7.53	13.39		13.09	5.86	0.00	6.95	0.00	0.00	0.00	1.45			0.00	3.80		58.72	0.00	0.00
14-May-24	1.51	0.00	10.85		7.25	0.00	0.00	2.54		0.75	0.00				0.00	1.61		24.51	0.00	0.00
15-May-24	0.54	6.20	10.34		3.52	0.00	0.00	1.61		1.03	0.00	0.35			0.00	1.37		24.96	0.00	0.00
16-May-24	3.08	0.00	12.17		5.73	0.00	0.00	2.77		0.75	0.00	0.75	0.48		0.00	1.08		26.81	0.00	0.00
17-May-24	2.95	0.00	12.05		7.74	0.00	0.00	3.23		0.83	0.00	0.85	0.00		0.00	1.73		29.38	0.00	0.00
18-May-24	2.30	0.00	10.70		7.85	0.31	0.00	4.02	0.00	0.54	0.00	0.94	0.48		0.00	2.25	0.70	30.09	0.00	0.00
19-May-24	4.55	0.65	11.99		11.25	4.61	0.00	6.30		0.00	0.18	0.82			0.00	3.34	2.34	46.03	0.00	0.00
20-May-24	4.97	5.49	7.36		13.54	5.94	0.17	7.41		0.00	0.08	0.72	0.48		0.00	3.98	3.44	53.58	0.00	0.00
21-May-24	4.82	0.58	11.19		10.89	0.82	0.00	5.25		0.00	0.00	0.81	0.48		0.00	3.01	2.34	40.19	0.00	0.00
22-May-24	4.50	2.44	6.98		11.50	4.33	0.00	5.82		0.14	0.00		0.00	0.66	0.00	3.51	2.43	42.31	0.00	0.00
23-May-24	3.54	0.40	5.30		7.90	1.33	0.00	4.51		0.28	0.00	0.43	0.00		0.00	2.70	1.19	27.58	0.00	0.00
24-May-24	3.13	1.05	6.40		9.95	0.00	0.00	3.56		0.54	0.00	0.35			0.00	2.65	1.15	28.78	0.00	0.00
25-May-24	0.00	0.00	5.50		4.85	0.00	0.00	2.01		0.93	0.00				0.00	1.90	0.00	15.19	0.00	0.00
26-May-24	1.20	0.00	6.32		3.35	2.58	0.00	2.66		0.48	0.00	0.05			0.00	1.48	0.00	18.12	0.00	0.00
27-May-24	6.50	7.87	12.00		11.00	12.33	0.90	8.78	0.00	0.00	0.00	0.86			0.00	4.50	4.50	69.24	0.00	0.00
28-May-24	5.31	3.64	7.65		8.99	12.86	1.98	7.87		0.00	0.32	0.92	0.00		0.00	2.11	3.30	54.95	0.00	0.00
29-May-24	2.67	0.00	9.41		7.78	10.18	0.00	6.41	1.28	0.22	0.00	0.45			0.00	1.23	1.28	40.91	0.00	0.00
30-May-24	2.80	0.00	8.70		8.72	8.87	0.00	7.39		0.58	0.00	0.40	0.00		0.72	2.03	0.95	41.16	0.00	0.00
31-May-24	7.60	7.10	10.45		14.50	14.53	3.83	11.74		0.00	0.34	1.22			0.00	4.52	5.43	81.26	0.00	0.00
Total	**133.89**	**114.34**	**293.69**	**0.00**	**288.47**	**135.70**	**10.98**	**164.75**	**1.28**	**16.74**	**1.97**	**26.80**	**2.88**	**0.66**	**0.72**	**81.16**	**29.05**	**1303.08**	**0.00**	**0.00**

Disclaimer:- Blank entry if the line under outage/shutdown on corresponding day

15.2 Import-Export of NR with ER & NER during May 2024

Date	Export of NR to ER (NR- ER)																		Export of NR to NER (NR-NER)	
	NR - ER 765 kV Sasaram - Fatehpur	NR - ER 765 kV Gaya - Varanasi 2*S/C	NR - ER 765 kV Gaya Balia	NR - ER HVDC Alipurduar - Agra	NR-ER 400 kV Patna - Balia D/C	NR - ER 400 kV Muzaffarpur - Gorakhpur D/C	NR - ER 400 kV Biharshariff - Balia D/C	NR - ER 400 kV Motihari - Gorakhpur D/C	NR - ER 400 kV Biharshariff - Varanasi D/C	NR-ER 400 kV Sasaram - Varanasi	NR-ER 400 kV Sasaram - Allahabad	NR - ER 220 kV Sahupuri-Karamnasa	NR - ER 132 kV Sahupuri - Karamnasa	NR - ER 132 kV Rihand - Nagar Untari	NR - ER 132 kV Rihand - Garhwa	NR-ER 400 kV Balia-Naubatpur D/C	NR-ER 400 kV Sahupuri-Biharshariff D/C	Total NR -ER	NR-NER HVDC Biswanath Chariali - Agra	Total NR-NER
1-May-24	0.25	8.93	0.00		0.00	5.77	7.29	0.00		0.00	0.60	0.00			0.87	0.00		**23.71**	20.13	**20.13**
2-May-24	0.00	10.01	0.00		0.00	7.42	7.24	1.01		0.00	0.84	0.00		0.00	0.83	1.31		**28.66**	17.57	**17.57**
3-May-24	0.00	8.48	0.00		0.00	5.01	6.65	0.00		0.00	0.75	0.00			0.74	0.00		**21.63**	16.06	**16.06**
4-May-24	0.00	6.23	0.00		0.00	3.75	6.35	0.00		0.00	0.68	0.00	0.00		0.62	0.00		**17.63**	16.50	**16.50**
5-May-24	0.00	0.61	0.00		0.00	0.12	4.60	0.00		0.00	0.45	0.00	0.00		0.82	0.00		**6.60**	16.12	**16.12**
6-May-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00		0.00	0.61	0.00			0.77	0.00		**1.38**	16.66	**16.66**
7-May-24	0.00	0.00	0.00		0.00	0.00	0.94	0.00		0.10	0.00	0.00			0.51	0.00		**1.55**	17.10	**17.10**
8-May-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00		0.45	0.00	0.00	0.00		0.67	0.00		**1.12**	16.55	**16.55**
9-May-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00		0.25	0.00	0.00			0.55	0.00		**0.80**	16.69	**16.69**
10-May-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00	0.00	0.00	0.06	0.00			0.69	0.00	0.00	**0.75**	16.91	**16.91**
11-May-24	0.00	0.00	0.00		0.00	0.00	1.19	0.00		0.00	0.41	0.00	0.00		0.76	0.00		**2.36**	17.12	**17.12**
12-May-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00		0.04	0.09	0.00			0.52	0.00	0.00	**0.65**	16.23	**16.23**
13-May-24	0.00	0.00	0.00		0.00	0.00	0.46	0.00	0.00	0.05	0.05	0.00			0.48	0.00		**1.04**	16.05	**16.05**
14-May-24	0.00	3.21	0.00		0.00	1.16	4.16	0.00		0.00	0.85				0.54	0.00		**9.92**	17.14	**17.14**
15-May-24	0.00	0.00	0.00		0.00	2.72	6.00	0.00		0.00	1.12	0.00			0.61	0.00		**10.45**	17.93	**17.93**
16-May-24	0.00	0.80	0.00		0.00	0.92	4.14	0.00		0.00	0.85	0.00	0.00		0.63	0.00		**7.34**	16.94	**16.94**
17-May-24	0.00	1.12	0.00		0.00	1.31	4.67	0.00		0.00	0.90	0.00	0.00		0.45	0.00		**8.45**	16.81	**16.81**
18-May-24	0.00	2.81	0.00		0.00	0.00	4.10	0.00	0.00	0.00	0.65	0.00	0.00		0.59	0.00	0.00	**8.15**	17.37	**17.37**
19-May-24	0.00	0.00	0.00		0.00	0.00	1.46	0.00		0.27	0.00	0.00			0.70	0.00	0.00	**2.43**	16.94	**16.94**
20-May-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00		0.00	0.00	0.00	0.00		0.69	0.00	0.00	**0.69**	13.36	**13.36**
21-May-24	0.00	0.00	0.00		0.00	0.00	2.98	0.00		0.05	0.06	0.00	0.00		0.67	0.00	0.00	**3.76**	8.82	**8.82**
22-May-24	0.00	0.00	0.00		0.00	0.00	1.40	0.00		0.00	0.20		0.75	0.00	0.67	0.00	0.00	**3.02**	8.90	**8.90**
23-May-24	0.00	0.00	0.00		0.00	0.00	2.11	0.00		0.00	0.39	0.00	0.00		0.61	0.00	0.00	**3.11**	8.35	**8.35**
24-May-24	0.00	0.00	0.00		0.00	1.11	4.49	0.00		0.00	0.14	0.00			0.67	0.00	0.00	**6.41**	8.48	**8.48**
25-May-24	0.21	6.13	0.00		0.00	3.12	6.85	0.00		0.00	0.98				0.67	0.00	1.25	**19.21**	8.78	**8.78**
26-May-24	0.00	1.51	0.00		0.00	0.00	5.20	0.00		0.00	0.39	0.00			0.71	0.00	0.35	**8.16**	15.31	**15.31**
27-May-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00	0.00	0.70	0.70	0.00			0.72	0.00	0.00	**2.12**	14.24	**14.24**
28-May-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00		0.36	0.00	0.00	0.00		0.76	0.00	0.00	**1.12**	10.87	**10.87**
29-May-24	0.00	3.60	0.00		0.00	0.00	2.48	0.00	0.00	0.00	0.35	0.00			0.61	0.00	0.00	**7.04**	12.65	**12.65**
30-May-24	0.00	2.73	0.00		0.00	0.00	1.80	0.00		0.00	0.70	0.00	0.48		0.00	0.00	0.00	**5.71**	17.20	**17.20**
31-May-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00		0.36	0.00	0.00			0.73	0.00	0.00	**1.09**	16.99	**16.99**
Total	**0.46**	**56.17**	**0.00**	**0.00**	**0.00**	**32.41**	**86.56**	**1.01**	**0.00**	**2.63**	**12.82**	**0.00**	**1.23**	**0.00**	**19.86**	**1.31**	**1.60**	**216.06**	**466.77**	**466.77**

Disclaimer:- Blank entry if the line under outage/shutdown on corrosponding day

15.3 Import and Export of ER with WR & NER during May 2024

Date	Export of WR to ER (WR-ER)								Import of WR from ER (ER - WR)								Export of ER to NER (ER-NER)				Import of ER from NER (NER-ER)			
	WR - ER 765 kV Dharamjaygarh - Ranchi 2xS/C	WR - ER 765 kV Dharamjaygarh - Jharsuguda Q/C	WR - ER 765 kV Durg - Jharsuguda D/C	WR - ER 400 kV Sipat - Ranchi D/C	WR - ER 400 kV Raigarh - Jharsuguda- 2xD/C	WR - ER 220 kV Korba - Budhipadar D/C	WR - ER 220 kV Raigarh - Budhipadar	Total WR - ER	ER - WR 765 kV Dharamjaygarh - Ranchi 2xS/C	ER-WR 765 kV Dharamjaygarh - Jharsuguda D/C	ER-WR 765 kV Durg - Jharsuguda D/C	ER - WR 400 kV Sipat - Ranchi D/C	ER - WR 400 kV Raigarh - Jharsuguda- 2xD/C	ER - WR 220 kV Korba - Budhipadar D/C	ER - WR 220 kV Raigarh - Budhipadar	Total ER - WR	ER - NER 400 kV Binaguri - Bongaigaon D/C	ER - NER 400 kV Alipurduar - Bongaigaon D/C	ER - NER 220 kV Birpara - Salakati D/C	Total ER - NER	NER - ER 400 kV Binaguri - Bongaigaon D/C	NER - ER 400 kV Alipurduar - Bongaigaon 2xD/C	NER - ER 220 kV Birpara - Salakati D/C	Total NER - ER
1-May-24	28.1	41.6	0.9	5.7	1.9	0.8	0.3	**79.37**	0.0	0.0	1.4	0.0	3.1	0.0	0.3	**4.82**	0.15	0.11	0.07	**0.33**	3.67	6.44	0.94	**11.05**
2-May-24	18.7	35.6	0.0	4.2	2.6	0.7	0.0	**61.76**	0.0	0.0	4.6	0.0	1.8	0.0	0.7	**7.10**	0.07	0.00	0.00	**0.07**	2.43	5.17	0.86	**8.47**
3-May-24	24.1	27.5	0.0	5.1	2.0	0.6	0.0	**59.34**	0.0	0.0	5.7	0.0	1.8	0.0	0.8	**8.37**	0.80	0.55	0.14	**1.49**	2.54	4.97	0.62	**8.13**
4-May-24	24.8	26.0	0.0	5.1	1.1	0.9	0.0	**57.97**	0.0	0.4	6.2	0.0	2.9	0.0	0.4	**9.96**	0.95	0.64	0.21	**1.79**	1.74	3.61	0.27	**5.63**
5-May-24	23.2	19.7	0.0	4.5	0.4	0.5	0.0	**48.33**	0.0	0.7	6.5	0.0	5.2	0.1	0.8	**13.28**	0.00	0.00	0.00	**0.00**	5.88	10.49	1.84	**18.21**
6-May-24	10.7	18.2	0.0	1.4	0.0	0.0	0.0	**30.38**	0.2	0.0	10.4	0.3	9.3	1.1	1.4	**22.61**	0.00	0.00	0.00	**0.00**	7.70	13.52	2.27	**23.49**
7-May-24	7.6	12.0	0.0	1.3	0.1	0.4	0.0	**21.37**	1.5	1.1	13.8	1.0	7.4	0.6	1.7	**26.99**	0.00	0.00	0.00	**0.00**	5.88	10.26	1.92	**18.06**
8-May-24	6.9	14.5	0.0	1.0	0.4	1.1	0.1	**23.88**	2.6	0.7	12.7	0.6	4.7	0.1	1.2	**22.57**	0.00	0.00	0.00	**0.00**	4.38	8.05	1.44	**13.87**
9-May-24	3.1	16.4	0.0	0.8	0.1	0.6	0.0	**21.09**	4.8	0.2	14.7	1.4	5.2	0.2	1.4	**27.83**	0.00	0.00	0.00	**0.00**	4.06	7.63	1.35	**13.05**
10-May-24	5.3	19.4	0.0	1.9	0.0	0.8	0.0	**27.44**	3.5	0.6	18.4	1.2	8.2	0.3	1.5	**33.89**	0.00	0.00	0.00	**0.00**	3.77	7.59	1.17	**12.53**
11-May-24	6.3	28.3	0.0	3.5	0.1	1.0	0.0	**39.37**	1.9	5.0	20.3	0.0	7.7	0.1	1.0	**36.03**	0.00	0.00	0.00	**0.00**	4.22	8.45	1.11	**13.79**
12-May-24	7.9	28.8	0.0	3.6	0.2	1.1	0.0	**41.54**	1.1	3.5	20.2	0.0	6.0	0.1	1.1	**32.12**	0.00	0.00	0.01	**0.01**	3.51	7.13	0.70	**11.34**
13-May-24	16.8	26.3	0.0	6.2	0.8	0.8	0.0	**50.94**	0.0	2.6	17.8	0.0	3.6	0.0	0.8	**24.89**	0.32	0.34	0.14	**0.80**	1.45	3.28	0.18	**4.91**
14-May-24	23.1	30.3	0.0	8.5	2.5	0.6	0.1	**65.11**	0.0	1.1	17.5	0.0	2.3	0.4	0.2	**21.56**	1.27	2.29	0.68	**4.24**	0.24	1.24	0.01	**1.49**
15-May-24	27.2	26.6	0.0	9.4	2.4	0.3	0.6	**66.71**	0.0	3.8	16.9	0.0	2.2	1.2	0.0	**24.06**	1.24	2.33	0.52	**4.10**	0.64	1.86	0.15	**2.66**
16-May-24	29.0	24.8	0.0	9.4	1.7	0.8	0.3	**66.03**	0.0	6.4	18.7	0.0	3.2	0.6	0.2	**29.12**	1.96	3.66	0.77	**6.39**	0.27	0.77	0.04	**1.08**
17-May-24	32.8	19.1	0.0	9.6	0.7	0.7	0.1	**62.89**	0.0	4.7	14.3	0.0	3.1	0.1	0.3	**22.52**	1.00	1.57	0.13	**2.71**	0.73	1.53	0.35	**2.61**
18-May-24	31.6	18.0	0.0	9.7	1.3	0.7	0.1	**61.35**	0.0	8.3	16.4	0.0	2.9	0.4	0.3	**28.41**	0.00	0.00	0.00	**0.00**	2.31	5.64	1.45	**9.40**
19-May-24	26.0	11.9	0.0	7.7	2.6	0.8	0.2	**49.14**	0.0	5.1	15.8	0.0	2.3	0.5	0.3	**23.95**	0.00	0.00	0.00	**0.00**	4.28	7.64	1.69	**13.61**
20-May-24	20.2	8.4	0.0	6.5	2.0	1.9	0.1	**38.96**	0.0	4.5	16.9	0.0	2.1	0.0	0.4	**23.88**	0.48	1.07	0.17	**1.71**	1.65	3.03	0.76	**5.44**
21-May-24	28.4	4.5	0.0	11.0	2.5	3.6	0.2	**50.15**	0.0	9.4	9.5	0.0	2.0	0.0	0.2	**21.11**	1.70	4.00	0.53	**6.23**	0.07	0.36	0.06	**0.50**
22-May-24	27.5	7.5	0.0	9.4	1.6	2.2	0.0	**48.18**	0.0	10.4	16.2	0.0	2.5	0.0	0.5	**29.62**	1.89	4.74	0.82	**7.44**	0.05	0.22	0.03	**0.30**
23-May-24	32.7	9.3	0.0	10.3	2.1	0.4	0.0	**54.80**	0.0	10.1	16.3	0.0	2.4	0.0	1.6	**30.40**	3.48	7.60	0.82	**11.90**	0.00	0.02	0.06	**0.08**
24-May-24	36.3	10.2	0.0	10.8	3.0	0.5	0.0	**60.75**	0.0	9.0	14.5	0.0	2.2	0.0	1.5	**27.18**	3.52	7.63	0.76	**11.91**	0.00	0.00	0.01	**0.01**
25-May-24	50.7	10.3	0.0	14.2	3.2	0.8	0.0	**79.29**	0.0	14.7	9.6	0.0	2.8	0.2	1.2	**28.45**	3.13	8.89	0.89	**12.92**	0.03	0.18	0.09	**0.30**
26-May-24	40.2	5.5	0.0	11.6	2.0	0.2	0.0	**59.61**	0.0	15.3	10.4	0.0	3.7	0.5	1.0	**30.91**	1.12	2.33	0.07	**3.51**	0.55	0.76	0.37	**1.67**
27-May-24	21.4	2.5	0.0	6.1	0.9	0.6	0.0	**31.47**	0.0	8.5	9.9	0.0	4.8	0.5	1.0	**24.66**	0.00	0.00	0.00	**0.00**	7.19	9.20	2.60	**19.00**
28-May-24	37.3	0.2	0.0	9.5	1.0	0.7	0.1	**48.87**	0.0	14.2	7.6	0.0	4.6	0.5	0.9	**27.82**	0.00	0.00	0.00	**0.00**	15.72	22.30	4.95	**42.96**
29-May-24	46.1	2.7	0.3	11.8	2.9	0.9	0.1	**64.77**	0.0	16.9	3.6	0.0	3.0	0.3	0.9	**24.63**	0.00	0.00	0.00	**0.00**	13.99	20.01	3.87	**37.87**
30-May-24	34.4	5.8	0.0	9.3	3.2	1.0	0.1	**53.77**	0.0	8.9	4.6	0.0	3.7	0.3	0.8	**18.23**	0.00	0.00	0.00	**0.00**	13.07	20.35	4.12	**37.55**
31-May-24	23.4	12.7	0.0	6.8	1.3	0.6	0.0	**44.92**	0.0	3.6	6.4	0.0	4.5	0.5	1.1	**16.14**	0.00	0.00	0.00	**0.00**	10.91	17.01	3.33	**31.24**
Total	**751.76**	**524.74**	**1.25**	**216.17**	**46.71**	**26.39**	**2.55**	**1569.56**	**15.64**	**169.79**	**377.94**	**4.57**	**120.88**	**8.92**	**25.32**	**723.06**	**23.09**	**47.75**	**6.73**	**77.57**	**122.93**	**208.72**	**38.63**	**370.28**

16. भूटान , नेपाल, बांग्लादेश एवं म्यान्मार के साथ अंतरराष्ट्रीय विद्युत विनिमय

अप्रैल 2024 से मार्च 2025 April 2024 to March 2025

अंतरराष्ट्रीय विद्युत विनिमय [भारत 'से निर्यात'/ 'को आयात']

Transnational Exchange ('Export from'/'Import to' India)

माह MONTH	भूटान BHUTAN		नेपाल NEPAL		बांग्लादेश BANGLADESH		म्यान्मार MYANMAR	
	Energy Exported (In MU)	Energy Imported (In MU)	Energy Exported (In MU)	Energy Imported (In MU)	Energy Exported (In MU)	Energy Imported (In MU)	Energy Exported (In MU)	Energy Imported (In MU)
अप्रैल 2024 APR'24	86.47	18.58	346.08	0.00	694.34	0.00	0.76	0.00
मई 2024 MAY'24	51.37	68.17	271.84	3.92	655.55	0.00	0.74	0.00
कुल Total	**137.84**	**86.75**	**617.91**	**3.92**	**1349.89**	**0.00**	**1.49**	**0.00**

* Based on daily operational data

16.1 Import from neighbouring countries during May 2024

(All figures in MU)

Date	Import from Bhutan								Import from Bangladesh		Import from Nepal			Import from Myanmar
	400 kV Tala-Binaguri I,II & IV	400 kV Binaguri-Malbase	220 kV Birpara-Chuka D/C	220 kV Birpara-Malbase	400 kV Punatsanchu-Alipurdwar D/C*	400 kV Jigmeling-Alipurdwar D/C	132 kV Rangia-Motanga	132 kV Salakati-Gelephu	400 kV Behrampur-Bheramara 1,2,3&4	132 kV Surjyamaninagar-Comilla D/C	132 kV Tanakpur-Mahendranagar	400 kV Muzaffarpur-Dhalkebar	From BIHAR Source	11 kV Moreh-Tamu
1-May-24	0.07	0.00	0.00	0.00	0.00	3.88	0.79	0.06	0.00	0.00	0.00	0.00	0.00	0.00
2-May-24	0.00	0.00	0.00	0.00	0.00	2.46	0.88	0.08	0.00	0.00	0.00	0.00	0.00	0.00
3-May-24	0.00	0.00	0.00	0.00	0.00	1.79	0.44	0.07	0.00	0.00	0.00	0.00	0.00	0.00
4-May-24	0.00	0.00	0.00	0.00	0.00	1.02	0.44	0.19	0.00	0.00	0.00	0.00	0.00	0.00
5-May-24	0.00	0.00	0.00	0.00	0.00	1.93	0.44	0.00	0.00	0.00	0.00	0.00	0.00	0.00
6-May-24	0.00	0.00	0.00	0.00	0.00	1.17	0.44	0.00	0.00	0.00	0.00	0.00	0.00	0.00
7-May-24	0.00	0.00	0.00	0.00	0.00	2.22	0.44	0.00	0.00	0.00	0.00	0.00	0.00	0.00
8-May-24	0.01	0.00	0.00	0.00	0.00	2.33	0.44	0.00	0.00	0.00	0.00	0.00	0.00	0.00
9-May-24	0.13	0.00	0.00	0.00	0.00	2.29	0.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00
10-May-24	0.02	0.00	0.00	0.00	0.00	1.08	0.33	0.00	0.00	0.00	0.00	0.00	0.00	0.00
11-May-24	0.05	0.00	0.00	0.00	0.00	0.72	0.45	0.00	0.00	0.00	0.00	0.00	0.00	0.00
12-May-24	0.18	0.00	0.00	0.00	0.00	1.02	0.45	0.00	0.00	0.00	0.00	0.00	0.00	0.00
13-May-24	0.13	0.00	0.00	0.00	0.00	1.75	0.45	0.00	0.00	0.00	0.00	0.00	0.00	0.00
14-May-24	0.00	0.00	0.00	0.00	0.00	1.38	0.45	0.00	0.00	0.00	0.00	0.00	0.00	0.00
15-May-24	0.35	0.00	0.00	0.00	0.00	1.81	0.63	0.00	0.00	0.00	0.00	0.00	0.00	0.00
16-May-24	0.50	0.00	0.00	0.00	0.00	2.75	0.83	0.02	0.00	0.00	0.00	0.00	0.00	0.00
17-May-24	0.00	0.00	0.00	0.00	0.00	3.18	0.45	0.00	0.00	0.00	0.00	0.00	0.00	0.00
18-May-24	0.00	0.00	0.00	0.00	0.00	3.72	0.26	0.00	0.00	0.00	0.00	0.00	0.00	0.00
19-May-24	0.96	0.00	0.00	0.00	0.00	6.19	0.25	0.00	0.00	0.00	0.00	0.00	0.00	0.00
20-May-24	1.77	0.00	0.00	0.00	0.00	3.86	0.41	0.12	0.00	0.00	0.00	0.00	0.00	0.00
21-May-24	1.30	0.00	0.00	0.00	0.00	4.01	0.25	0.16	0.00	0.00	0.00	0.00	0.00	0.00
22-May-24	2.16	0.00	0.00	0.00	0.00	4.50	0.25	0.00	0.00	0.00	0.00	0.00	0.00	0.00
23-May-24	1.15	0.00	0.00	0.00	0.00	4.07	1.03	0.10	0.00	0.00	0.00	0.00	0.00	0.00
24-May-24	0.83	0.00	0.00	0.00	0.43	2.51	1.01	0.03	0.00	0.00	0.00	0.00	0.00	0.00
25-May-24	0.62	0.00	0.00	0.00	1.38	2.08	1.09	0.05	0.00	0.00	0.00	0.00	0.00	0.00
26-May-24	1.95	0.00	0.00	0.00	2.85	4.28	1.21	0.06	0.00	0.00	0.00	1.54	0.00	0.00
27-May-24	2.21	0.06	0.00	0.00	3.09	4.64	1.21	0.00	0.00	0.00	0.00	0.70	0.00	0.00
28-May-24	3.03	0.51	0.00	0.00	3.41	5.10	0.69	0.00	0.00	0.00	0.00	5.56	0.00	0.00
29-May-24	3.87	0.95	0.00	0.00	4.52	6.75	0.54	0.00	0.00	0.00	0.00	3.67	0.00	0.00
30-May-24	3.11	0.00	0.00	0.00	2.46	3.67	0.34	0.00	0.00	0.00	0.00	2.55	0.00	0.00
31-May-24	2.27	0.12	0.00	0.00	2.32	3.57	0.54	0.00	0.00	0.00	0.00	1.20	0.00	0.00
Total	**26.69**	**1.64**	**0.00**	**0.00**	**20.46**	**91.73**	**17.83**	**0.94**	**0.00**	**0.00**	**0.00**	**15.24**	**0.00**	**0.00**

Based on SEM/Energy meter data for links where available (*Mangdechu generation reciept at APD through a bypassed arrangement at 400kV Punatsanchu station)

16.2 Export to neighbouring countries during May 2024

(All figures in MU)

Date	Export to Bhutan								Export to Bangladesh		Export to Nepal			Export to Myanmar
	400 kV Tala-Binaguri I,II & IV	400 kV Binaguri-Malbase	220 kV Birpara-Chuka D/C	220 kV Birpara-Malbase	400 kV Punatsanchu-Alipurdwar D/C*	400 kV Jigmeling-Alipurdwar D/C	132 kV Rangia-Motanga	132 kV Salakati-Gelephu	400 kV Behrampur-Bheramara 1,2,3&4	132 kV Surjyamaninagar-Comilla D/C	132 kV Tanakpur-Mahendranagar	400 kV Muzaffarpur-Dhalkebar	From BIHAR Source	11 kV Moreh-Tamu
1-May-24	0.00	1.03	1.73	0.91	0.00	0.00	0.00	0.00	20.56	2.30	1.20	6.67	3.25	0.03
2-May-24	0.60	1.85	2.98	1.01	0.00	0.00	0.00	0.00	20.50	2.60	1.42	8.15	3.78	0.02
3-May-24	0.71	1.95	2.76	0.97	0.00	0.00	0.00	0.00	20.62	2.09	1.23	7.50	3.83	0.02
4-May-24	0.61	1.75	2.25	1.03	0.00	0.00	0.00	0.00	20.45	2.68	0.94	5.53	2.97	0.03
5-May-24	0.50	1.53	2.53	1.41	0.00	0.00	0.00	0.28	19.95	2.12	1.26	6.89	3.75	0.02
6-May-24	0.18	1.17	2.34	1.55	0.00	0.00	0.00	0.30	20.01	1.97	1.10	6.00	3.47	0.00
7-May-24	0.33	1.36	2.60	1.55	0.00	0.00	0.00	0.34	18.31	2.15	1.11	6.36	2.79	0.00
8-May-24	0.00	1.12	2.68	1.38	0.00	0.00	0.00	0.31	17.89	2.38	0.95	5.11	2.53	0.01
9-May-24	0.00	1.09	3.75	1.11	0.00	0.00	0.00	0.22	10.89	2.54	1.07	6.53	2.94	0.02
10-May-24	0.00	1.19	2.31	1.22	0.00	0.00	0.00	0.33	9.84	2.34	1.10	6.18	3.25	0.02
11-May-24	0.00	1.11	2.26	1.13	0.00	0.00	0.00	0.33	19.12	2.29	1.12	6.16	2.58	0.02
12-May-24	0.00	1.00	2.60	1.21	0.00	0.00	0.00	0.32	20.39	2.37	1.18	3.04	2.86	0.02
13-May-24	0.00	1.14	2.47	0.98	0.00	0.00	0.00	0.30	20.20	2.48	1.14	4.51	2.24	0.03
14-May-24	0.14	1.45	2.40	0.83	0.00	0.00	0.00	0.31	20.02	2.60	1.25	5.99	3.13	0.03
15-May-24	0.00	0.67	3.13	0.91	0.00	0.00	0.00	0.23	20.34	2.58	1.26	5.81	3.46	0.03
16-May-24	0.00	0.35	4.04	0.83	0.00	0.00	0.00	0.15	20.52	2.58	1.18	6.24	3.37	0.04
17-May-24	0.04	1.26	2.42	0.96	0.00	0.00	0.00	0.23	20.52	2.58	1.26	7.12	3.63	0.01
18-May-24	0.07	1.44	2.54	1.23	0.00	0.00	0.00	0.03	20.43	2.37	1.21	5.70	2.82	0.03
19-May-24	0.00	0.57	2.19	1.32	0.00	0.00	0.00	0.06	20.50	2.31	1.08	4.19	3.05	0.03
20-May-24	0.00	0.26	1.94	1.03	0.00	0.00	0.00	0.00	20.47	2.28	0.67	3.62	3.06	0.03
21-May-24	0.00	0.59	1.78	0.72	0.00	0.00	0.00	0.00	20.49	2.19	0.64	4.50	3.18	0.02
22-May-24	0.00	0.26	1.90	0.71	0.00	0.00	0.00	0.22	20.51	1.95	1.03	3.59	2.92	0.03
23-May-24	0.00	0.72	1.73	0.59	0.00	0.00	0.00	0.03	20.50	2.04	0.99	3.42	2.93	0.03
24-May-24	0.00	0.89	1.77	0.55	0.00	0.00	0.00	0.03	20.45	1.80	1.18	4.77	3.06	0.04
25-May-24	0.00	1.01	1.75	0.44	0.00	0.00	0.00	0.00	20.41	1.85	1.16	3.45	2.81	0.02
26-May-24	0.00	0.28	1.36	0.54	0.00	0.00	0.00	0.00	20.34	1.93	1.00	0.00	3.12	0.03
27-May-24	0.00	0.00	1.59	1.15	0.00	0.00	0.00	0.15	11.01	0.57	0.90	0.00	2.05	0.06
28-May-24	0.00	0.00	1.38	1.20	0.00	0.00	0.00	0.34	12.81	1.13	0.65	0.00	1.56	0.00
29-May-24	0.00	0.00	1.27	1.26	0.00	0.00	0.00	0.09	20.38	1.80	0.99	0.00	2.07	0.01
30-May-24	0.00	1.24	2.11	1.72	0.00	0.00	0.00	0.28	20.51	2.02	1.19	0.00	1.84	0.03
31-May-24	0.00	0.00	1.92	1.35	0.00	0.00	0.00	0.26	20.16	1.74	0.99	0.00	1.68	0.03
Total	**3.18**	**28.25**	**70.50**	**32.81**	**0.00**	**0.00**	**0.00**	**5.14**	**589.09**	**66.62**	**33.44**	**137.08**	**90.01**	**0.74**

Based on SEM/Energy meter data for links where available (*Mangdechu generation reciept at APD through a bypassed arrangement at 400kV Punatsanchu station)

17. एसटीओए (द्विपक्षीय एवं सामूहिक) एवं डी एस एम बिलिंग का ब्योरा – मई 2024

STOA (BILATERAL AND COLLECTIVE) & DSM BILLING DETAILS - MAY 2024

द्विपक्षीय एसटीओए BILATERAL SHORT TERM OPEN ACCESS

मई 2024 MAY 2024			Apr'24- May'25	
नोडल क्षे.भा.प्रे.के. का नाम Name of Nodal RLDC	अनुमोदित लेन – देनों की संख्या No. of Approved Transactions	अनुमोदित ऊर्जा (मि.यु.) Energy Approved(MU)	अनुमोदित लेन – देनों की संख्या No. of Approved Transactions	अनुमोदित ऊर्जा (मि.यु.) Energy Approved (MU)
ऊ. क्षे. NR	971	2153	1258	2485
प.क्षे. WR	415	754	795	1573
द.क्षे. SR	406	2698	892	6137
पू.क्षे. ER	344	1610	656	2928
पूर्वोत्तर क्षे. NER	122	25	222	38
कुल TOTAL	2258	7240	3823	13161

एसटीओए SHORT TERM OPEN ACCESS

	सामूहिक एसटीओए Collective STOA		द्विपक्षीय एसटीओए Bilateral STOA	
माह MONTH	क्रेताओं / विक्रेताओं की स. NO. of Buyers/Sellers	अनुमोदित ऊर्जा (मि.यु.) Approved Energy (MU)	अनुमोदित लेन – देनों की संख्या No. of Approved Transactions	अनुमोदित ऊर्जा (मि.यु.) Approved Energy (MU)
अप्रैल 2024 Apr'24	9860	7128	1565	5921
मई 2024 May'24	497	8268	2258	7240
कुल TOTAL	10357	15396	3823	13161

मासिक डी एस एम बिलिंग का ब्योरा* 2024-25
MONTHLY DSM BILLING DETAILS* 2024-25

अनंतिम आँकड़े
Provisional data subject to change

करोड़ रु. मे (RS. IN CRORES)

क्षेत्र REGION ⟹ / सप्ताह WEEK ⇩	उत्तरी क्षेत्र NORTH	पश्चिमी क्षेत्र WEST	दक्षिणी क्षेत्र SOUTH	पूर्वी क्षेत्र EAST	पूर्वोत्तर क्षेत्र NORTH EAST
29.04.24 to 05.05.24	345.13	356.79	351.31	314.18	80.49
06.05.24 to 12.05.24	104.88	225.64	255.15	152.96	60.13
13.05.24 to 19.05.24	316.83	360.89	269.37	252.18	64.61
20.05.24 to 26.05.24	362.31	389.50	441.38	283.28	36.91
27.05.24 to 02.06.24	264.77	336.39	483.21	277.64	63.56

* Amount shown is Payble to DSM pool ^Provisional Data

18. पावर मार्केट की सूचना (स्रोत : आई.ई.एक्स. एवं पी.एक्स.आई.एल.)
POWER MARKET INFORMATION (Source IEX & PXIL)
पावर एक्स्चेंज के माध्यम से विनिमय – माह:- मई 2024
EXCHANGES THROUGH POWER EXCHANGES -MAY 2024

क्र. स. S. No.	क्षेत्रीय इकाई Regional Entity	क्षेत्र Region	पावर एक्स्चेंज के माध्यम से (मि.यु. मे) Through Power Exchange in MU	
			(DAM+HP DAM+RTM)	
1	AD HYDRO POWER LIMITED	उत्तरी क्षेत्र NR	81.38	0.00
2	ADANI SOLAR ENERGY JAISALMER TWO PRIVATE LIMITED(Project-2)		36.01	0.00
3	ADANI SOLAR ENERGY JODHPUR TWO LIMITED		12.76	0.00
4	ADANI SOLAR ENERGY RJ TWO PRIVATE LIMITED		48.45	0.00
5	ADANI SOLAR ENERGY RJ TWO PRIVATE LIMITED (infirm)		4.54	0.00
6	ADANI SOLAR ENERGY RJ TWO PRIVATE LIMITED(Phalodi)		32.39	0.00
7	Adept Renewable Technologies Private Limited		33.75	0.00
8	Amp Energy Green Six Private Limited		30.28	0.00
9	Amplus Ages Private Limited		20.60	0.00
10	Ayana Renewable Power Three Private Limited		30.26	0.00
11	Budhil HEP (Greenko Budhil Hydro Power Pvt. Ltd.)		4.62	0.00
12	Chandigarh (UT)		16.30	12.81
13	Delhi		184.63	208.85
14	Grian Energy Private Limited		28.22	0.00
15	Haryana		173.53	315.41
16	Himachal Pradesh		533.92	79.31
17	IGSTPS JHAJJAR		9.53	0.00
18	Jammu Kashmir		68.57	57.50
19	KARCHAM WANGTOO HYDRO ELECTRIC PLANT.		105.24	0.00
20	NEA Nepal Upper Chameliya Hydropower Project		0.07	0.00
21	NEA Nepal Upper Kalangagad Hydropower Project		0.56	0.00
22	NEPAL ELECTRICITY AUTHORITY TANAKPUR		0.00	18.15
23	NTPC Dadri Stage I		6.47	0.00
24	NTPC Dadri Stage II		5.20	0.00
25	NTPC Rihand stage I		10.46	0.00
26	NTPC Rihand stage II		12.62	0.00
27	NTPC Rihand stage III		11.16	0.00
28	NTPC Singrauli		6.13	0.00
29	NTPC Tanda Stage II		6.33	0.00
30	NTPC Unchahar Stage I		2.98	0.00
31	NTPC Unchahar Stage II		4.25	0.00
32	NTPC Unchahar Stage III		1.50	0.00
33	NTPC Unchahar Stage IV		4.26	0.00
34	North Central Railway Prayagraj		0.00	44.61
35	Onevolt Energy Private Limited		28.17	0.00
36	Punjab		72.27	707.62
37	RENEW SURYA PRATAP PRIVATE LIMITED		47.05	0.00
38	RENEW SURYA ROSHNI PRIVATE LIMITED-Fatehgarh-III PS		15.18	0.00
39	RENEW SURYA ROSHNI PRIVATE LIMITED-Fatehgarh-III PS_Infirm		20.35	0.00
40	Rajasthan		42.76	1491.37
41	ReNew Surya Ravi Private Limited		55.38	0.00
42	SHREE CEMENT LIMITED TPS		1.66	8.18
43	Sainj HEP		29.89	0.85
44	Serentica Renewables India 4 Private Limited		32.63	0.00

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क्र. स. S. No.	क्षेत्रीय इकाई Regional Entity	क्षेत्र Region	पावर एक्स्चेंज के माध्यम से (मि.यु. मे) Through Power Exchange in MU	
			(DAM+HP DAM+RTM)	
45	Serentica Renewables India 4 Private Limited- Infirm		0.58	0.00
46	Singoli Bhatwari HEP		20.78	0.00
47	Sorang HEP (Himachal Sorang Power Pvt. Ltd.))		28.14	0.00
48	TPSL 200MW TPTCL Banderwala		44.39	0.00
49	TPSL Banderwala SECI 100_INF		18.67	0.00
50	Transition Energy Services Private Limited		8.13	0.00
51	Transition Green Energy Private Limited		23.64	0.00
52	Transition Sustainable Energy Services Private Limited		12.29	0.00
53	Uttar Pradesh		150.45	828.14
54	Uttarakhand		32.23	299.93

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			(DAM+HP DAM+RTM)	
55	ACB (INDIA) LIMITED	पश्चिमी क्षेत्र WR	63.91	1.33
56	ADANI GREEN ENERGY TWENTY SIX A LIMITED_infirm		0.48	0.00
57	ADANI RENEWABLE ENERGY FIFTY FIVE LIMITED_Infirm		13.20	0.00
58	ADANI RENEWABLE ENERGY HOLDING FOUR LIMITED		170.38	0.00
59	ADANI WIND ENERGY KUTCHH FOUR Ltd. Nakhatrana		77.34	0.00
60	ADANI WIND ENERGY KUTCHH FOUR Ltd. Nakhatrana (infirm)		13.77	0.00
61	Adani Green Energy Twenty Five A Limited_Infirm		4.53	0.00
62	Adani Green Energy Twenty Five B Limited_infirm		71.35	0.00
63	Adani Green Energy Twenty Six B Limited_infirm		12.17	0.00
64	Adani Power Limited - Raigarh TPP		70.07	0.00
65	Adani Power Limited-Raipur TPP		201.17	9.90
66	ArcelorMittal Nippon Steel India Limited		0.00	92.02
67	BHARAT ALUMINIUM COMPANY LTD		25.49	0.00
68	BHARAT ALUMINIUM COMPANY LTD (Bulk Consumer)		0.00	15.09
69	CONTINUUM POWER TRADING (TN) PRIVATE LIMITED		10.68	0.00
70	Chhattisgarh		725.58	2.51
71	D B Power Limited		142.81	0.00
72	DGEN MEGA POWER PROJECT		11.54	0.00
73	DHARIWAL STU OTH		8.94	0.00
74	Daman and Diu - Dadra and Nagar Haveli		0.45	90.99
75	Dhariwal ISTS		14.31	0.00
76	GMR Warora Energy Limited		2.38	0.00
77	Goa WR		43.49	3.54
78	Gujarat		167.27	713.39
79	Jaypee Nigrie Super Thermal Power Plant		137.67	0.00
80	Jhabua Power Limited		18.49	1.11
81	Jindal Power Limited, Stage-1		83.35	2.07
82	Jindal Power Limited, Stage-2		213.00	0.00
83	Jindal Steel & Power Ltd , DCPP		69.24	6.19
84	KSK MAHANADI POWER COMPANY LIMITED		0.30	0.00
85	MAHAN ENERGEN LIMITED.		170.34	11.86
86	MB POWER (MADHYA PRADESH) LIMITED		25.68	5.46
87	Madhya Pradesh		384.72	78.81
88	Maharashtra		239.43	430.60
89	Maruti Clean Coal and Power Limited		10.20	0.31
90	NREL-Dayapar Wind		13.39	0.00
91	NTPC Gadarwara		12.18	0.00
92	NTPC Jhanor Gandhar GPS		0.00	0.28
93	NTPC JhanorGandhar GPS RLNG		1.00	0.00
94	NTPC Kawas GPS		0.00	0.15
95	NTPC Kawas GPS RLNG		1.25	0.00
96	NTPC Korba Stage I &II		3.79	0.00
97	NTPC Korba Stage III		0.88	0.00
98	NTPC Lara Stage I		2.12	0.00

18. पावर मार्केट की सूचना (स्रोत : आई.ई.एक्स. एवं पी.एक्स.आई.एल.)
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पावर एक्स्चेंज के माध्यम से विनिमय – माह:- मई 2024
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क्र. स. S. No.	क्षेत्रीय इकाई Regional Entity	क्षेत्र Region	पावर एक्स्चेंज के माध्यम से (मि.यु. मे) Through Power Exchange in MU	
			(DAM+HP DAM+RTM)	
99	NTPC Mouda Stage I		2.99	0.00
100	NTPC Mouda Stage II		4.05	0.00
101	NTPC SAIL POWER COMPANY LIMITED		0.71	0.00
102	NTPC Sipat Stage I		5.89	0.00
103	NTPC Sipat Stage II		2.73	0.00
104	NTPC Solapur		5.16	0.00
105	NTPC Vindhyachal Stage I		2.39	0.00
106	NTPC Vindhyachal Stage II		2.22	0.00
107	NTPC Vindhyachal Stage III		2.76	0.00
108	NTPC Vindhyachal Stage IV		3.04	0.00
109	NTPC Vindhyachal Stage V		1.52	0.00
110	NTPC khargone		5.89	0.00
111	Nani Virani Wind Energy Private Limited		4.55	0.00
112	R.K.M POWERGEN PRIVATE LIMITED		18.72	0.00
113	Ratnagiri Gas & Power Private Limited		0.00	0.41
114	SKS Power Generation Chhattisgarh Limited		42.48	5.42
115	Sasan Power Limited		5.39	0.00
116	TRN ENERGY PRIVATE LIMITED		1.68	0.00
117	The Tata Power Co Ltd (MTPS)		23.41	0.00
118	WIND FIVE RENERGY LIMITED		12.99	0.00

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			(DAM+HP DAM+RTM)	
119	Andhra Pradesh	दक्षिणी क्षेत्र SR	177.22	249.62
120	COASTAL ENERGEN PRIVATE LIMITED		2.34	0.00
121	Goa SR		0.01	20.85
122	IL&FS TAMIL NADU POWER COMPANY LIMITED		8.59	0.00
123	JINDAL POWER LIMITED SIMHAPURI		27.24	0.13
124	JSW RENEW ENERGY LIMITED INFIRM		4.37	0.00
125	Karnataka		481.17	79.73
126	Kerala		153.66	41.94
127	Lanco Kondapalli Power Limited STG2		5.78	0.00
128	MEENAKSHI ENERGY LIMITED		0.50	0.00
129	NLC INDIA LIMITED NEYVELI NEW THERMAL POWER STATION		10.42	0.00
130	NLC INDIA LIMITED THERMAL POWER STATION I EXPANSION		4.83	0.00
131	NLC INDIA LIMITED THERMAL POWER STATION II EXPANSION		2.32	0.00
132	NLC INDIA LIMITED THERMAL POWER STATION II STAGE I		0.66	0.00
133	NLC INDIA LIMITED THERMAL POWER STATION II STAGE II		1.44	0.00
134	NLC Tamilnadu Power Limited		4.69	0.00
135	NTECL VALLUR		4.40	0.00
136	NTPC KUDGI		13.43	0.00
137	NTPC Ramagundam Stage I &II		9.65	0.00
138	NTPC Ramagundam Stage III		1.33	0.00
139	NTPC Simhadri Stage I		8.06	0.00
140	NTPC Simhadri Stage II		6.08	0.00
141	NTPC Telangana		9.48	0.00
142	OSTRO KANNADA POWER PRIVATE LIMITED		14.12	0.00
143	Pondicherry UT		50.59	0.07
144	RENEW SURYA OJAS PRIVATE LIMITED-Wind		47.43	0.00
145	RENEW SURYA OJAS PRIVATE LIMITED_INFIRM		10.40	0.00
146	RENEW SURYA ROSHNI PRIVATE LIMITED Koppal PS		0.61	0.00
147	Ramagundam Floating solar		10.03	0.00
148	SEIL ENERGY INDIA LIMITED		102.46	0.00
149	SEIL Energy India Limited Project II		27.69	0.00
150	Serentica Renewables India 1 Private Limited- Koppal (Infirm)		2.36	0.00
151	Simhadri FSP 15 MW		2.66	0.00
152	Simhadri Floating solar (10 MW)		1.82	0.00
153	Tamil Nadu		55.64	421.32
154	Telangana		133.99	506.44

18. पावर मार्केट की सूचना (स्रोत : आई.ई.एक्स. एवं पी.एक्स.आई.एल.)
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			(DAM+HP DAM+RTM)	
155	ADHUNIK POWER & NATURAL RESOURCES LIMITED	पूर्वी क्षेत्र ER	9.76	0.00
156	Basochhu Hydropower Plant Bhutan		1.43	0.00
157	Bihar		224.34	190.89
158	Chilime Hydro Power Project NEPAL ELECTRICITY AUTHORITY		0.30	0.00
159	Chuzachen HEP		0.03	0.00
160	DEVIGHAT NEPAL ELECTRICITY AUTHORITY		0.25	0.00
161	Damodar Valley Corporation		48.34	86.78
162	Druk Green Power Corporation Limited (DGPCL-Bhutan)		0.00	78.51
163	GMR KAMALANGA ENERGY LTD-CTU		2.83	0.00
164	IND BARATH ENERGY UTKAL LIMITED		21.52	0.00
165	JORETHANG LOOP HEP, DANS ENERGY PRIVATE LIMITED		0.02	0.00
166	Jharkhand		43.08	86.81
167	Jindal India Thermal Power Limited		88.04	0.00
168	KALI GANDAKI NEPAL ELECTRICITY AUTHORITY		14.28	0.00
169	KANTI BIJLEE UTPADAN NIGAM LIMITED		13.14	0.00
170	Kabeli B-1 Hydro Power Project NEPAL ELECTRICITY AUTHORITY		0.46	0.00
171	LIKHU-iV NEA		0.83	0.00
172	Lower Modi Hydro Power Project NEPAL ELECTRICITY AUTHORITY		0.36	0.00
173	MARSYANGDI NEPAL ELECTRICITY AUTHORITY		4.03	0.00
174	NABINAGAR POWER GENERATING COMPANY LIMITED		25.00	0.00
175	NEA - Bihar STU		0.00	88.27
176	NEA Nepal Dordi Khola Hydropower Project		0.16	0.00
177	NEA Nepal Solu Khola (Dudhkoshi) Hydropower Project		7.13	0.00
178	NEA Nepal Upper Dordi A Hydropower Project		0.23	0.00
179	NEPAL ELECTRICITY AUTHORITY MUZAFFARPUR		0.00	132.91
180	NEPAL ELECTRICITY AUTHORITY-MIDDLE MARSYANGDI		3.96	0.00
181	NTPC BARH Stage I		9.62	0.00
182	NTPC BARH Stage II		27.19	0.00
183	NTPC Darlipali		4.37	0.00
184	NTPC Farakka stage I		15.02	0.00
185	NTPC Farakka stage III		12.93	0.00
186	NTPC Kahalgaon stage I		7.75	0.00
187	NTPC Kahalgaon stage II		12.25	0.00
188	NTPC North Karanpura STPS		12.25	0.00
189	NTPC Talcher Stage I		7.10	0.00
190	NTPP BRBCL		9.59	0.00
191	Odisha		98.49	296.53
192	Sikkim		5.06	8.51
193	Solu Hydropower Project NEPAL ELECTRICITY AUTHORITY		0.06	0.00
194	TRISHULI NEPAL ELECTRICITY AUTHORITY		0.50	0.00
195	West Bengal		280.81	310.42

18. पावर मार्केट की सूचना (स्रोत : आई.ई.एक्स. एवं पी.एक्स.आई.एल.)
POWER MARKET INFORMATION (Source IEX & PXIL)
पावर एक्स्चेंज के माध्यम से विनिमय – माह:- मई 2024
EXCHANGES THROUGH POWER EXCHANGES -MAY 2024

क्र. स. S. No.	क्षेत्रीय इकाई Regional Entity	क्षेत्र Region	पावर एक्स्चेंज के माध्यम से (मि.यु. मे) Through Power Exchange in MU	
			(DAM+HP DAM+RTM)	
196	AGARTALA GAS BASED POWER STATION	पूर्वोत्तर क्षेत्र NER	0.71	0.00
197	ASSAM GAS BASED POWER STATION		2.19	0.00
198	Arunachal Pradesh		19.46	4.66
199	Assam		156.37	59.39
200	Bongaigaon Thermal Power Station NTPC		5.26	0.00
201	KAMENG HYDRO POWER STATION		19.73	0.00
202	Kopili 2 IN		2.64	0.00
203	Manipur		10.93	18.10
204	Meghalaya		15.13	12.94
205	Mizoram		5.84	0.00
206	Nagaland		2.94	9.61
207	Palatana Plant		2.60	0.00
208	Tripura		16.15	19.85
Total			8268	8268

19. INFORMATION ABOUT RENEWABLE ENERGY CERTIFICATE MECHANISM

MONTH : MAY 2024

RE Source & Unit wise break up (01.05.2024-31.05.2024)

Sr.No	Source Wise	Accreditation		Registration	
		Capacity (MW)	Unit	Capacity (MW)	Unit
1	Wind	0	0	0	0
2	Urban or Municipal Waste	0	0	12	1
3	Solar Thermal	0	0	0	0
4	Solar PV	20	5	3	2
5	Small Hydro	0	0	24	1
6	Others	0	0	0	0
7	Geothermal	0	0	0	0
8	Biomass	0	0	0	0
9	Bio-fuel cogeneration	0	0	0	0
	Total	**20**	**5**	**39**	**4**

RECs Issued (01.05.2024-31.05.2024)

Sr.No.	Non Solar	Solar	Total
1	2629272	86561	2715833

Redemption of REC (01.05.2024-31.05.2024)

Sr.No.	Non Solar	Solar	Total
1	1659073	42454	1701527

19. INFORMATION ABOUT RENEWABLE ENERGY CERTIFICATE MECHANISM

MONTH : MAY 2024

RE Source & Unit wise break up (Apr'24-May'24)

Sr.No	Source Wise	Accreditation		Registration	
		Capacity (MW)	Unit	Capacity (MW)	Unit
1	Wind	1	1	1	1
2	Urban or Municipal Waste	0	0	12	1
3	Solar Thermal	0	0	0	0
4	Solar PV	20	5	4	3
5	Small Hydro	24	1	24	1
6	Others	0	0	0	0
7	Geothermal	0	0	0	0
8	Biomass	0	0	0	0
9	Bio-fuel cogeneration	0	0	0	0
	Total	**45**	**7**	**41**	**6**

RECs Issued (Apr'24-May'24)

Sr.No.	Non Solar	Solar	Total
1	7814951	143413	7958364

Redemption of REC (Apr'24-May'24)

Sr.No.	Non Solar	Solar	Total
1	2729162	52448	2781610

19. INFORMATION ABOUT RENEWABLE ENERGY CERTIFICATE MECHANISM

MONTH : MAY 2024

RE Source & Unit wise break up Since Inception to May'24

Sr.No	Source Wise	Accreditation		Registration	
		Capacity (MW)	Unit	Capacity (MW)	Unit
1	Wind	3134	531	2977	532
2	Urban or Municipal Waste	12	1	12	1
3	Solar Thermal	0	0	0	0
4	Solar PV	1651	578	1471	482
5	Small Hydro	869	43	881	45
6	Others	4	2	3	1
7	Geothermal	0	0	0	0
8	Biomass	402	37	378	35
9	Bio-fuel cogeneration	823	91	383	55
	Total	**6895**	**1283**	**6105**	**1151**

RECs Issued since Inception to May'24

Sr.No.	Non Solar	Solar	Total
1	122919989	13683791	136603780

Redemption of REC since Inception to May'24

Sr.No.	Non Solar	Solar	Total
1	81233491	12501705	93735196

REC Closing balance as on 31.05.2024

Sr.No.	Non Solar	Solar	Total
1	41686498	1182086	42868584

20.Details of Grid Events during the Month of May 2024 in Northern Region

Sl No.	Category of Grid Event (GI 1or GI 2/ GD-1 to GD-5)	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
						Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
1	GD-1	Rajasthan	01-05-2024 09:31	01-05-2024 12:11	02:40	244	0	0.439	0.000	55573	57450	i)Generation of 220kV SBE6PL station evacuates through 220 KV Bhadla(PG)-ESUCRL SL_BHD_PG (ESUCRL) (ESUCRL) Ckt. During antecedent condition, SBE6PL station was generating approx. 244MW (as per PMU). ii)As reported, at 09:31hrs, 220 KV Bhadla(PG)-ESUCRL SL_BHD_PG (ESUCRL) (ESUCRL) Ckt tripped from Bhadla(PG) end only on B-N phase to earth fault (fault sensed in zone-1 at Bhadla(PG) end) with fault current of 7.8kA and fault distance of 9.8km from Bhadla(PG) end. iii)As per PMU at Bhadla(PG), Y-N phase to earth fault is observed with delayed fault clearing time of 360ms (Phase sequence issue is observed at Bhadla(PG) end).	1) 220 KV Bhadla(PG)-ESUCRL SL_BHD_PG (ESUCRL) (ESUCRL) Ckt
2	GD-1	Punjab	04-05-2024 07:10	04-05-2024 09:20	02:10	0	90	0.000	0.158	47911	57078	i)As reported, at 07:10 hrs, 220kV side R-Ph CT of 220/132kV Auto T/F-1 at Sarna(PS) blasted which created bus fault at both the 220kV buses at Sarna(PS). ii)Bus-bar protection is not available at Sarna(PS). Hence, all the 220kV lines connected to Sarna(PS) tripped on zone-4 protection operation at Sarna(PS) end and lines tripped from remote ends on zone-2 protection operation. From DR at Sarna(PS), it was observed that zone-4 operated after a delay of ~500ms. iii)Due to tripping of all the 220kV lines connected to Sarna(PS), complete blackout occurred at iv)As per PMU at Kishenpur(PG), R-N phase to earth fault with fault clearing time of 120ms followed by R-B phase to phase fault converted to R-Y phase to phase fault with delayed fault clearing time of 560ms is observed. v)As per SCADA, no change in demand is observed in Punjab control area. But as reported by SLDC-Punjab, load loss of approx. 90MW occurred in Punjab Control area. vi)As per SCADA, change in demand of approx. 40MW is observed in J&K control area. But as reported by SLDC-J&K, no load loss occurred in J&K Control area.	1)220 KV Sarna(PS)-Dasuya(PS) (PG) Ckt-1 2)220 KV Sarna(PS)-Dasuya(PS) (PG) Ckt-2 3)220 KV Sarna(PS)-Udhampur(PDD) (PDD) Ckt 4)220 KV Sarna(PS)- Hiranagar(PDD) (PG) Ckt 5)220 KV Kishenpur(PG)-Sarna(PS) (PG) Ckt-1 6)220 KV Kishenpur(PG)-Sarna(PS) (PG) Ckt-2 7)220 KV Sarna(PS)- Wadala(PS) Ckt-1 8)220 KV Sarna(PS)- Wadala(PS) Ckt-2 9)220 KV Sarna(PS)- Wadala(PS) Ckt-3 10)220 KV Sarna(PS)- Wadala(PS) Ckt-4 11)220 KV Sarna(PS)- RSDPH(PS) Ckt-1 12)220 KV Sarna(PS)- RSDPH(PS) Ckt-2 13)220 KV Sarna(PS)- RSDPH(PS) Ckt-3 14)220 KV Sarna(PS)- RSDPH(PS) Ckt-4 15)220 KV Sarna(PS)- Railway Ckt
3	GI-1	Punjab	05-05-2024 08:11	05-05-2024 09:30	01:19	615	0	1.220	0.000	50430	58722	i)220/132kV Ropar GGSTPS(PS) has main and transfer bus scheme at 220kV level. ii)As per SCADA, during antecedent condition, 210 MW Guru Gobind Singh TPS (Ropar) – UNIT 3, 4 & 6 were carrying approx. 186MW, 148MW & 198 MW respectively. Unit-3, Unit-6, 220kV feeders to Kharar, Mohali, Bassi Pathana & Gobindgar1 ckt-2 & 3 were connected to 220kV main Bus section-III. Rest of the elements were connected to main Bus section-I & II. iii)As reported at 08:11 hrs, during the boxing up (due to boiler tube leakage) of Unit-4, blue phase pole lagged in opening of GT breaker which led to LBB operation. Due to LBB operation, all 220kV feeders (Gobindgar1 ckt-2 & 3, Bassi Pathana ckt, Kharar ckt, Mohali ckt) and Unit-6 tripped as Unit-4 was running on the same section (Section-III). iv)Further, as repoted, during the same time, due to tripping of ST-3 (exact reason yet to shared) all allied auxiliaries of Unit-3 tripped causing tripping of Unit-3. v)As per PMU at Jalandhar(PG), no fault is observed in the system. vi)As per SCADA, generation loss of approx. 615MW occurred at Ropar GGSTPS. vii)As per SCADA no load loss is observed in Punjab control area. viii)As reported, corrective action has been taken and both the trip coils of blue phase of GT breaker of Unit-4 were replaced.	1)210 MW Guru Gobind Singh TPS (Ropar) - UNIT 3 2)210 MW Guru Gobind Singh TPS (Ropar) - UNIT 4 3)210 MW Guru Gobind Singh TPS (Ropar) - UNIT 6 4)220kV GGSTPS(PS) – Mohali(PS) ckt 5)220kV GGSTPS(PS) – Kharar(PS) ckt 6)220kV GGSTPS(PS) – Bassi Pathana(PS) ckt 7)220kV GGSTPS(PS) – Gobindgar1(PS) ckt-2 8)220kV GGSTPS(PS) – Gobindgar1(PS) ckt-3
4	GD-1	Uttarakhand	06-05-2024 16:47	06-05-2024 17:30	00:43	0	13	0.000	0.021	55223	63163	i)As reported, at 16:47 Hrs, 220 KV Pithoragarh-Jauljivi(PG) ckt-1 tripped during bad weather condition on R-N phase to earth fault with fault distance of ~17km from Pithoragarh end. A/R is kept off in the line as there is cable in some part of the line. ii)At the same time, 220kV Pithoragarh-Jauljivi ckt-2 also tripped from Jauljivi end only on maloperation of D60 relay. However, R-phase pole of ckt-2 at Jauljivi end got stuck and didn't open. Due to unavailability of one phase, there was unbalance in system and 220/132kV ICT-1&2 at Pithoragarh(PG) tripped on DEF protection. iii)With the tripping of 220/132kV ICT-1&2, 132kV Pithoragarh(PG)-Pithorgarh(Utt) ckt and 132kV Pithoragarh(PG)-Almora(Utt) ckt became dead due to loss of supply. iv)132kV Pithoragarh(Utt) was radially connected to Pithoragarh(PG) hence, load of Pithoragarh(Utt) got affected which restored at 17:42hrs on revival of 220/132kV ICTs at Pithoragarh(PG). v)132kV Almora have further connectivity from 132kV Bhawali (connected to 220/132kV Haldwani) and 132kV Ranikhet and Bageshwar (having small hydro at 33kV level). Hence, supply to 132kV Almora didn't affect during the event. vi)As per PMU, R-N phase to earth fault with fault clearing time of 80ms is observed. vii)As per SCADA, no load loss is observed in Uttarakhand control area. However, 13MW load loss at Pithoragarh(Utt) S/s is reported by Uttarakhand SLDC.	1)220 KV Pithoragarh-Jauljivi (PG) Ckt-1 2)220 KV Pithoragarh-Jauljivi (PG) Ckt-2 3)220/132 kV 100 MVA ICT 1 at Pithoragarh(PG) 4)220/132 kV 100 MVA ICT 2 at Pithoragarh(PG)
5	GI-1	Himachal Pradesh	06-05-2024 06:18	06-05-2024 08:49	02:31	124	0	0.260	0.000	47693	60730	i)During antecedent condition, 66MW Unit-1, 2 & 3 at Pong HEP were running and generating approx. 59MW, 65MW and 65MW respectively (as per SCADA). 66MW Unit-4, 5 & 6 at Pong HEP were not in service. Unit-1 & 3, 220/66kV 40MVA Transformer and 220kV feeders to Bairasiul, Jalandhar ckt-1 and Dasuya ckt-1 were connected at 220kV Bus-1 and Unit-2 & 220kV feeders to Jessore, Jalandhar ckt-2 and Dasuya ckt-2 were connected at 220kV Bus-2 . ii)As reported, during the stopping process of 66MW Unit-1 (as per schedule change at Pong(BB)), blue phase pole of Unit-1 CB lagged in opening and couldn't open properly which caused flashover in CB. iii)During the same time, LBB operated which led to tripping of all 220kV feeders (Bairasiul ckt, Jalandhar ckt-1 and Dasuya ckt-1), 66MW Unit-3 & 220kV 40MVA ICT connected to Bus-1 and Bus-1 became dead. iv)IIAs per PMU at Jalandhar(PG), no fault is observed in the system. v)As per SCADA, generation loss of approx. 124MW occured at Pong(BB) HEP and no load loss is observed in HP control area.	1)66 MW Pong HPS - UNIT 1 2)66 MW Pong HPS - UNIT 3 3)220 KV Bairasiul(NH)-Pong(BB) (PG) Ckt 4)220 KV Pong(BB)-Dasuya(PS) (BBMB) Ckt-1 5)220 KV Jalandhar-Pong (BB) Ckt-1 6)220KV Bus 1 at Pong(BB) 7)220/66kV 40MVA ICT-1 at Pong(BB)
6	GD-1	Uttarakhand	06-05-2024 16:59	06-05-2024 18:09	01:10	0	0	0.000	0.000	54673	62854	i)400kV Tehri(THDC) has double main bus scheme. ii)During antecedent condition, 250 MW TEHRI HPS – UNIT- 1, 2, 3 & 4 were not in service. So, total generation at TEHRI HPS was 0 MW. Power flow from 400 KV Tehri(THDC)-Koteshwar(PG) (PG) Ckt-1 & Ckt-2 was also 0MW. iii)As reported, at 16:59 hrs, Y-B phase to phase fault is observed at a distance of 3.7km from Tehri(THDC) end on 400kV Tehri(THDC)-Koteshwar(PG)(PG) ckt-1. From Tehri(THDC) end, only 400kV Tehri(THDC)-Koteshwar(PG)(PG) ckt-1 tripped however both ckt-1 & 2 tripped from the Koteshwar(PG) end. iv)As per DR of 400kV Tehri(THDC)-Koteshwar(PG)(PG) ckt-1, directional E/F pick up is observed during Y-N phase to earth fault; however zone-1 operated at both ends after conversion of fault into Y-B phase to phase fault with fault current of Iy=~5.7kA, Ib=~5.1kA (from Tehri end) and Iy=~6.0kA, Ib=~5.1kA (from Koteshwar end). Delayed fault clearance of 880ms is observed. v)As per DR of 400kV Tehri(THDC)-Koteshwar(PG)(end)(PG) ckt-2, directional E/F pick up is observed during Y-N phase to earth fault; however zone-1 operated at Koteshwar end after conversion of fault into Y-B phase to phase fault with fault current of Iy=~4.5kA and Ib=~3.8kA from Koteshwar end. Delayed fault clearance of 880ms is observed. vi)As per PMU at Koteshwar(PG), Y-N phase to earth fault converted into Y-B phase to phase fault is observed with delayed fault clearance time of 880ms is observed. vii)As per SCADA no load loss/generation loss is observed during the tripping event (there was no generation at Tehri HEP during the event).	1)400 KV Tehri(THDC)-Koteshwar(PG) (PG) Ckt-1 2)400 KV Tehri(THDC)-Koteshwar(PG) (PG) Ckt-2

20.Details of Grid Events during the Month of May 2024 in Northern Region

Sl No.	Category of Grid Event (GI 1or GI 2/ GD-1 to GD-5)	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
						Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
7	GI-1	Punjab	07-05-2024 14:30	07-05-2024 19:10	04:40	500	100	0.830	0.150	60250	66629	i)During antecedent condition, 220 kV Sultanpur(PS)-Chohla Sahib(PS) Ckt, 220 kV Sultanpur(PS)-Patti(PS) Ckt and 220 kV Chohla Sahib(PS)- Patti(PS) Ckt were not in service. ii)As reported, at 14:30 hrs, a piece of stray flexible conductor came within induction zone of 220 KV Sultanpur(PS)- Goindwal TPS(PS) Ckt-2 as it fell on tower location no. 16 which led to tripping of this circuit with fault distance of 5km from Goindwal TPS end. iii)Distance Protection Relay (DPR) at Goindwal TPS end sensed the fault in zone-1 and line tripped immediately from Goindwal TPS end. But Distance Protection Relay at Sultanpur end detected power swing scenario and due to PSD block of DPR for 2 seconds (de-blocking time) at Sultanpur end, fault continued to feed through 220 KV Sultanpur(PS)- Badshahpur(PS) Ckt, 220 KV Sultanpur(PS)- Jamsher(PS) Ckt and 220 KV Sultanpur(PS)- Goindwal TPS(PS) Ckt-1. iv)Badshahpur and Jamsher end DPRs issued trip command in zone-3 after 800ms and fault feeding stopped from these ends. But fault feeding continued through 220 KV Sultanpur(PS)- Goindwal TPS(PS) Ckt-1 as Goindwal TPS end DPR did not pick up the fault in zone-3. v)This led to tripping of 270 MW Goindwal(GVK) - UNIT 1 & 2 on earth-fault protection operation (51 NGT) after 1 sec. vi)As per PMU at Amritsar(PG), R-N phase to earth fault converted to 3-phase fault with delayed fault clearing time of 2120ms is observed. vii)As per SCADA, change in demand of approx. 100MW is observed in Punjab control area. viii)As reported by SLDC-Punjab, generation loss of approx. 500MW occurred at Goindwal TPS(PS).	1)220 KV Sultanpur(PS)- Goindwal TPS(PS) Ckt-2 2)220 KV Sultanpur(PS)- Badshahpur(PS) Ckt 3)220 KV Sultanpur(PS)- Jamsher(PS) Ckt 4)270 MW Goindwal(GVK) - UNIT 1 5)270 MW Goindwal(GVK) - UNIT 2
8	GD-1	Haryana	07-05-2024 11:16	07-05-2024 12:24	01:08	0	270	0.000	0.421	61025	64157	i)220/132/33kV Hissar(BB) S/s has double main bus scheme at 220kV level. ii)During antecedent Condition, all the 220kV feeders (Chirawa ckt, Jindal Steel ckt, Sangrur ckt-1 & 2, Bhiwani ckt-1 & 2, Hissar I/A ckt-1 & 2) and 220/132kV 100MVA ICT-1, 2 & 3 were connected to 220kV Bus-1. 220kV Bus-2 was not in service. iii)As reported, at 11:16 hrs R-ph jumper of 220 KV Hissar-Sangrur (BB) Ckt-1 snapped from common point of 220kV Bus isolator no. 223 & 224 at Hissar(BBMB) S/s which caused R-N phase to earth fault. iv)As reported, on this fault, Bus bar relay at 220kV Hissar(BBMB) didn't operate. 220 KV Hissar(BB)-Hissar IA(HV) (HVPNL) Ckt-1 & 2 tripped in Zone-4 from Hissar(BB) end and other all 220kV feeders (Chirawa ckt, Jindal Steel ckt, Sangrur ckt-1 & ckt-2, Bhiwani ckt-1 & ckt-2) tripped from the remote ends in Zone-2 protection operation. v)During the same time 220 KV Barnala-Sangrur(BB) Ckt also tripped (exact reason of tripping need to be shared). vi)As per DR of 220 KV Hissar(BB)-Hissar IA(HV)(end) (BBMB) Ckt-2, R-N phase to earth fault (fault current Ir=~7.4kA) is observed in zone-4 with fault clearing time of 360ms. vii)As per DR of 220 KV Hissar-Sangrur(end) (BB) Ckt-1, R-N phase to earth fault converted into R-Y phase to phase fault (fault current Ir=~1.3kA, Iy=~1.3kA) is observed in zone-2 with fault clearing time of 560ms. viii)As reported, Isolator status of 220 KV Hissar-Sangrur (BB) Ckt-2 provided in the Bus bar relay settings were incorrect (status of isolator no. 227 & 228 were incorrect). Due to this, false differential current generated in Bus Bar Z-2 and differential current supervision feature of relay picked up which led to blocking of Bus bar relay. ix)Due to tripping of all the elements connected to 220kV Bus-1 (220kV Bus-2 was not in service) at Hissar(BB), the complete 220/132/33kV Hissar(BB) S/s became dead. x)As per PMU at Hissar(PG), R-N phase to earth fault with fault clearing time of 360ms (delayed) is observed.	1)220 KV Hissar(BB)-Hissar IA(HV) (HVPNL) Ckt-1 2)220 KV Hissar(BB)-Hissar IA(HV) (HVPNL) Ckt-2 3)220 KV Bhiwani-Hissar (BB) Ckt-1 4)220 KV Bhiwani-Hissar (BB) Ckt-2 5)220 KV Hissar-Sangrur (BB) Ckt-1 6)220 KV Hissar-Sangrur (BB) Ckt-2 7)220 KV Hissar(BB)-Jindal Steel(HR) (HVPNL) Ckt 8)220 KV Hissar(BB)-Chirawa(RS) (BB) Ckt 9)220 KV Barnala-Sangrur(BB) Ckt 10)220KV Bus 1 at Hissar(BB)
9	GI-2	Uttar Pradesh	07-05-2024 16:18	07-05-2024 16:45	00:27	0	185	0.000	0.281	56440	65748	i)400/220/132kV Muzaffarnagar(UP) has double main and transfer bus scheme at 220kV level. ii)During antecedent condition, 400/220 KV 315 MVA ICT-3 & 500 MVA ICT-4 at Muzaffarnagar(UP), 220/132kV 160MVA ICT-4 at Muzaffarnagar(UP), 220 KV Muzaffarnagar(UP)– Badhni Kalan Ckt, 220 KV Muzaffarnagar(UP)– Jansath Ckt and 220 KV Muzaffarnagar(UP)– Khatauli Ckt were connected to 220kV Bus-1 at Muzaffarnagar(UP) and rest of the elements were connected to 220kV Bus-2 at Muzaffarnagar(UP). iii)As reported, at 16:18hrs, Y-ph CT at Muzaffarnagar(UP) end of 220 KV Muzaffarnagar(UP)– Badhni Kalan Ckt blasted which led to bus-bar protection operation at 220kV Bus-1 at Muzaffarnagar(UP). iv)Due to bus-bar protection operation, all the elements connected to 220kV Bus-1 at Muzaffarnagar(UP) tripped and Bus-1 became dead. v)As per PMU at Muzaffarnagar(UP), Y-N phase to earth fault with fault clearance time of 120ms is observed. vi)As per SCADA, change in demand of approx. 185 MW is observed in UP control area.	1)400/220 KV 315 MVA ICT-3 at Muzaffarnagar(UP) 2)400/220 KV 500 MVA ICT-4 at Muzaffarnagar(UP) 3)220/132kV 160MVA ICT-4 at Muzaffarnagar(UP) 4)220 KV Muzaffarnagar(UP)– Badhni Kalan Ckt 5)220 KV Muzaffarnagar(UP)– Jansath Ckt 6)220 KV Muzaffarnagar(UP)– Khatauli Ckt
10	GD-1	Himachal Pradesh	07-05-2024 16:17	07-05-2024 23:29	07:12	30	0	0.054	0.000	55751	65721	i)Total generated power of Sainj HEP(HP), Parbati_2(NH) and parbati_3(NH) evacuates through 400 kV Parbati_2(NH)- Banala(PG) (PKTCL) Ckt and 400 kV Parbati_3(NH)- Banala(PG) (PKTCL) Ckt via 400 KV Parbati_2(NH)-Sainj(HP) (PKTCL) Ckt and 400 KV Parbati_3(NH)-Sainj(HP) (PKTCL) Ckt. ii)During antecedent condition, only 50MW Unit-1 at Sainj HEP(HP) was running (generating approx. ~30MW) and 50MW Unit-2 at Sainj HEP(HP), all four units (4*250MW) at Parbati_2(NH) and all four units (4*130MW) at Parbati_3(NH) were not in service. iii)As reported, at 16:17hrs, 400 KV Parbati_3(NH)-Sainj(HP) (PKTCL) Ckt tripped on B-N phase to earth fault with fault distance of 1.808km from. Sainj end. From initial patrolling it was observed that a Pine Tree admeasuring approx. 50m fell from an uphill position outside the transmission line corridor. The tree weight resulted in simultaneous failing of cross arm of Tower Location No. 7 of 400 KV Parbati_3(NH)-Sainj(HP) (PKTCL) Ckt as well as snapping of conductor leading to tripping of line. iv)As per DR of 400 KV Parbati_3(NH) (end)-Sainj(HP) (PKTCL) Ckt, B-N phase to earth fault is observed with delayed fault clearance time of 400ms and fault current of 7.105kA from Parbati_3(NH) end. Fault was sensed in zone-2 from Parbati_3(NH). Due to non-receipt of carrier signal from remote end, three phase trip command issued after Z2 time delay. v)As further reported, 400 KV Parbati_2(NH)-Sainj(HP) (PKTCL) Ckt also tripped at the same time (exact reason of the same yet to be shared). As per DR (time sync issue observed), B-N phase to earth fault with fault current of 1.212kA from Parbati_2(NH) end is observed. vi)During the same time, 400 KV Parbati_2(NH)- Banala(PG) (PKTCL) Ckt also tripped from Parbati_2(NH) end only on B-N phase to earth fault with fault distance of 22.5km and fault current of 1.304kA from Banala(PG) end. As per DR, fault was sensed in zone-1 at Banala(PG) end and carrier sent to Parbati_2(NH) end and line successfully auto-reclosed from Banala(PG). But as reported by NHPC, fault was sensed in zone-2 at Parbati_2(NH) end and upon receipt of carrier signal from Banala(PG) B-phase CB opened. But reclosing command was not extended to B-phase CB after dead time, hence, line tripped from Parbati_2(NH) on operation of Pole discrepancy relay. vii)As per PMU at Banala(PG), B-N phase to earth fault is observed with delayed fault clearing time of 400ms. viii)As per SCADA, generation loss of approx. 30 MW at Sainj HEP (HP) is observed.	1)400 KV Parbati_2(NH)-Sainj(HP) (PKTCL) Ckt 2)400 KV Parbati_3(NH)-Sainj(HP) (PKTCL) Ckt 3)400 KV Parbati_2(NH)- Banala(PG) (PKTCL) Ckt 4)50 MW Unit-1 at Sainj HEP(HP)
11	GD-1	Rajasthan	08-05-2024 12:18	08-05-2024 17:19	05:01	138	0	0.237	0.000	58188	63043	i)Generation of 220kV ASER2PH(Phalodi) station evacuates through 220 KV Bhadla(PG)- ASER2PH(Phalodi) Ckt. During antecedent condition, ASER2PH(Phalodi) station was generating approx. 138MW (as per SCADA). ii)As reported, at 12:18hrs, 220 KV Bhadla(PG)- ASER2PH(Phalodi) Ckt tripped due to snapping of conductor at distance of 9.8km from Bhadla(PG) end (exact nature of protection operated yet to be shared). iii)As per PMU at Bhadla(PG), B-N phase to earth fault is observed with fault clearing time of 80ms. iv)As per SCADA, solar generation loss of approx. 138 MW is observed at ASER2PH (Phalodi).	1)220 KV Bhadla(PG)- ASER2PH(Phalodi) Ckt
12	GI-2	Uttar Pradesh	09-05-2024 13:40	09-05-2024 21:49	08:09	860	0	1.424	0.000	60409	64571	i)During antecedent condition, 500 MW Rihand STPS - UNIT 1 & 2 were generating approx. 414MW & 443 MW respectively. ii)As reported at 13:40 hrs, four running circulating water pumps pertaining to 500 MW Rihand STPS - UNIT 1 & 2 tripped on Band screen (filter) high DP protection> 800 mmwc which led to tripping of 500 MW Rihand STPS - UNIT 1 & 2. iii)Vegetation debris in cooling water intake channel was causing choking of Band screen(filter) of CW pumps. According to Protection philosophy implemented for auxiliary equipments at Rihand(NTPC), CW pump will trip on Band screen high DP protection> 800 mmwc to save CW pumps. iv)As per PMU at Rihand(NT), no fault is observed in the system. v)As per SCADA, generation loss of approx. 860MW occurred at Rihand (NTPC). vi)As per SCADA, no load loss is observed in UP control area. vii)As further reported, as remedial action taken, multi-layer screen/ net was installed to prevent vegetation debris.	1)500 MW Rihand-I STPS - UNIT 1 2)500 MW Rihand-I STPS - UNIT 2

20.Details of Grid Events during the Month of May 2024 in Northern Region

Sl No.	Category of Grid Event (GI 1or GI 2/ GD-1 to GD-5)	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
						Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
13	GI-2	Uttar Pradesh	09-05-2024 17:06	09-05-2024 19:49	02:43	585	0	1.085	0.000	53903	63512	i)During antecedent condition, 200 MW Singrauli STPS - UNIT 1, 3 & 5, 400 KV Singrauli(NT)-Lucknow(UP) (PG) Ckt and 400 kV Singrauli(NT)-Allahabad(PG) (PG) Ckt-1 were connected to 400KV Bus 2 at Singrauli(NT). As per SCADA, 200 MW Singrauli STPS - UNIT 1, 3 & 5 were generating approx. 187MW, 191MW & 183 MW respectively. ii)As reported at 17:06 hrs, busbar protection operated at 400kV Bus-2 at Singrauli(NT) as due to bad weather condition wave trap fallen over current transformer which in turn created bus fault. Due to busbar protection operation, all the elements connected to Bus-2 tripped and Bus-2 became dead. iii)As per DR of 400 kV Singrauli(NT)-Lucknow(UP) (end) (PG) Ckt, Y-B phase to phase fault is observed with fault current of Iy=~952A and Ib=~1405A from Lucknow(UP) end; fault sensed in in zone-2 at Lucknow(UP) end. However, line tripped on DT received at Lucknow(UP) end. iv)As per PMU at Singrauli(NT), Y-B phase to phase fault with fault clearing time of 80 ms is observed. v)As per SCADA, generation loss of approx. 585MW occurred at Singrauli (NTPC). vi)As per SCADA no load loss is observed in UP control area.	1)200 MW Singrauli STPS - UNIT 1 2)200 MW Singrauli STPS - UNIT 3 3)200 MW Singrauli STPS - UNIT 5 4)400 kV Singrauli(NT)-Lucknow(UP) (PG) Ckt 5)400 kV Singrauli(NT)-Allahabad(PG) (PG) Ckt-1 6)400KV Bus 2 at Singrauli(NT)
14	GI-2	Haryana	10-05-2024 19:35	10-05-2024 21:18	01:43	1072	0	2.109	0.000	50841	63360	i)Generation of 600MW Unit-1 & 2 at Khedar TPS (total ~1072MW) was evacuating through 400kV Khedar(HR)-Fatehabad(PG) ckt (carrying ~858MW), 400kV Khedar-Nuhiyawali (HR) ckt (carrying ~174MW) only. ii)At 19:35:24:255 hrs, R-N phase to earth fault occurred on 400kV Khedar-Nuhiyawali (HR) ckt. As per DR of 400kV Khedar-Nuhiyawali (HR) ckt, distance protection relay at both end sensed R-N fault in Z-1 (Khedar end: Ir=12kA, 75km). R-ph A/R started from both ends. iii)At 19:35:24:291 hrs, as per DR of 400kV Khedar(HR)-Fatehabad(PG) ckt, distance protection relay at Khedar end sensed B-N fault in Z-1(Khedar end: Ib=1.1kA, 8.5km) and initiated A/R in B-ph at Khedar end. Fatehabad(PG) end distance protection relay didn't sense this B-N fault and no operation occurred at Fatehabad end. iv)At 19:35:24:758 hrs, as per DR of 400kV Khedar(HR)-Fatehabad(PG) ckt, distance protection relay at Khedar end initiated 3-ph tripping command and R & Y ph pole also opened. v)As R-ph autoreclosing was also going on in 400kV Khedar-Nuhiyawali (HR) ckt at this time and all 3-ph pole of 400kV Khedar(HR)-Fatehabad(PG) ckt opened, all the power now started evacuating through Y & B phase of 400kV Khedar-Nuhiyawali (HR) ckt. Current in Y & B phase started increasing, it increased to ~1850A by 19:35:25:153 hrs. vi)At 19:35:25:376 hrs, R-ph pole of 400kV Khedar-Nuhiyawali (HR) ckt closed from both ends and line successfully autoreclosed. vii)At 19:35:25:421 hrs, all 3-ph pole at Khedar end of 400kV Khedar(HR)-Fatehabad(PG) ckt closed and line successfully autoreclosed. viii)At the same time, power swing also observed in DR of Unit-1&2 and evacuating lines. ix)At 19:35:25:944hrs, 600MW Unit-1 tripped on pole slip protection operation. x)At 19:35:25:974hrs, 600MW Unit-2 tripped on pole slip protection operation. xi)At 19:35:30:309 hrs and 19:35:30:350 hrs, over frequency protection also operated of Unit-1 & 2 respectively. xii)600MW Unit-1 at Khedar TPS revived at 04:26hrs on 11th May 2024 and Unit-2 have been kept out for inspection purpose.	1)400kV Fatehabad(PG)-Nuhiyawali(HR) ckt 2)400kV Khedar-Kirori (HR) ckt-1 3)400kV Khedar-Kirori (HR) ckt-2 4)400kV Khedar(HR)-Fatehabad(PG) ckt 5)400kV Khedar-Nuhiyawali ckt 6)600MW Unit-1 at Khedar (RGTPS) 7)600MW Unit-2 at Khedar (RGTPS)
15	GI-2	Haryana	10-05-2024 19:41	11-05-2024 02:24	06:43	0	0	0.000	0.000	50214	63513	i)During antecedent condition, 800kV HVDC Champa-Kurukshetra was carrying total 3000MW power from Champa to Kurukshetra (Pole-1=1500MW, Pole-2 & Pole-4: 750MW each). The weather condition at the time of incident was very worse accompanying thunderstorms & very heavy lightning along with wind was prevailing at Champa. ii)As reported, the sequence of event is as follows: a.At 19:41:46:169 hrs, 800 kV HVDC Kurukshetra (PG) Pole-02 blocked from Champa end on activation of SSAD Instability detection & DC Undercurrent Protection. There was SSAD Instability detected in Pole-2 Lane-1 which was the Active Lane. Then Lane Changeover from Lane-1 to Lane-2 as per Control Philosophy was facilitated by Pole-2 Control system. Again, DC Undercurrent Minor Fail alarm latched in Pole-2 & eventually Pole-2 blocked on latching of DC Undercurrent Protection at Champa. b.At 19:41:50:576 hrs, 800 kV HVDC Kurukshetra (PG) Pole-04 also blocked due to on latching of T-Zone Protection. As heavy flashover was detected at the instance of tripping of Pole-2 & Pole-4, there was noticeable increase in Idl (up to 3KA) in Pole-2 which was also seen by Pole-4 Idl & Ihl which generated T-Zone Protection operation in Pole-4 along with generation of CAT B alarm due to which Pole-4 got blocked. iii)As further reported, from logs it was observed that optical communication between Bipole 1 & Bipole 2 got unhealthy. Due to unhealthy optical communication, abnormal values of Pole Idl current resulted in latching of T-zone protection. After re-inserting optical fibres in Bipole 1 lane 1 CIB-4, optical communication got healthy. iv)As per SCADA, due to tripping of two poles (Pole-02 and Pole-04), power order reduced from 2900MW to 1980MW. But, as reported by POWERGRID, Pole Power was compensated in only one Pole in service and power order reduced from 3000MW to 1500MW. v)As per PMU, fluctuation in voltage was observed. vi)As per SCADA, no change in demand is observed in Haryana control area.	1.800 kV HVDC Kurukshetra(PG) Pole-02 2.800 kV HVDC Kurukshetra(PG) Pole-04
16	GI-1	Jammu and Kashmir	10-05-2024 13:06	10-05-2024 18:10	05:04	0	130	0.000	0.196	62886	66490	i)220/132kV Ziankote S/s have two buses at 220kV side i.e., main bus & reserve bus. ii)During antecedent condition, 220kV Amargarh(INDIGRID) –Ziankote(JK) D/C (carrying 70MW each) was feeding Ziankote load. iii)As reported, at 13:06 hrs, 220 KV Amargarh(INDIGRID)-Ziankote(JK) (PDD JK) Ckt-1 tripped on R-Y phase to phase fault with fault current of Ir=~2.11kA and Iy=~2.363kA and fault distance of 4.714km from Ziankote(JK) end; fault sensed in zone-1 at Ziankote(JK) end. (Exact reason yet to be shared) iv)As further reported, at the same time, 220 KV Amargarh(INDIGRID)-Ziankote(JK) (PDD JK) Ckt-2 also tripped on over-current protection operation. v)As per PMU at Amargarh(INDIGRID), R-Y phase to phase fault with fault clearing time of 120ms is observed. vi)As per SCADA, change in demand of approx. 130MW is observed in J&K control area.	1)220 KV Amargarh (INDIGRID)-Ziankote(JK) (PDD JK) Ckt-1 2)220 KV Amargarh (INDIGRID)-Ziankote(JK) (PDD JK) Ckt-2
17	GI-2	Jammu and Kashmir	12-05-2024 07:26	12-05-2024 08:12	00:46	390	0	0.787	0.000	49537	55574	i)During antecedent condition, 130MW Unit-1, 2 & 3 at Dulhasti HEP were running (generating approx. ~122MW, ~131MW & ~129MW respectively) and total generated power was evacuating through 400 kV Dulhasti(NH)-Kishenpur(PG) (PG) Ckt-1 only. 400 kV Dulhasti(NH)-Kishenpur(PG) (PG) Ckt-2 was not in service (planned shutdown since Jan, 2024). ii)As reported, at 07:26hrs, 400 kV Dulhasti(NH)-Kishenpur(PG) (PG) Ckt-1 successfully auto-reclosed from both the ends end on transient R-N phase to earth fault. Fault was sensed in zone-1 and fault current was Ir=~1.82kA at Dulhasti(NH) end. iii)Due to the opening of R-phase pole of CB the three phase currents of all the three running units got unbalanced that lead to flow of unbalanced current in the GT neutral. iv)As further reported, all the three units got tripped on operation of three phase current monitoring protection. v)As per DR, 74CT (alarm relay) and 50/51N (non-directional earth fault relay) operated. 74CT operated after 1200ms which coincides with A/R dead time (as also confirmed from PMU). vi)As per PMU at Kishenpur (PG), R-N phase to earth fault is observed with delayed fault clearing time of 120ms. vii)Generation loss of approx. 373 MW at Dulhasti HEP (NHPC) (as per SCADA). However, NHPC reported generation loss of 390 MW at Dulhasti HEP (NHPC).	1)130MW Unit-1 at Dulhasti HEP 2)130MW Unit-2 at Dulhasti HEP 3)130MW Unit-3 at Dulhasti HEP
18	GI-1	Rajasthan	12-05-2024 12:44	12-05-2024 15:10	02:26	156	0	0.250	0.000	62430	61010	i)Generation of 220kV Mahoba Solar(Adani) station evacuates through 220 KV Bhadla(PG)- Mahoba Solar(Adani) Ckt which is connected to 220/33 kV 150 MVA ICT 1 & 2 at Mahoba Solar(Adani) carrying approx. 155MW and 152MW respectively. During antecedent condition, Mahoba Solar(Adani) station was generating approx. 307MW (as per SCADA). ii)As reported, at 12:44hrs, 220/33 kV 150 MVA ICT-1 at Mahoba Solar (Adani) tripped due to Y phase cable end kit failure (exact nature of protection operated yet to be shared). On inspection it was found that fire caught in Y-phase cable. iii)As per PMU at Bhadla(PG), Y-N phase to earth fault is observed with fault clearing time of 120ms. iv)As per SCADA, solar generation loss of approx. 156 MW is observed at Mahoba Solar(Adani).	1)220/33 kV 150 MVA ICT 1 at Mahoba Solar(Adani)

20.Details of Grid Events during the Month of May 2024 in Northern Region

Sl No.	Category of Grid Event (GI 1or GI 2/ GD-1 to GD-5)	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
						Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
19	GI-1	Himachal Pradesh	12-05-2024 03:40	12-05-2024 05:55	02:15	64	0	0.143	0.000	44636	56495	i)During antecedent condition, 66MW Unit-1, 2 & 3 at Pong HEP were running and generating approx. 58MW, 62MW and 64MW respectively (as per SCADA). 66MW Unit-4, 5 & 6 at Pong HEP were not in service. Unit-1 & 3, 220/66kV 40MVA Transformer and 220kV feeders to Bairasiul, Jalandhar ckt-1 and Dasuya ckt-1 were connected at 220kV Bus-1 and Unit-2 & 220kV feeders to Jessore, Jalandhar ckt-2 and Dasuya ckt-2 were connected at 220kV Bus-2. ii)As reported, at 03:40 hrs, heavy sparking was observed on Y-phase PT of 220kV Bus-2 at Pong(BB). Due to this sparking cables in trench caught fire. iii)During the same time, bus bar protection operated which led to tripping of all 220 kV feeders (Jessore ckt, Dasuya ckt-2, Jalandhar ckt-2), 66MW Unit-2 and Bus coupler between 220kV Bus-1 & Bus-2 connected to Bus-2 and Bus-2 became dead. iv)During the same time, 220/66kV 40MVA Transformer (connected to Bus-1) also tripped due to over current protection operation (exact reason yet to be shared). v)As per PMU at Jalandhar(PG), Y-N phase to earth fault is observed with fault clearing time of 120ms. vi)As per SCADA, generation loss of approx. 62 MW at Pong HEP (BB) and no load loss is observed in HP control area. However, BBMB reported 64MW generation loss at Pong HEP.	1)220 KV Jessore(HP)-Pong(BB) (PG) Ckt 2)220 KV Pong(BB)-Dasuya(PS) (BBMB) Ckt-2 3)220 KV Jalandhar-Pong (BB) Ckt-2 4)220KV Bus 2 at Pong(BB) 5)66 MW Pong HPS - UNIT 2 6)220/66kV 40MVA ICT-1 at Pong(BB)
20	GD-1	Uttar Pradesh	13-05-2024 17:27	13-05-2024 20:11	02:44	0	5	0.000	0.008	50404	62451	i)During antecedent condition, load at 400kV Agra South(UP) was approx. 100MW (was evacuating through 132kV feeders). Power incoming at Agra South(UP) S/s was through 400 KV Agra Fatehbad-Agra South (UP) Ckt (approx. 76MW) and 400 KV Agra South(UP)-Firozabad(PJFTL) (UP) Ckt (approx. 24MW). ii)As reported, at 17:27 hrs, during inclement weather condition, B-N phase phase to earth fault occurred. (Exact location of fault yet to be received). iii)Due to B-N phase to earth fault, both the lines 400kV Agra South-Agra Fatehabad (UP) & 400kV Agra South -Firozabad (UP) tripped. iv)As per DR of Firozabad end of 400 KV Agra South(UP)-Firozabad(PJFTL) (UP) Ckt, B-N phase to earth fault is observed (Ib=~845A). Line tripped on zone-1 distance protection operation. v)As per DR of both ends of 400 KV Agra Fatehabad-Agra South (UP) Ckt, B-N phase to earth fault is observed (Ib=~796A, Ib=~6.2kA from Agra South & Agra Fatehabad end respectively). No A/R operation is observed from Agra Fatehabad end. vi)Due to tripping of both 400kV feeders connected to 400kV Agra South(UP), Agra South(UP) S/s lost its connectivity from grid and the complete 400/132kV Agra South(UP) S/s became dead. vii)As per PMU at Agra(PG), multiple B-N phase to earth faults are observed with fault clearing time of 80ms. viii)As per SCADA, no load loss is observed in UP control area. However, SLDC-UP reported 5MW load loss at 132kV Kheragarh S/s in UP control area.	1)400 KV Agra Fatehabad-Agra South (UP) Ckt 2)400 KV Agra South(UP)-Firozabad(PJFTL) (UP) Ckt)
21	GI-2	Rajasthan	13-05-2024 07:11	13-05-2024 08:46	01:35	0	136	0.000	0.218	53784	62451	i)400/220kV Bhiwadi(PG) has one and half breaker bus arrangement at 400kV side and double main and transfer bus scheme at 220kV side. ii)During antecedent condition, 500 KV HVDC Balia-Bhiwadi (PG) ckt-1 & ckt-2 were carrying approx. 200 MW each. iii)As reported, at 07:11hrs, B-N phase to earth fault occurred (exact location of fault yet to be received). iv)As per DR of Bhiwadi(RS) end, 220 KV Bhiwadi(PG)-Bhiwadi(RS) (RS) Ckt-1 and 220 KV Bhiwadi(PG)-Kushkhera(RS) (RS) Ckt tripped from Bhiwadi(RS) end only on back up E/F protection operation. Line didn't trip from Bhiwadi(PG) end. v)At the same time, 400 KV Khetri (PKTSL)-Bhiwadi(PG) (PBTSL) D/C, 400KV Bhiwadi-Neemrana (PG) Ckt-1, 400 KV Bassi-Bhiwadi (PG) Ckt, 500 KV HVDC Balia-Bhiwadi (PG) D/C, 220 KV Neemrana(PG)-Neemrana(RS) (RS) Ckt-1 and 220kV lines from Bhiwadi(PG) to Rewari(HV) (HV) D/C, HSIIDC Bawal(HV), Mau(HV) also tripped (exact reason for multiple elements tripping yet to be received). vi)As per PMU at Bhiwadi(PG), at 07:11 hrs, B-N phase to earth fault with no A/R operation is observed with delayed fault clearing time of 1480ms. vii)As per SCADA, change in demand of approx. 136MW in Rajasthan control area is observed.	1)400 KV Khetri (PKTSL)-Bhiwadi(PG) (PBTSL) Ckt-2 2)400 KV Khetri (PKTSL)-Bhiwadi(PG) (PBTSL) Ckt-1 3)400KV Bhiwadi-Neemrana (PG) Ckt-1 4)400 KV Bassi-Bhiwadi (PG) Ckt 5)500 KV HVDC Balia-Bhiwadi (PG) Ckt-1 6)500 KV HVDC Balia-Bhiwadi (PG) Ckt-2 7)220 KV Bhiwadi(PG)-Bhiwadi(RS) (RS) Ckt-1 8)220 KV Bhiwadi(PG)-Rewari(HV) (HV) Ckt-1 9)220 KV Bhiwadi(PG)-Rewari(HV) (HV) Ckt-2 10)220 KV Bhiwadi(PG)-HSIIDC Bawal(HV) (HVPNL) Ckt 11)220 KV Bhiwadi(RS)-Mau(HV) (HV) Ckt 12)220 KV Bhiwadi(PG)-Kushkhera(RS) (RS) Ckt 13)220 KV Neemrana(PG)-Neemrana(RS) (RS) Ckt-1
22	GD-1	Uttarakhand	15-05-2024 11:29	15-05-2024 14:52	03:23	0	200	0.000	0.285	65685	70164	i)220/132kV Jhajhra(Utt) S/s has double main bus scheme. 220kV Jhajhra is connected with 220kV IIP Harrawala, 400/220kV Dehradun(PG), 220kV Vyasi HEP and 220kV KHodri HEP. 220kV IIP Harrawala is load station which is further connected to 220kV Rishikesh. ii)During antecedent conditon, 220kV Khodri-Jhjhra ckt was in open condition and 220kV Jhajhra-Dehradun ckt, 220kV Jhajra-Vyasi ckt were under shutdown. 220kV Jhajra was connected to 220kV IIP Harrawala and source was through 220kV Rishikesh-Harrawal-Jhajhra. As only one 220kV circuit was available at Jhajra S/s, partial load (120-130MW) were already shifted to 400/220/132kV Rishikesh. iii)As reported, at 11:29hrs, 220kV Rishikesh-IIP Harrawala ckt tripped on B-N phase to earth fault. At tower location no. 115, 11kV line came in contact with the line due to less spacing. 220kV Rishikesh-IIP Harrawala ckt was carrying ~200MW so clearance issue due to sag is also suspected. iv)As per PMU at Roorkee(PG), B-N phase to earth fault cleared within 120msec is observed. v)Due to tripping of 220kV Rishikesh-IIP Harrawala ckt, supply to 220/132kV Jhajhra lost and S/s became dead. vi)As per SCASA, change in demand of approx. 200MW is observed in Uttarakahnd control area. vii)As reported, PTCUL has communicated respective DISCOM to make the 11kV line underground.	1) 220kV Rishikesh-IIP Harrawala ckt
23	GI-1	Haryana	17-05-2024 16:21	17-05-2024 17:15	00:54	Data not avaialbale due to SCADA issue						i)During antecedent condition, 500 KV HVDC Mundra-Mahindergarh(APL) bipole was carrying total ~1500MW. ii)As reported, at 16:21 hrs, 500 KV HVDC Mundra-Mahindergarh(APL) bipole blocked due to RPC No AC Filter alarm raised at Mohindergarh end. After thorough investigation, it was observed that RCI changeover has been initiated from RCI B to RCI A ACTIVE and after that "RPC SET RANGE EXCEED" event triggered followed by RPC NO AC FILTER, which caused blocking of both the Poles. iii)As per PMU at Mahinedergarh(PG), no fault in system is observed, fluctuation voltage is observed. iv)Due to blocking of both the poles of 500 KV HVDC Mundra-Mahindergarh(APL), there was power order reduction of ~1500MW. As per HVDC Mundra-Mahendergarh SPS, SPS case-3 would have operated and as per action in this case, load relief in UP, Haryana, Punjab, Rajasthan & Delhi and generation relief at Mundra Stage-III is desired. v) Details of load relief not received from SLDCs. Communication has been sent to all the SLDCs to share the quantum of load relief occurred in their respective control area due to SPS operation. SCADA data at NRLDC was not healthy during the event time. vi) As per detail BCU log of Mahindergarh end, DTPC fail alarm is recorded except Dhanonda. Any communciation related issue need to be rectified at the earliest to ensure proper SPS operation. vii) Both RCI System was restarted and the system was normalized	1)500 KV HVDC Mahindergarh(APL) Pole-1 2)500 KV HVDC Mahindergarh(APL) Pole-1
24	GD-1	Haryana	15-05-2024 23:12	16-05-2024 00:55	01:43	0	870	0.000	1.259	55540	69099	i)220/132kV Mohana S/s has six (06) 220kV lines i.e., 220 KV Mohana(HV)-Sonipat(PG) (HVPNL) D/C, 220kV Mohana-Samalkha (HV) D/C and 220kV Mohana-Sampla (HV) D/C. Samalkha and Sampla are load centre. Power source is through 220kV Sonipat only. Samalkha(HV) S/s had connectivity from grid through 220kV Mohana-Samalkha (HV) D/C only and 220kV Smapla is connected with 220kV Mohana and 400/220kV Kabulpur also. ii)During antecedent condition, 220 KV Mohana(HV)-Sonipat(PG) (HVPNL) D/C was carrying total ~300MW. Loading at 400/220 kV 315 MVA ICT 1 & 2 at Kabulpur(HV) was 282 MW & 279 MW respectively. iii)As reported, at 23:12 hrs, 220 KV Mohana(HV)-Sonipat(PG) (HVPNL) D/C tripped. (Exact reason of tripping yet to be received). iv)As per PMU at Bahadurgarh(PG), no fault is observed. v)At the same time, 400/220 kV 315 MVA ICT 1 & 2 at Kabulpur(HV) also tripped. vi)It is suspected that, after tripping of 220 KV Mohana(HV)-Sonipat(PG) (HVPNL) D/C, loading of 220kV Sampla (HV) would have shifted to 400/220kV Kabulpur ICTs and 400/220 kV 315 MVA ICT 1 & 2 at Kabulpur(HV) tripped on overcurrent protection operation. vii)Due to tripping of 220 KV Mohana(HV)-Sonipat(PG) (HVPNL) D/C, 220/132kV Mohana(HV) S/s lost its connectivity from grid and the complete 220/132kV Mohana(HV) S/s became dead. viii)As per SCADA, 870 MW load loss is observed in Haryana control area.	1)220 KV Mohana(HV)-Sonipat(PG) (HVPNL) Ckt-1 2)220 KV Mohana(HV)-Sonipat(PG) (HVPNL) Ckt-2 3)400/220 kV 315 MVA ICT 1 at Kabulpur(HV) 4)400/220 kV 315 MVA ICT 2 at Kabulpur(HV)

20.Details of Grid Events during the Month of May 2024 in Northern Region

Sl No.	Category of Grid Event (GI 1or GI 2/ GD-1 to GD-5)	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
						Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
25	GD-1	Uttarkhand	17-05-2024 17:21	17-05-2024 18:09	00:48	0	0	0.000	0.000	56219	67302	i)400kV Tehri(THDC) has double main bus scheme. ii)During antecedent condition, 250 MW TEHRI HPS – UNIT- 1, 2, 3 & 4 were not in service. iii)As reported, at 17:21 hrs, 400kV Tehri(THDC)-Koteshwar(PG)(PG) ckt-1 tripped on Y-B phase to phase fault. Fault distance was ~2.8km from Tehri(THDC) end. At the same time, 400kV Tehri(THDC)-Koteshwar(PG)(PG) ckt-2 tripped from Koteshwar(PG) end only. iv)As per DR of Tehri end of 400kV Tehri(THDC)-Koteshwar(PG)(PG) ckt-1, directional earth fault started followed by sensing Y-B-N fault in Z-1. Fault current was Iy=~4.9kA, Ib=~4.2kA. Total fault clearance time was ~760ms.. v)As per DR of Tehri end of 400kV Tehri(THDC)-Koteshwar(PG)(PG) ckt-2, distance protection sensed Y-B-N fault in Z-4, no tripping was initiated. Fault current was Iy=~4.8kA, Ib=~4.0kA. vi)As per PMU at Koteshwar(PG), Y-N phase to earth fault converted into Y-B phase to phase fault is observed with delayed fault clearance time of 760ms is observed. vii)As per SCADA no load loss/generation loss is observed during the tripping event (there was no generation at Tehri HEP during the event).	1)400 KV Tehri(THDC)-Koteshwar(PG) (PG) Ckt-1 2)400 KV Tehri(THDC)-Koteshwar(PG) (PG) Ckt-2
26	GD-1	Himachal Pradesh	18-05-2024 21:32	18-05-2024 22:00	00:28	30	540	0.053	0.717	56182	75336	i)During antecedent condition, 220/66kV 100MVA Transformer-1,2&3 at Baddi(HP) were carrying approx. 70MW each. ii)As reported, at 21:32 hrs, R-ph CT at 220kV side of 220/66kV 100MVA transformer-3 damaged. Deberis of damaged CT fell on B-ph CT of 220/66kV 100MVA transformer-2 and it also damaged. iii)As per PMU at Panchkula(PG), R-N fault followed by B-N fault with delayed clearance in 400msec is observed. iv)On this fault multiple 220kV lines at 220kV Kunihar, Baddi complex triped. v)220kV lines tripped during the event are 220kV Baddi(HP)-Pinjore(HV) (HPPTCL) D/C, 220 kV Baddi-Kunihar(HP) D/C, 220 kV Baddi-Upperla Nangal(HP) Ckt, 220 kV Baddi-Madhala(HP) Ckt, 220kV Upperla Nangal-Baddi ckt and 220 kV Wangtoo-Kunihar ckt. vi)Due to tripping of aforementioned elements, 220/66kV Baddi(HP) became dead. Supply to 220/132kV Kunihar (HP) also affected. 132kV bus at Kunihar S/s was operated at split mode, due to which supply to 132kV Barotiwala remained healty via 132kV Kangoo. vii)As per SCADA, change in demand of approx. 540MW in HP control area is observed. viii)Damaged B-ph CT of 220/66kV 100 MVA transformer-2 was repaced with B-ph CT of transformer-3 and transformer-2 was restored at 22:00hrs on the same day. ix)R & B phase CT of transformer-3 was replaced with the spare CT from other stations and transformer-3 at Baddi was restored at ~05:30hrs on 19th May 2024.	1)220/66kV 100MVA Transformer-2 at Baddi(HP) 2)220/66kV 100MVA Transformer-3 at Baddi(HP) 3)220kV Baddi(HP)-Pinjore(HV) (HPPTCL) Ckt-1 4)220kV Baddi(HP)-Pinjore(HV) (HPPTCL) Ckt-2 5)220 kV Baddi-Kunihar(HP) Ckt-1 6)220 kV Baddi-Kunihar(HP) Ckt-2 7)220 kV Baddi-Upperla Nangal(HP) Ckt 8)220 kV Madhala -Upperla Nangal(HP) Ckt 9)220 kV Baddi-Madhala (HP) Ckt 10)220 kV Wangtoo-Kunihar(HP) ckt
27	GD-1	Uttar Pradesh	18-05-2024 17:25	18-05-2024 18:00	00:35	0	860	0.000	1.243	56695	69211	i)400/220kV Gr. Noida S/s has double main transfer bus scheme and feeds 220/132kV Noida Sec20, Noida RC Green, Noida Sec 129 substations through 220kV feeders. ii)As reported, at 17:25hrs, B-ph CT at Gr. Noida end of 220kV Gr. Noida – Noida sec 20 ckt-1 damaged and R-ph isolator arm of 220kV Gr. Noida – Noida sec 20 ckt-1 also broked. iii)As per PMU at Dadri Thermal(NTPC), B-N and R-N fault with delayed clearance in ~600msec is observed. iv)On this fault, 220kV Gr. Noida – Noida sec 20 D/C tripped and supply to Noida Sec 20 S/s lost. v)At the same time, 400/220 kV 315 MVA ICT 6 at Gr.Noida(UP) trippped on of PRD (Pressure Relief Device) and OSR (Oil Surge Relay) protection operation. vi)As per SCADA SOE at NRLDC, 220kV Gr. Noida – Noida Sec 129 ckt and 220kV Greater Noida - Jalpura ckt also tripped at the same time. vii)As per SCADA, total change in demand of approx. 860MW occurred in UP contro, area. viii)Partial supply of Noida Sec 20 restored at 18:00 hrs from 132kV Noida Sec 45 and at 18:06 hrs from Noida Sec 66.	1)400/220 kV 315 MVA ICT 6 at Gr.Noida(UP) 2)220kV Gr. Noida – Noida Sec 20 ckt-1 3)220kV Gr. Noida – Noida Sec 20 ckt-2 4)220kV Gr. Noida – Noida Sec 129 ckt 5)220kV Greater Noida - Jalpura ckt
28	GI-2	Uttar Pradesh	19-05-2024 00:55	19-05-2024 10:12	09:17	1758	0	3.239	0.000	54284	73440	i)During antecedent condition, 660MW Unit-1, 2 & 3 at Bara TPS was carrying ~600MW, ~604MW & ~554MW respectively. Generation was evacuating through 765kV Bara-Mainpuri ckt (carrying ~1095MW) and 765/400kV 1500MW ICT-1 & 2 (carrying ~330MW each). ii)At 00:55:30hrs, 660MW Unit-2 at Bara TPS tripped. As reported, overall differential protection of GT of Unit-2 operated on B-N fault occurred due to surface tracking at bushing of GT of Unit-2. iii)As per PMU at Mainpuri(PG) at 00:55:30hrs, B-N phase to earth fault cleared within 80msec is observed. iv)At the same time, fluctuation (ramp down and up) in active power of Unit-1 at Bara TPS also started. As per SCADA data, MW reduced from 600MW to ~150MW and again ramped up to ~600MW. v)Further at 00:57:27 hrs, 660MW Unit-3 at Bara TPS also tripped. As reported, overall differential protection of GT of Unit-3 operated on B-N fault occurred due to surface tracking at bushing of GT of Unit-2. vi)As per PMU at Mainpuri(PG) at 00:57:27hrs, B-N phase to earth fault cleared within 80msec is observed. vii)Further at 00:57:35hrs, 660MW Unit-1 at Bara TPS also tripped. As reported, unit-1 tripped due to turbine extraction pressure ratio protection operation. viii)As per SCADA plot of Bara TPS generation, Unit-2 tripped at 00:55hrs and subsequently fluctuation in power of Unit-1 started. At 00:57 hrs, Unit-3 tripped followed by tripping of Unit-1. ix)Tripping of Unit-1 not recorded in SCADA SOE. x)As reported, bushing cleaning and IR test of GT-2&3 has been done.	1)660 MW Bara TPS UNIT 2 2)660 MW Bara TPS UNIT 3 3)660 MW Bara TPS UNIT 1
29	GD-1	Punjab	19-05-2024 20:04	19-05-2024 21:57	01:53	0	90	0.000	0.131	52155	68557	i)220kV Dhandarikalan(PS) S/s has single Bus scheme. It has four (4) 220 kV lines i.e. 220 KV Jamalpur(BB)-DandhariKalanl(PS) (PSTCL) D/C & 220 KV DandhariKalanl(PS)-Ludhiana(PG) (PSTCL) D/C. ii)During antecedent condition, 220 KV DandhariKalanl(PS)-Ludhiana(PG) (PSTCL) ckt-2 was not in service (As per SCADA). iii)As reported, at 20:04 hrs, B-phase CT of 220 KV Jamalpur(BB)-DandhariKalanl(PS) (PSTCL) Ckt-1 burst which created a bus fault at 220 KV DandhariKalanl(PS). Further all the elements (220 KV Jamalpur(BB)-DandhariKalanl(PS) (PSTCL) D/C, 220 KV DandhariKalanl(PS)-Ludhiana(PG) (PSTCL) ckt-1 and 220/66kV ICT-1 & 2) connected at 220kV Bus tripped due to Bus bar protection operation at DandhariKalanl(PS) S/s. iv)At the same time 220 KV Ganguwal-Jamalpur(BB) Ckt-2, 220 KV Jamalpur(BB)-Sangrur(PS) (BB) Ckt-2 & 220KV Bus 2 at Jamalpur(BB) also tripped (Exact reason of tripping yet to be received). v)As per DR of Sangrur(PS) end of 220 KV Jamalpur(BB)-Sangrur(PS)(BB) Ckt-2, multiple faults (B-N & Y-N phase to earth) are observed. B-N phase to earth fault was sensed in zone-2. vi)Due to Bus bar protection operation, 220 KV DandhariKalanl(PS) S/s lost its connectivity from grid and the complete 220 KV DandhariKalanl(PS) S/s became dead. vii)As per PMU at Ludhiana(PG), B-N phase to earth fault is observed. viii)As per SCADA, 216 MW load loss is observed in Punjab control area. However SLDC-Punjab reported load loss of 90 MW in Punjab control area.	1)220 KV Jamalpur(BB)-DandhariKalanl(PS) (PSTCL) Ckt-1 2)220 KV Jamalpur(BB)-DandhariKalanl(PS) (PSTCL) Ckt-2 3)220 KV DandhariKalanl(PS)-Ludhiana(PG) (PSTCL) Ckt-1 4)220/66kV ICT-1 at Dhandarikalan(PS) 5)220/66kV ICT-2 at Dhandarikalan(PS) 6)220 KV Ganguwal-Jamalpur (BB) Ckt-2 7)220 KV Jamalpur(BB)-Sangrur(PS) (BB) Ckt-2 8)220KV Bus 2 at Jamalpur(BB)

20.Details of Grid Events during the Month of May 2024 in Northern Region

Sl No.	Category of Grid Event (GI 1or GI 2/ GD-1 to GD-5)	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
						Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
30	GD-1	Delhi	20-05-2024 00:46	20-05-2024 01:32	00:46	605	700	1.100	0.970	54981	72168	i)During antecedent condition, 400kV Interconnectors between 400kV Bawana(DTL) and 400kV CCGT Bawana were in open position. 400/220kV 315MVA ICT-1, 4, 5 & 6 were connected to CCGT Bawana and 315 MVA ICT-2 & 3 were connected to Bawana(DTL). 216 MW Unit-1 (GTG-1), Unit-2 (GTG-2) & Unit-5 (STG-1) at CCGT Bawana were generating approx. 194MW, 197MW and 218MW respectively. 400/220kV 315MVA ICT-1,4,5&6 at Bawana(DTL) (connected at CCGT Bawana) were carrying total ~740MW. ii)As reported, at 21:22 hrs of 19th May 2024, 400 KV SF6 Lockout alarm appeared on Main CB of 400 KV CCGT Bawana– Bhiwani ckt at CCGT Bawana end (connected at 400kV Bus-1). CCGT Bawana requested to take 400kV Bus-1 on shutdown to isolate the bus and attend the issue. iii)Further at 23:52hrs, 400 KV CCGT Bawana– Bhiwani ckt was opened from Bhiwani end and DT signal received at CCGT Bawana end. Tie CB at CCGT Bawana end opened on DT signal however, Main CB which was under lockout didn't open. iv)Further, the maintenance staff tried to isolate the lockout Main CB by opening the line Isolator however, huge spark along with blast sound emanated from the contacts of the isolator. v)As per PMU at Bhiwani(PG), Y-B fault converted into R-Y-B-N fault with delayed clearance of ~560msec is observed. vi)On this fault, 400 KV Bawana CCGTB(DTL)-Bahadurgarh(PG) (PG) Ckt tripped from CCGT Bawana end on Z-4 distance protection operation with the delay of ~500msec and TEED protection operated on 400 KV CCGT Bawana –Bhiwani Line at CCGT Bawana end. (DR file attached) vii)With the tripping of 400 KV Bawana CCGTB(DTL)-Bahadurgarh(PG) (PG) Ckt, generation at CCGT Bawana got islanded with the load of 400/220kV 315MVA ICT-1,4,5&6 at Bawana(DTL) (connected at CCGT Bawana). viii)As 400/220kV 315MVA ICT-1,4,5&6 at Bawana(DTL) (connected at CCGT Bawana) were carrying total ~740MW and running units at CCGT Bawana were generating ~605MW, there was load generation imbalance. Further, 216 MW Unit-1 (GTG-1), Unit-2 (GTG-2) & Unit-5 (STG-1) at CCGT Bawana tripped on back up impedance protection operation. ix)With the tripping of running units at CCGT Bawana, 400/220kV 315MVA ICT-1,4,5&6 at Bawana(DTL) got dead and the load of 220kV Shalimar Bagh, 220kV SGTN,	1) 400 KV Bawana CCGTB(DTL)-Bahadurgarh(PG) (PG) Ckt 2) 216 MW Bawana GPS - UNIT 1 (GTG-1) 3) 216 MW Bawana GPS - UNIT 2 (GTG-2) 4) 253 MW Bawana GPS - UNIT 5 (STG-1)
31	GD-1	Rajasthan	20-05-2024 21:30	21-05-2024 02:00	04:30	0	148	0.000	0.199	54630	74367	i)220/132kV Khushkhera(RS) has Double main Bus arrangement at 220kV side. ii)During antecedent condition, 220/132kV 160MVA ICT-1, 220kV lines from Khushkhera to Bhiwadi(PG), Alwar(RS), Kishangarh(RS) were not in service. Only 220 KV Neemrana(PG)-Khushkhera(RS) (RS) Ckt and 220/132kV 160MVA ICT-2 were in service. iii)As reported, at 21:30hrs, during the charging of 220/132kV 160MVA ICT-1, jumper of the same ICT snapped which caused Y-B phase to phase fault which led to tripping of 220 KV Neemrana(PG)-Kushkhera(RS) (RS) Ckt on zone-2 distance protection operation from the Neemrana(PG) end. 220 KV Neemrana(PG)-Khushkhera(RS) (RS) Ckt was only source available at 220kV Kushkhera(RS). iv)As per DR of Neemrana(PG) end of 220 KV Neemrana(PG)-Kushkhera(RS) (RS) Ckt, Y-B phase to phase fault is observed with fault current of Iy~=3.6kA, Ib~=3.4kA and line tripped on zone-2 distance protection operation from Neemrana end. v)Due to tripping of 220 KV Neemrana(PG)-Kushkhera(RS) (RS) Ckt, 220/132kV Khushkhera(RS) S/s lost its connectivity from grid and the complete 220/132kV Khushkhera(RS) S/s became dead. vi)As per PMU at Bhiwadi(PG), at 21:30 hrs, Y-B phase to phase fault is observed with delayed fault clearing time of 560ms. vii)As per SCADA, change in demand of approx. 148MW in Rajasthan control area is observed.	1) 220 KV Neemrana(PG)-Khushkhera(RS) (RS) Ckt
32	GD-1	Punjab	21-05-2024 01:48	21-05-2024 05:01	03:13	0	38	0.000	0.054	52398	71010	i)220/132/66kV Baghapurana(PS) has single main and transfer bus scheme at 220kV level. ii)As reported, at 01:48hrs, 220 kV Y-phase line CT and R-phase limb of 220 kV SF-6 CB of 220 kV Baghapurana(PS)-Bajakhana(PS) Ckt-1 got damaged at Baghapurana(PS) end which created bus fault at 220kV main bus at Baghapurana(PS). iii)Due to this, bus bar protection operated at 220kV main bus of Baghapurana(PS). Hence, all the 220kV circuits connected to Baghapurana(PS), i.e., 220 kV Baghapurana(PS)-Bajakhana(PS) Ckt 1 & 2 and 220 kV Baghapurana(PS)-Moga(PS) Ckt-1 & 2 tripped and complete blackout occurred at Baghapurana(PS) S/s. iv)As per PMU at Ludhiana(PG), Y-N phase to earth fault followed by R-B phase to phase fault followed by R-Y phase to phase fault with fault clearance time of respectively 120ms, 200ms and 480ms (delayed) are observed. v)As per SCADA, change in demand of approx. 430MW is observed in Punjab control area. But as reported by SLDC Punjab, load loss of approx. 38MW occurred during the event.	1) 220 kV Baghapurana(PS)-Bajakhana(PS) Ckt-1 2) 220 kV Baghapurana(PS)-Bajakhana(PS) Ckt-2 3) 220 kV Baghapurana(PS)-Moga(PS) Ckt-1 4) 220 kV Baghapurana(PS)-Moga(PS) Ckt-2
33	GI-2	Rajasthan	23-05-2024 13:02	23-05-2024 14:29	01:27	0	255	0.000	0.324	69145	78813	i)During antecedent condition, MVA power flows of 400/220 kV 315 MVA ICT 1 and 500 MVA ICT 2 at Kankani(RS) were approx. 248MVA and 388MVA respectively as per SCADA. ii)As reported and as confirmed from DR, at 13:02hrs, 400/220 kV 315 MVA ICT 1 at Kankani(RS) tripped due to PRD-1 (Pressure Relief Device) relay operation (exact reason of relay operation yet to be shared). iii)Due to this tripping, 400/220 kV 500 MVA ICT 2 at Kankani(RS) got overloaded and tripped due to over-current protection operation. iv)As per SCADA SOE, 220/132kV 100MVA ICT 1 & 2 at Bilara(RS) tripped during the same time (exact reason of tripping yet to be shared). v)As per PMU at Jodhpur(RS), no fault is observed in the system. But voltage dip of approx. 9.275kV (ph) was observed at the same time. vi)As per SCADA, change in demand of approx. 255MW is observed in Rajasthan control area.	1) 400/220 kV 315 MVA ICT 1 at Kankani(RS) 2) 400/220 kV 500 MVA ICT 2 at Kankani(RS)
34	GD-1	Uttar Pradesh	23-05-2024 03:28	23-05-2024 04:49	01:21	0	245	0.000	0.341	53019	71891	i)220/132/33kV Kanpur Naubasta(UP) has single main and transfer bus scheme at both 220kV and 132kV level. ii)During antecedent condition, MVA power flows of 220/132kV 160MVA ICT-1 & 2 at Kanpur Naubasta(UP) were approx. 83MVA and 85MVA respectively as per SCADA. iii)As reported, at 03:28hrs, B-N phase to earth fault occurred on 220 KV Kanpur(PG)-Kanpur Naubasta(UP) (PG) Ckt-1 (exact location of fault yet to be shared). Line auto-reclosed successfully from Kanpur(PG) end. As per DR at Kanpur Naubasta(UP) end, fault was sensed in zone-4 and fault current was approx. 8.64kA. Fault clearing time was approx. 55ms. iv)During the same time, bus bar protection operated at 220kV bus of Kanpur Naubasta(UP) which led to tripping of all the elements connected to 220kV bus and complete blackout occurred at 220/132/33kV Kanpur Naubasta(UP) (exact reason of bus bar protection operation yet to be shared). v)As per PMU at Kanpur(PG), B-N phase to earth fault with fault clearing time of 120ms is observed. vi)As per SCADA, change in demand of approx. 245MW is observed in Uttar Pradesh control area.	1) 220 KV Kanpur(PG)-Kanpur Naubasta(UP) (PG) Ckt-1 2) 220 KV Fatehpur(PG)- Kanpur Naubasta(UP) (PG) Ckt-1 3) 220/132kV 160MVA ICT-1 at Kanpur Naubasta(UP) 4) 220/132kV 160MVA ICT-2 at Kanpur Naubasta(UP)
35	GD-1	Jammu and Kashmir	23-05-2024 14:49	23-05-2024 16:13	01:24	0	235	0.000	0.289	69934	81359	i)220/132kV Pampore(PDD) has Double main Bus arrangement at 220kV side. ii)During antecedent condition, power flow from Wagoora(PG) S/s to Pampore(PDD) S/s was approx. 200 MW through 220 KV Wagoora(PG)-Pampore(PDD) (PG) D/C. 220kV Pampore-Mirbazar(PDD) D/C was not in service. iii)As reported, at 14:49 hrs, 220 KV Wagoora(PG)-Pampore(PDD) (PG) D/C tripped from Wagoora(PG) end on R-N phase to earth fault on 220 KV Wagoora(PG)-Pampore(PDD) (PG) Ckt-2 with fault distance of 2.38km from Wagoora(PG) end. From Pampore(PDD JK) end, only 220 KV Wagoora(PG)-Pampore(PDD) (PG) Ckt-2 tripped. iv)As per DR of Wagoora(PG) end of 220 KV Wagoora(PG)-Pampore(PDD) (PG) ckt-1 & ckt-2, R-N phase to earth fault is observed in zone-2 and zone-1 for 220 KV Wagoora(PG)-Pampore(PDD) (PG) ckt-1 & ckt-2 respectively. Fault current observed is Ir=~7.9kA & Ir=~12.9kA for 220 KV Wagoora(PG)-Pampore(PDD) (PG) ckt-1 & ckt-2 respectively. From relay flags, it is suspected that fault was on 220 KV Wagoora(PG)-Pampore(PDD) (PG) ckt-2 and ckt-1 tripped from Wagoora(PG) end due to delayed clearance from Pampore end. Details of Pampore end yet to be received. v)As per PMU at Wanpoh(PG), R-N phase to earth fault is observed with delayed fault clearing time of 520ms. vi)As per SCADA, change in demand of approx. 235MW is observed in J&K control area.	1) 220 KV Wagoora(PG)-Pampore(PDD) (PG) Ckt-1 2) 220 KV Wagoora(PG)-Pampore(PDD) (PG) Ckt-2
36	GD-1	Punjab	24-05-2024 10:08	24-05-2024 10:27	00:19	0	460	0.000	0.577	69088	79682	i)220/66kV Laltonkalan(PS) has double main bus scheme at 220kV level. ii)During antecedent condition, 220kV Bus-1 at Laltonkalan(PS) was under shutdown for maintenance. All the 220kV lines and 220/66kV ICTs were connected to Bus-2. iii)As reported, at 10:08hrs, Y-phase CT of 220/66kV 100MVA ICT-4 at Laltonkalan(PS) blasted which created bus fault at 220kV Bus-2 at Laltonkalan(PS). iv)Due to this, bus bar protection operated at 220kV Bus-2 of Laltonkalan(PS). Hence, all the 220kV circuits connected to Laltonkalan(PS) along with three 220/66kV ICTs tripped and complete blackout occurred at Laltonkalan(PS) S/s. v)As per PMU at Jalandhar(PG), Y-N phase to earth fault with fault clearance time of respectively 80ms is observed. vi)As per SCADA, change in demand of approx. 460MW is observed in Punjab control area.	1)220kV Laltonkalan(PS)-Hambran(PS) Ckt 2)220kV Laltonkalan(PS)-Dandharikalan(PS) Ckt 3)220kV Laltonkalan(PS)-Jagraon(PS) Ckt 4)220kV Laltonkalan(PS)-Ferozepur Road LDH(PS) Ckt 5)220 KV Ludhiana(PG)-Laltokalan(PS) (PSTCL) Ckt-1 6)220 KV Ludhiana(PG)-Laltokalan(PS) (PSTCL) Ckt-2 7)220/66kV 160MVA ICT-1 at Laltonkalan(PS) 8)220/66kV 100MVA ICT-4 at Laltonkalan(PS) 9)220/66kV 100MVA ICT-5 at Laltonkalan(PS) 10)66/11kV 20MVA ICT-2 at Laltonkalan(PS) 11)66/11kV 20MVA ICT-3 at Laltonkalan(PS)

20.Details of Grid Events during the Month of May 2024 in Northern Region

Sl No.	Category of Grid Event (GI 1or GI 2/ GD-1 to GD-5)	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
						Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
37	GI-2	Haryana, Punjab, UP, Delhi & Rajasthan	25-05-2024 12:46	25-05-2024 14:16	01:30	1100	2200	1.556	2.737	70679	80392	i)During antecedent condition, 800kV HVDC Champa-Kurukshetra Bipole-1&2 was running on its full capacity (carrying approx. 6000MW). ii)At 12:46:14:160hrs, Y-B fault converted into R-Y-B after ~220msec occurred on 220 KV Hissar(BBMB)-Hissar IA(HV) (HVPNL) Ckt-1. Line CB at Hissar(BBMB) end opened instantaneously on Y-B fault in Z-1. However, CB at Hissar_IA(HR) end opened with delay of ~440msec in Z-2. At the same time, line CB at Hissar(BBMB) end of 220 KV Hissar(BBMB)-Hissar IA(HV) (HVPNL) Ckt-2 also opened. Fault current was in the range of ~25kA. Snapshot of DR of 220 KV Hissar(BBMB)-Hissar IA(HV) (HVPNL) D/C attached in Annexure. iii)As reported, fault occurred due to vegetation fire incident in line corridor at distance ~3km from Hissar_IA(HR) end. Initially Y-B fault occurred and after ~220msec R-ph jumper at Hissar_IA end of 220 KV Hissar(BBMB)-Hissar IA(HV) (HVPNL) Ckt-1 snapped led to R-Y-B fault. iv)As per PMU plot of phase voltages, Y-B fault converted into R-Y-B fault with delayed clearance in ~480msec is observed. v)On this fault, generation reduction at some of the RE stations observed. As per SCADA, reduction of ~1100MW RE generation at Rajasthan occurred, out of which ~700MW recovered within 2.5minutes. As per PMU phase voltage at POI end of 220kV line of RE stations, voltage was above 0.9pu during the fault. vi)At 12:46:14:706hrs, 800kV HVDC Champa-Kurukshetra pole-1 blocked on DC overcurrent protection operation at Kurukshetra end. DC overcurrent protection operated due to multiple commutation failure in pole-1. vii)There was variation in frequency during the incident. From 12:46:15:800hrs, frequency started ramping down from 49.90Hz and reached 49.874Hz at 12:46:16:120hrs. During this frequency reduction, multiple feeders at distribution level in Northern Region tripped on df/dt (ROCOF) stage-1 (0.1Hz/sec) operation. List of feeders tripped on df/dt operation in UP, Punjab, Delhi and Rajasthan is attached in Annexure. DR file of df/dt relay of Lodhi Road S/s (Delhi control area) is also attached in Annexure. viii)As per SCADA, change in Northern Region demand of approx. 2200MW (Punjab:1140MW; Rajasthan:180MW; Haryana:520MW; Delhi:60MW) observed. As reported by SLDCs, load loss of ~172MW in UP, ~82MW in Delhi, ~1375MW in Punjab and ~140MW in Rajasthan occurred due to df/dt operation.	1) 220 KV Hissar(BBMB)-Hissar IA(HV) (HVPNL) Ckt-1 2) 220 KV Hissar(BBMB)-Hissar IA(HV) (HVPNL) Ckt-2 3) 800 KV HVDC Kurukshetra(PG) Pole-1
38	GI-1	Delhi	25-05-2024 16:10	25-05-2024 17:20	01:10	195	211	0.295	0.267	66081	79118	i)220/66kV Pragati(DV) has Double main Bus arrangement at 220kV side. ii)During antecedent condition, power generation of 104.6MW UNIT-1, 2 and 121.2MW UNIT-3 was 85MW, 80MW & 107MW respectively at 220/66kV Pragati(DV). iii)As reported, at 16:10 hrs, UNIT-1 & 3 tripped on internal fault of UNITs (exact reason of tripping yet to be received). iv)As per PMU at Maharanibagh(PG), Y-B phase to phase fault is observed with delayed fault clearing time of 400ms. v)As per SCADA, change in demand of approx. 211MW is observed in Delhi control area. Generation loss of approx. 195MW is observed at 220/66kV Pragati(DV) (as per SCADA).	1) 104.6 MW Pragati Gas Turbines - UNIT 1 2) 121.2 MW Pragati Gas Turbines - UNIT 3
39	GD-1	Himachal Pradesh	26-05-2024 13:30	26-05-2024 14:12	00:42	185	0	0.271	0.000	68338	77854	i)During antecedent condition (before event at @13:30hrs), 96 MW Unit-1 & 2 at ADHPL(IP) were generating approx. 93MW and 92MW respectively as per SCADA. ii)As reported, at 13:21 hrs, 220 KV Phozal(HP)-Nallagarh(PG) (ADHPL) Ckt tripped on B-N phase to earth fault with fault distance of 121km from Nallagarh(PG) end and 37.8km from Phozal(HP) end (as per DR). As per DR, fault sensed in zone-1 at Nallagarh(PG) end and zone-2 at Phozal(HP) and fault current was ~1.164kA from Nallagarh(PG). DT received at Phozal(HP) end. iii)As per PMU at Nallagarh(PG), B-N phase to earth fault with fault clearing time of 240ms is observed. iv)As per SCADA, generation loss of approx. 40MW is observed at ADHPL(IP). v)Further at 13:30hrs, R-N phase to earth fault occurred on 220 KV AD hydro(AD)-Nallagarh(PG) (ADHPL) Ckt with fault current of ~6.07kA from Nallagarh(PG) end and ~1.164kA from AD hydro end (as per DR). As reported, there was forest fire near 400/220kV Nallagarh(PG) S/s. vi)As per DR at Nallagarh(PG) end, fault was sensed in zone-1 at Nallagarh(PG) and line successfully auto-reclosed from Nallagarh(PG) end. vii)As per DR at AD hydro end, fault was sensed in zone-2 and carrier received at AD hydro, but zone-2 got reset. Y and B-ph current increased upto ~1.53kA and ~1.47kA and A/R lockout operated and line tripped from AD hydro end. viii)Again, B-N phase to earth fault occurred on 220 KV AD hydro(AD)-Nallagarh(PG) (ADHPL) Ckt within the reclaim time with fault current of ~1.07kA from Nallagarh(PG) end (as per DR); fault was sensed in zone-1 and tripped from Nallagarh(PG) end. ix)Due to tripping of 220 KV Phozal(HP)-Nallagarh(PG) (ADHPL) Ckt and 220 KV AD hydro(AD)-Nallagarh(PG) (ADHPL) Ckt, island formed with 96MW Unit-1 & 2 at AD hydro(AD) and 220 KV AD hydro(AD)-Phozal(HP) (ADHPL) Ckt and generation-load imbalance occurred. After this, 96MW Unit-1 & 2 at AD hydro(AD) tripped due to over-speeding. x)After tripping of 96MW Unit-1 & 2 at AD hydro(AD), 220 KV AD hydro(AD)-Phozal(HP) (ADHPL) Ckt got de-energized and complete blackout occurred at 220kV ADHPL(IP). xi)As reported, line CB at AD hydro end of 220 KV AD hydro(AD)-Phozal(HP) (ADHPL) Ckt was opened manually at 13:48 hrs. xii)As per PMU at Nallagarh(PG), B-N fault and then R-N phase to earth fault with fault clearing time of 120ms and 120ms are observed.	1) 220 KV Phozal(HP)-Nallagarh(PG) (ADHPL) Ckt 2) 220 KV AD hydro(AD)-Nallagarh(PG) (ADHPL) Ckt 3) 220 KV AD hydro(AD)-Phozal(HP) (ADHPL) Ckt 4) 96MW Unit-1 at AD hydro(AD) 5) 96MW Unit-1 at AD hydro(AD)
40	GD-1	Rajasthan	26-05-2024 10:45	26-05-2024 11:45	01:00	85	260	0.134	0.339	63418	76627	i)During antecedent condition, MW power flows of 220 KV Bassi(PG)-Bagru(RS) (PG) Ckt and 220 KV Phulera(RS)-Bagru(RS) Ckt were approx. 286MW and 217MW respectively as per SCADA. ii)As reported, at 10:45hrs, 220 KV Bassi(PG)-Bagru(RS) (PG) Ckt and 220 KV Phulera(RS)-Bagru(RS) Ckt tripped due to over-loading (exact reason, nature and location of fault yet to be shared). iii)Due to this tripping, complete blackout occurred at 220/132kV Bagru(RS) S/s. iv)During the same time, 90MW Anta Solar plant also tripped (exact reason of tripping yet to be shared). v)As per PMU at RAPP-C(NPC), R-N phase to earth fault with fault clearing time of 80ms is observed. vi)As per SCADA, change in demand of approx. 260MW is observed in Rajasthan control area. vii)As per SCADA, change in NR solar generation of approx. 85MW is observed.	1) 220 KV Bassi(PG)-Bagru(RS) (PG) Ckt 2) 220 KV Phulera(RS)-Bagru(RS) Ckt
41	GI-1	Rajasthan	27-05-2024 16:40	27-05-2024 16:51	00:11	0	165	0.000	0.209	62324	78769	i)220/132kV Heerapura(RS) has double main and transfer bus arrangement at 220kV side. ii)During antecedent condition, 220/132kV 100MVA ICT-1 at Heerapura220(RS) was not in service. iii)As reported, at 16:40hrs, during the charging of 132 KV Heerapura(RS)-Chambal(RS) Ckt, line CB pole at Heerapura220(RS) end bursted. iv)During the same time, B-C-G bus coupler (Bus B and C coupled by B-C-G Bus coupler), 220 KV Heerapura(RS)-Mansarovar(RS) Ckt, 220 KV Heerapura(RS)-VKIA(RS) Ckt, 220/132kV 160MVA ICT-1, 2 and 3 at Heerapura220(RS) also tripped (exact reason and nature of protection operated yet to be shared). v)As per PMU at Bassi(PG), 3-phase to earth fault (delayed fault clearance in B-ph) is observed with delayed fault clearing time of 720ms. vi)As per SCADA, change in demand of approx. 165MW in Rajasthan control area is observed.	1) 132 KV Heerapura(RS)-Chambal(RS) Ckt 2) 220 KV Heerapura(RS)-Mansarovar(RS) Ckt 3) 220 KV Heerapura(RS)-VKIA(RS) Ckt 4) 220/132kV 160MVA ICT-1 at Heerapura220(RS) 5) 220/132kV 160MVA ICT-2 at Heerapura220(RS) 6) 220/132kV 160MVA ICT-4 at Heerapura220(RS)
42	GI-2	Haryana	27-05-2024 14:36	27-05-2024 15:53	01:17	665	1060	0.959	1.249	69347	84892	i)400kV Jhajjar(APCPL) has one and half breaker arrangement at 400kV side. ii)During antecedent condition, 500MW Unit 1, 2 and 3 at Jhajjar(APCPL) were generating approx. 280MW each. iii)As reported, at 14:36hrs, 400 KV Jhajjar(APCL)-Mundka(DV) (APCL) Ckt-2 tripped on R-Y-N double phase to earth fault with fault distance of 55.1km from Mundka(DV) end. As per DR, fault current was Ir=~8.263kA and Iy=~8.258kA from Mundka(DV); zone-1 distance protection operated at Mundka(DV). iv)During the same time, 400 KV Jhajjar(APCL)-Daulatabad(HV) (HV) Ckt-2 also tripped (exact reason and nature of protection operated yet to be shared). v)As per PMU at Gurgaon(PG), R-Y-N double phase to earth fault is observed with fault clearing time of 80ms. vi)As per SCADA, change in demand of approx. 540MW in Haryana, 280MW in Delhi, 140MW in UP and 100MW in Uttarakhand control area are observed (details from states regarding any frequency related protection (df/dt) operation yet to be received). vii)As per SCADA, change in NR solar generation of approx. 665MW among which almost 495MW revived within 3 minutes.	1) 400 KV Jhajjar(APCL)-Mundka(DV) (APCL) Ckt-2 2) 400 KV Jhajjar(APCL)-Daulatabad(HV) (HV) Ckt-2

20.Details of Grid Events during the Month of May 2024 in Northern Region

Sl No.	Category of Grid Event (GI 1or GI 2/ GD-1 to GD-5)	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
						Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
43	GI-2	Rajasthan	28-05-2024 05:37	28-05-2024 06:38	01:01	0	620	0.000	0.845	51468	73388	i)During antecedent condition, 400/220 kV 315 MVA ICT 1 and 500 MVA ICT 2 at Bikaner(RS) was carrying approx. 294MVA and 296MVA respectively. 220kV Bikaner-Dungargarh (RS) line was in open condition from Bikaner end. ii)As reported, at 05:37 hrs, 400/220 kV 315 MVA ICT 2 at Bikaner(RS) tripped on WTI (Winding Temperature Indicator) protection operation. iii)As per PMU at Bhadla(PG), no fault in system is observed. iv)Due to tripping of ICT-2, SPS implemented at Bikaner(RS) S/s related to overloading of remaining ICTs after tripping of any ICT operated. As per SPS scheme, 220kV Bikaner-Nokha (RS) line and 220kV Bikaner-Dungargarh (RS) line has to open. 220kV Bikaner-Nokha (RS) line (carrying ~140MW) opened on SPS operation. v)With the tripping of 400/220kV ICT-2, load shifted on 400/220kV ICT-1 and as 220kV Bikaner-Dungargarh (RS) line was already in open condition, after SPS operation sufficient load relief was not achieved. Thus, loading of ICT-1 remained in the range of ~400MVA. vi)Further, 400/220 kV 315 MVA ICT 1 at Bikaner(RS) also tripped on overcurrent protection operation. vii)As per SCADA, change in demand of approx. 620MW is observed in Rajasthan control area. viii)As reported, setting of WTI in ICT-2 has been revised from 85-90°C to 95-100°C.	1) 400/220 kV 315 MVA ICT 1 at Bikaner(RS) 2) 400/220 kV 315 MVA ICT 2 at Bikaner(RS)
44	GI-2	Rajasthan	28-05-2024 10:43	28-05-2024 11:31	00:48	0	495	0.000	0.607	65751	81584	i)During antecedent condition, 400/220 kV 315 MVA ICT 1 and 500 MVA ICT 2 at Bikaner(RS) were carrying approx. 300MVA. 220kV Bikaner-Dungargarh (RS) line was in open condition from Bikaner end. ii)As reported, at 10:43 hrs, 400/220 kV 315 MVA ICT 2 at Bikaner(RS) tripped on overcurrent protection operation. iii)From DR of 400/220 kV 315 MVA ICT 2 at Bikaner(RS) of HV side, it is observed that phase current were started increasing from 430A and reached till ~590A due to sudden increase in loading of 220kV Bikaner-Nokha (RS) line. Reason of sudden increase in loading of line is not identified yet. iv)As per PMU at Bhadla(PG), no fault in system is observed. v)Due to tripping of ICT-2, SPS implemented at Bikaner(RS) S/s related to overloading of remaining ICTs after tripping of any ICT operated. As per SPS scheme, 220kV Bikaner-Nokha (RS) line and 220kV Bikaner-Dungargarh (RS) line has to open. 220kV Bikaner-Nokha (RS) line opened on SPS operation. As per DR of 220kV Bikaner-Nokha (RS) line, phase current of line before tripping was in the range of ~680A. vi)With the tripping of 400/220kV ICT-2, load shifted on 400/220kV ICT-1 and as 220kV Bikaner-Dungargarh (RS) line was already in open condition, after SPS operation sufficient load relief was not achieved. Thus, loading of ICT-1 remained high (phase current increased from ~590A to ~770A after tripping of ICVT-2). vii)Further, 400/220 kV 315 MVA ICT 1 at Bikaner(RS) also tripped on overcurrent protection operation. viii)As per SCADA, change in demand of approx. 495MW is observed in Rajasthan control area.	1) 400/220 kV 315 MVA ICT 1 at Bikaner(RS) 2) 400/220 kV 315 MVA ICT 2 at Bikaner(RS)
45	GD-1	Haryana	28-05-2024 18:27	28-05-2024 19:50	01:23	0	222	0.000	0.314	53002	70596	i)220/132kV Mohana S/s has six (06) 220kV lines i.e., 220 KV Mohana(HV)-Sonipat(PG) (HVPNL) D/C, 220kV Mohana-Samalkha (HV) D/C and 220kV Mohana-Sampla (HV) D/C. Samalkha and Sampla are load centre. Power source is through 220kV Sonipat only. Samalkha(HV) S/s is having connectivity from grid through 220kV Mohana-Samalkha (HV) D/C only and 220kV Sampla has connectivity with 220kV Mohana and 400/220kV Kabulpur also. iii)During antecedent condition, 220 KV Mohana(HV)-Sonipat(PG) (HVPNL) D/C was carrying total ~228MW. Approx. 116MW was evacuating through 220kV Mohana-Samalkha (HV) D/C and remaining power was evacuating through 132KV load feeders from Mohana(HV) S/s. 220kV Mohana-Sampla (HV) D/C was not in service during the event (opened from Sampla end). iii)As reported, at 18:27 hrs, R-phase CT of 220 KV Mohana(HV)-Sonipat(PG) (HVPNL) Ckt-1 blasted at Mohana(HV) end which resulted in tripping of 220 KV Mohana(HV)-Sonipat(PG) (HVPNL) D/C from Sonipat(PG) end and only 220 KV Mohana(HV)-Sonipat(PG) (HVPNL) Ckt-1 tripped from Mohana(HV) end. (exact location of fault during 2nd instant yet to be received). iv)As per DR at Sonipat(PG) end of 220 KV Mohana(HV)-Sonipat(PG) (HVPNL) Ckt-1, R-N phase to earth fault with fault current of Ir=~14.3kA is observed. Line tripped instantaneously on distance protection operation in zone-1. As per relay flag at Mohana end of 220 KV Mohana(HV)-Sonipat(PG) (HVPNL) Ckt-1, line tripped instantaneously on distance protection operation in zone-1. v)As per DR at Sonipat(PG) end of 220 KV Mohana(HV)-Sonipat(PG) (HVPNL) Ckt-2, R-N phase to earth fault with fault current of Ir=~13.4kA is observed. Line tripped on distance protection operation in zone-2 with fault clearing time of 400msec. vi)Due to tripping of 220 KV Mohana(HV)-Sonipat(PG) (HVPNL) D/C, 220kV Mohana-Samalkha (HV) D/C became dead and 220/132kV Mohana(HV) S/s also lost its connectivity from grid and the complete 220/132kV Mohana(HV) S/s became dead. vii)As per PMU at Sonipat(PG), multiple R-N phase to earth fault is observed with fault clearing time of 160msec & 400msec. viii)As per SCADA, approx. 222 MW load loss is observed in Haryana control area.	1) 220 KV Mohana(HV)-Sonipat(PG) (HVPNL) Ckt-1 2) 220 KV Mohana(HV)-Sonipat(PG) (HVPNL) Ckt-2
46	GI-2	Uttar Pradesh	29-05-2024 15:57	29-05-2024 17:11	01:14	0	100	0.000	0.121	66380	82732	i)400/132kV Masoli(UP) S/s has one and half breaker bus scheme at 400kV voltage level side. ii)During antecedent condition, loading of 400/132 kV 200 MVA ICT 1,2&3 at 400/132kV Masoli(UP) was approx. 162MW (approx. 54MW for each ICT) (As per SCADA). iii)As reported, at 15:57 hrs, during inclement weather condition, towers of 132kV feeders to Naini and Karchhana from Masoli(UP) damaged which created B-N phase to earth fault on 132kV Masoli-Naini (UP) ckt followed by Y-N phase to earth fault on 132kV Masoli-Karchhana (UP) ckt. iv)B-N phase to earth fault on 132kV Masoli-Naini (UP) ckt cleared instantaneously (within 120msec as per PMU). CB of 132kV Masoli-Karchhana (UP) ckt could not open from Masoli(UP) end on Y-N phase to earth fault. v)As CB of 132kV Masoli-Karchhana (UP) ckt failed to open, fault cleared with the tripping of 400/132 kV 200 MVA ICT 1,2&3 and 125 MVAR Bus Reactor at Masoli(UP) tripped on O/C E/F protection operation. vi)As per PMU at Allahabad(PG), B-N followed by Y-N phase to earth fault is observed with fault clearing time of 120msec and 840msec respectively. vii)As per SCADA, change in demand of approx. 94MW is observed in UP control area. However, 100MW load loss is reported by SLDC-UP in UP control area.	1) 400/132 kV 200 MVA ICT 1 at Masoli(UP) 2) 400/132 kV 200 MVA ICT 2 at Masoli(UP) 3) 400/132 kV 200 MVA ICT 3 at Masoli(UP) 4) 125 MVAR Bus Reactor No 1 at 400 KV Masoli(UP) 5) 132kV Masoli-Karchhana (UP) ckt 6) 132kV Masoli-Naini (UP) ckt
47	GI-2	Haryana	30-05-2024 09:07	30-05-2024 09:53	00:46	0	0	0.000	0.000	66977	77406	i)During antecedent condition, 800kV HVDC Champa-Kurukshetra was carrying total 6000MW power from Kurukshetra to Champa (carrying ~1500MW each). ii)As reported, Bipole-2 (Pole-3&4) of 800kV HVDC Champa-Kurukshetra blocked due to non-availability of minimum AC filter. Further, as per detail received sequence of event are as follows: a.At 08:49:08.042hrs, B-type filter (HACQ-52) of bipole-2 tripped on capacitor unbalance stage-1 protection. b.Now, Bipole-2 was running with 3A (HACQ-41, HACQ-51, HACQ61) +1C+6E type filter at 3000MW without satisfying minimum filter requirement. c.B-type filter (HACQ-42) was available for energization but was not taken in service by Reactive Power Controller (RPC) as E-Type filters were connected for Voltage Regulation which had resulted in generation of alarm "Fltr Ctrl Incorrect Element Energized". (As per M/s GE, RPC philosophy, when D/E type Filters are connected, then RPC will not follow the filter table and RPC shall not allow for energization/de-energization of A/B type filter). d.For satisfying the harmonic performance of HVDC System, one B-Type filter was additionally supposed to be in service which could also satisfy the minimum filter requirement. e.At 09:08:03:182hrs, the operator tried to take B-type filter (HACQ-42) in manual mode. As soon as HACQ42 was taken in manual mode, Pole-3 & Pole-4 both got blocked on controlled blocking due to non-availability of minimum B type filter. iii)As per SCADA, due to outage of two poles (Pole-03 and Pole-04), power order reduced from 6000MW to 3000MW. iv)As per PMU, fluctuation in voltage was observed. v)As per SCADA, no change in demand is observed in Haryana control area.	1) 800 kV HVDC Kurukshetra(PG) Pole-03 2) 800 kV HVDC Kurukshetra(PG) Pole-04

20.Details of Grid Events during the Month of May 2024 in Northern Region

Sl No.	Category of Grid Event (GI 1or GI 2/ GD-1 to GD-5)	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
						Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
48	GI-2	Rajasthan	30-05-2024 10:16	31-05-2024 10:33	00:17	1435	0	2.014	0.000	71257	81922	i)During antecedent condition, 400 kV Sikar(PG)-Ratangarh(RS) (PG) Ckt-1, 400 kV Sikar(PG)-Bikaner(RS) (PG) Ckt-1 & 2 and 400 kV Ratangarh(RS)-Merta(RS) (RS) Ckt were not in service. Active power flow of 400 kV Sikar(PG)-Ratangarh(RS) (PG) Ckt-1 was approx. 470MW from Ratangarh(RS) to Sikar(PG). ii)As reported, at 10:16 hrs, 400 KV Sikar(PG)-Ratangarh(RS) (PG) Ckt-1 tripped on R-Y-N double phase to earth fault with fault distance of 26.592km from Sikar(PG) end. Y-Ph conductor snapped between location 83-84. iii)As per DR, zone-1 distance protection operated at Sikar(PG) end. Fault current was Ir=~11.986kA and Iy=~12.735kA from Sikar(PG). iv)As per SCADA SOE, 220kV Ratangarh(RS)-Khetri(RS) Ckt-2 tripped during the same time (exact reason yet to be shared). v)As per PMU at Sikar(PG), R-Y-N double phase to earth fault with fault clearing time of 80ms is observed. Voltage dipped upto 0.746 p.u. at Sikar(PG). vi)As per SCADA, change in NR solar generation of approx. 1290MW is observed which revived within 3.5 minutes (change in Rajasthan solar generation of approx. 66MW is observed which revived within 2 minutes). vii)As per SCADA, change in Rajasthan wind generation of approx. 145MW is observed.	1) 400 KV Sikar(PG)-Ratangarh(RS) (PG) Ckt-1
49	GD-1	Haryana and Himachal Pradesh	31-05-2024 11:25	31-05-2024 14:54	03:29	0	160	0.000	0.194	72346	82633	i)220kV Pinjore(HR) S/s has double main bus arrangement at 220kV side. ii)During antecedent condition, Incoming power at Pinjore(HR) S/s was approx. 230 MW through 220 KV Panchkula(PG)-Pinjore (HR) (HVPNL) D/C out of which approx. 188 MW was evacuating through 220 KV Baddi(HP)-Pinjore (HR) (HPPTCL) D/C and approx. 42MW was evacuating through 66kV feeders at Pinjore(HR) S/s. 220 KV Baddi(HP)-Pinjore (HR) (HPPTCL) D/C and some 66kV feeders connected to 220kV Bus-2 at Baddi(HP) and no other 200kV line connected at the same Bus. (As per SCADA). iii)As reported, at 11:25 hrs, due to forest fire, B-N phase to earth occurred on 220 KV Panchkula(PG)-Pinjore (HR) (HVPNL) ckt-2 which led to tripping of 220 KV Panchkula(PG)-Pinjore (HR) (HVPNL) D/C from Panchkula(PG) end and only 220 KV Panchkula(PG)-Pinjore (HR) (HVPNL) ckt-2 tripped from Pinjore(HR) end. iv)At the same time, Bus coupler of 220kV Bus-1 & Bus-2 at Baddi(HP) S/s opened due to which 220kV Bus-2 & 220 KV Baddi(HP)-Pinjore (HR) (HPPTCL) D/C became dead. Detail of protection operation on which bus coupler opened is yet to be received from HP. v)As per DR of Panchkula(PG) end of 220 KV Panchkula(PG)-Pinjore (HR) (HVPNL) Ckt-1, B-N phase to earth fault is observed in Zone-2 and line tripped from Panchkula end with the delay of ~200msec. vi)As per DR at Panchkula(PG) end of 220 KV Panchkula(PG)-Pinjore (HR) (HVPNL) Ckt-2, initially B-N phase to earth fault in zone-1 with fault current of 1.3kA is observed. On this fault A/R operation started but before A/R reclosing Y-N phase to earth fault occurred which led to 3-phase tripping of the line. vii)As 220 KV Baddi(HP)-Pinjore (HR) (HPPTCL) D/C became dead and due to tripping of 220 KV Panchkula(PG)-Pinjore (HR) (HVPNL) D/C, 220 Pinjore(HR) lost its connectivity from Grid and complete Pinjore(HR) S/s became dead. viii)As per PMU at Abdullapur(PG), B-N phase to earth fault followed by Y-N fault is observed with delayed fault clearing time of 1240msec & 80msec respectively. ix)As per SCADA, change in demand of approx. 33MW & 98MW is observed in Haryana & HP control area respectively. However, approx. 60MW & 100MW load loss is reported by SLDC-Haryana & SLDC-HP in Haryana & HP control area respectively.	1) 220 KV Panchkula(PG)-Pinjore (HR) (HVPNL) Ckt-1 2) 220 KV Panchkula(PG)-Pinjore (HR) (HVPNL) Ckt-2

20.Details of Grid Events during the Month of May 2024 in Western Region

Sl No.	Category of Grid Event (GI 1or GI 2/ GD-1 to GD-5)	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
						Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
1	GD-1	WR	23:21 / 01-05-2024	00:04 / 02-05-2024	00:43	167	-	0.20%	-	84090	66152	At 23:21 Hrs / 01-05-2024, 220 kV Bhuj-Baranda-1 tripped on 3 phase to ground fault due to falling of spare conductor of 220 kV Bhuj-Dayapar-D/C on 220 kV Bhuj-Baranda-1 near Bhuj gantry . At the same time 220 kV Bhuj-Gadhsisa-1 tripped at Gadhsisa end only. Generation loss of 132 MW and 35 MW occurred at Gadhsisa (Renew Power) and Baranda (Avikiran) respectively due to loss of evacuation path.	Tripping of following Elements: 1. 220 kV Bhuj-Baranda-1 2. 220 kV Bhuj-Gadhsisa-1
2	GI-2	WR	10:46 / 03-05-2024	14:15 / 03-05-2024	03:29	250	-	0.31%	-	81200	68933	At 10:46 Hrs / 03-05-2024, busbar protection operation in 400 kV Satpura-Bus-1 (Double main and transfer bus scheme) resulted in tripping of all elements connected to 400 kV Satpura-Bus-1 (400 kV Satpura-Koradi-1, 400 kV Satpura-Indirasagar-1, 400 kV Satpura-Itarsi-1, 400 kV Satpura-Astha-1 and Satpura Unit-11 (250 MW)). Busbar protection maloperated as there was no fault signature present in nearby PMUs. Detailed investigation of relay maloperation is being done by MPPTCL. Generation loss of 250 MW occurred due to the tripping of Satpura Unit-11 during the event.	Tripping of following Elements: 1. 400 kV Satpura-Bus-1 2. 400 kV Satpura-Koradi-1 3. 400 kV Satpura-Indirasagar-1 4. 400 kV Satpura-Itarsi-1 5. 400 kV Satpura-Astha-1 6. Satpura Unit-11 (250 MW)
3	GD-1	WR	10:49 / 10-05-2024	13:34 / 10-05-2024	02:45	248	-	0.31%	-	80062	69092	At 10:49 Hrs / 10-05-2024, 220 kV Bhavsingpura-Khandwa tripped from Khandwa end only due to relay maloperation. Generation loss of 248 MW occurred at Bhavsingpura and Kanwani (Masaya Solar) Solar power plants respectively, due to the loss of evacuation path.	Tripping of following Elements: 1. 220 kV Bhavsingpura-Khandwa 2. 220 kV Bhavsingpura-Kanwani
4	GI-2	WR	15:47 / 13-05-2024	16:26 / 13-05-2024	00:39	-	348	-	0.51%	80866	68076	At 15:47 Hrs / 13-05-2024, 220 kV Kalwa-Bus-2 and all connected elements tripped due to LBB operation of 220 kV Kalwa-DCHI due to non-quenching of arc of Circuit Breaker. At the same time, 220 kV Kharghar-Khandeshwar tripped on differential protection operation on R-B-E phase to phase to earth fault. At 15:53 Hrs / 13-05-2024, 400/220 kV Kharghar ICT-2 tripped on the operation of the Oil Surge Relay, resulting in LTS operation in 400/220 kV Kharghar ICT-1. At the time of tripping heavy winds and rains were reported and many elements at Kalwa and Kharghar region tripped on tranisent faults. +/- 500 kV HVDC-Chandrapur-Padghe-1&2 tripped at 16:07 and 16:10 hrs / 13-05-2024 respectively due to voltage dip on transient faults. Load loss of 348 MW occurred at Mumbai and MMR (Mumbai Metropolitan Region) due to LTS operation.	Tripping of following Elements: 1. 220 kV Kalwa-DCHI 2. 220kV Kalwa-Bus-2 3. 220 kV Kalwa-Salsette-3 4. 220 kV Kalwa-Mulund-1&2 5. 220 kV Kalwa-Borivali 6. 220kV Kalwa-TIFIL 7. 400/220 kV Kalwa-ICT-1&2 (500 MVA) 8. 400/220 kV Kharghar-ICT-2 (315 MVA) 9. +/- 500 kV HVDC-Chandrapur-Padghe-1&2 10. 220 kV Kharghar-Khandeshwar-2 11. 220 kV Kharghar-Borivali-2
5	GI-2	WR	19:33 / 13-05-2024	22:13 / 13-05-2024	02:40	-	-	-	-	78514	58752	At 19:33 Hrs / 13-05-2024, 400 kV Charanka-Varsana-1 tripped on Y-E fault from Charanka end only and auto recloser was successful at Varsana end. At the same time, 400 kV Veloda-Charanka-1 tripped on R-E fault. 400 kV Veloda-Charanka-2 tripped from Veloda end on R-E fault and the Circuit Breaker at Charanka end didn't open due to problem in breaker. Due to this, 400 kV Charanka-Kansari-1&2 tripped on Z-3 operation from Kansari end. 400/220 kV Charanka-ICT-1&2 tripped due to Backup earth fault protection operation. With these trippings 400 kV Charanka became dead and 220 kV Charanka was intact. At the time of tripping heavy wind and thunderstorms were reported. No Generation or Load loss occurred due to the event.	Tripping of following Elements: 1. 400 kV Charanka-Varsana-1 2. 400 kV Veloda-Charanka-1&2 3. 400 kV Charanka-Kansari-1&2 4. 400/220 kV Charanka-ICT-1&2
6	GI-1	WR	19:52 / 13-05-2024	21:58 / 13-05-2024	02:06	72	-	0.09%	-	78168	58222	At 19:52 Hrs / 13-05-2024, 220 kV CHPH-RBPH-1&2 tripped due to a lightning strike on the ATR gantry at SSP. At the same time, 400/220 kV SSP-ICT-1 tripped on differential protection operation due to the grounding of the lightning strike.400/220 kV SSP-ICT-2 tripped on neutral over current protection operation. Generation loss of 72 MW occurred due to tripping of SSP-CHPH-Unit-1&2 (50 MW) due to loss of evacuation path.	Tripping of following Elements: 1. 220 kV CHPH-RBPH-1&2 2. 400/220 kV SSP-ICT-1&2 3. SSP-CHPH-Unit-1&2 (50 MW)
7	GD-1	WR	16:26 / 15-05-2024	04:58 / 16-05-2024	12:32	68	-	0.08%	-	83355	67892	At 16:26 Hrs / 15-05-2024, 220 kV Bhuj II-Nakhatrana tripped on Y-E fault due to insulator decapping at Tower location 37. Generation loss of 68 MW occurred at Nakhatrana and Dedhiya Wind Power Plant (Adani) due to the loss of evacuation path.	Tripping of following Elements: 1. 220 kV Bhuj II-Nakhatrana 2. 220 kV Nakhatrana-Dedhiya
8	GD-1	WR	22:12 / 15-05-2024	01:51 / 16-05-2024	03:39	85	-	0.10%	-	84685	65854	At 22:12 Hrs / 15-05-2024, 220 kV Bhuj-Baranda tripped on R-E fault and at same time 220 kV Bhuj-Gadhsisa tripped from Gadhsisa end only on R-E fault due to falling of earth wire of of 220 kV Bhuj-Dayapar at 220 kV Bhuj. Generation loss of 85 MW (71 MW at Gadhsisa (Renew Power) and 14 MW at Baranda (Avikiran) occurred at Gadhsisa and Baranda Wind Power Plant respectively due to loss of evacuation path.	Tripping of following Elements: 1. 220 kV Bhuj-Gadhsisa 2. 220 kV Bhuj-Baranda
9	GD-1	WR	16:54 / 16-05-2024	20:03 / 16-05-2024	03:09	75	-	0.09%	-	83142	65635	At 16:54 Hrs / 16-05-2024, 220 kV Bhuj-Kotda Madh tripped due on B-E fault. During patrolling no abnormality was found. Generation loss of 75 MW occurred at Kotda Madh (Alfanar) due to loss of evacuation path.	Tripping of following Elements: 1. 220 kV Bhuj-Kotda Madh
10	GI-1	WR	23:34 / 20-05-2024	02:26 / 21-05-2024	02:52	-	140	-	0.20%	86685	68796	At 23:34 Hrs / 20-05-2024, Y phase suspension string of flyover span of line side CT of 220 kV Kondhwa-Jejuri flashed over at Jejuri end, the fault was not cleared in zone-1 protection and was being fed from all lines and ICTs connected at 220 kV Jejuri substation. Due to this bus coupler tripped on backup to busbar protection operation after 200 msec (definite time delay) isolating 220 kV Jejuri-Bus-1&2. Y phase of 220 kV Lonand-Jejuri-1 tripped on Z-1 distance relay operation at Jejuri end although the fault was in reverse direction and later R phase tripped after 500 msec. This tripping was undesirable and the relay maloperated.After the isolation of 220 kV Jejuri buses 220 kV Kondhwa-Jejuri was connected through 220 kV Jejuri Bus-1, but all the lines connected to 220 kV Jejuri-Bus-2 (220 kV Jejuri-Barmati, 220 kV Jejuri-Phursangi-2, 220 kV Jejuri-Jejuri(II)-2) tripped on Z-1 distance protection operation (the relays over reached) and autorecloser cycle was initiated. Further tripping of 220 kV Jejuri-Phursangi-2 initiated LBB protection and after 200 msec all elements connected to 220 kV Jejuri-Bus-2 tripped. The operation of LBB was undesirable. Remote ends of the lines 220 kV Jejuri-Phursangi-1 and 220 kV Lonand-Jejuri-1 tripped on Z-2 distance protection. After 550 msec of the first fault appearance, a fault occurred in the R phase of 220 kV Kondhwa-Jejuri resulting in Z-1 R-Y phase-phase fault operation. With the tripping of 220 kV Kondhwa-Jejuri, 220 kV Parvati-Phursangi tripped on overcurrent protection operation due to overloading and LTS stage-1 operation. Only 220 kV Jejuri-Jejuri(II)-1 and 400/220 kV Jejuri-ICT-1 were in service load loss of 140 MW occurred at Kothrud, NCL, and Kondhwa region due to LTS operation. Also, the correct auto-closing of the 220 kV Lonikhand-Lonikhand(II) interconnector saved the LTS operation in 400/220 kV Lonikhand(I)-ICTs which were overloaded due to these trippings.	Tripping of following Elements: 1. 220 kV Jejuri Bus 2 2. 220 kV Jejuri-Barmati 3. 220 kV Jejuri-Jejuri(II)-2 4. 220 kV Jejuri-Parvati 5. 220 kV Jejuri-Phursangi-1&2 6. 220 kV Kondhwa-Jejuri 7. 220 kV Lonand-Jejuri-1 8. 400/220 kV Jejuri-ICT-2&3
11	GI-1	WR	02:02 / 22-05-2024	03:40 / 22-05-2024	01:38	-	-	-	-	85177	66501	At 02:02 Hrs / 22-05-2024, Bus protection mal-operated in 220 kV Chhhegaon-Bus-1 due to fault in 220 kV Chhegaon-Saktapur circuit resulting in tripping of all connected elements. On Investigation it was found that the polarity of connection use for Bus bar protection from 220 kV Chhegaon-Saktapur was reversed, resulting in bus bar protection operation. No load loss occurred, as all other elements were in service through 220 kV Chhhegaon-Bus-2.	Tripping of following Elements: 1. 220 kV Chhhegaon-Bus-1 2. 220 kV Chhegaon-Saktapur 3. 220 kV Chhegaon-Singaji-1 4. 220 kV Chhegaon-Nimrani-1 5. 220 kV Chhegaon-Khandwa-1
12	GD-1	WR	17:59 / 28-05-2024	20:18 / 28-05-2024	02:19	-	1045	-	1.58%	83170	66310	At 17:59 Hrs / 28-05-2024, Failure of R phase Bus-1 side isolator chamber at 220 kV Gorai GIS led to bus bar protection operation in 220 kV Gorai-Bus-1&2 and tripping of all connected elements. Load loss of 99 MW was reported by SLDC Maharashtra in the Gorai area due to the event. However, from the demand curves of Maharashtra and Mumbai from SCADA, the instantaneous load loss of around 2000 MW and 1000 MW was observed in the Maharashtra and Mumbai control area, out of which around 500 MW recovered within 5 minutes of the tripping. The same was evident from the PMU plots, wherein frequency rose to 0.15 Hz during the above incident and the Phase voltage of Kalwa dipped from 230 kV to 110 kV. It is suspected that induction machines during severe voltage dip might have stalled resulting in demand reduction and after fault clearance might have started automatically (demand started increasing in a few seconds itself after fault clearance). Also, some machines might have stopped and had to be started manually as restoration took longer than normal. SLDC Maharashtra has reported load drop of 1045 MW in the Mumbai Control area. The load loss calculated using ICTs loading at the Maharashtra control area was 1587 MW.	Tripping of following Elements: 1. 220 kV Gorai-Versova 2. 220 kV Gorai -Ghodbunder 3. 220 kV Gorai-Borivali-1&2 4. 220 kV Gorai-Bus-1&2 5. 220/33 kV Gorai-ICT-1&2 (125 MVA)

20.Details of Grid Events during the Month of May 2024 in Western Region

Sl No.	Category of Grid Event (GI 1or GI 2/ GD-1 to GD-5)	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
						Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
13	GI-2	WR	00:21 / 29-05-2024	03:22 / 29-05-2024	03:01	-	-	-	-	89119	68822	At 00:21 Hrs / 29-05-2024, Failure of Bus bar CVT at 400 kV Raigarh- led to tripping of 400 kV Raigarh-Bus-2 on Busbar protection operation. During the Busbar operation, R phase pole of 400 kV Raigarh-Jharsuguda-3 stuck and didn't open and fumes from failed CVT led to tripping of 400 kV Raigarh-Jharsuguda-3 on Local Breaker Backup (LBB) protection operation. At same time, fumes from CVT created flashover on line side of 400 kV Raigarh-Jharsuguda-1 resulting in its tripping. No generation / load loss occurred due to the event	Tripping of following Elements: 1. 400 kV Raigarh-Bus-2 2. 400 kV Raigarh-Jharsuguda-1&3
14	GI-2	WR	12:02 / 29-05-2024	12:33 / 29-05-2024	00:31	408	-	0.48%	-	85160	67605	At 12:02 Hrs / 29-05-2024, 400 kV Chorania-Asoj-2 tripped due to wavetrap jumper broken at Chorania end. At the same time 400 kV Chorania-Amreli-1 and 400 kV Chorania-Hadala-1 tripped from remote ends only due to overreaching of relays. During investigation of the relays at Amreli and Hadala by Gujarat the compensation factor was found to be higher same has been changed and under investigation. Wind generation loss of around 408 MW occurred in Chorania area due to the event.	Tripping of following Elements: 1. 400 kV Chorania-Amreli-1 2. 400 kV Chorania-Hadala-1 3. 400 kV Chorania-Asoj-2
15	GI-1	WR	22:48 / 29-05-2024	23:16 / 29-05-2024	00:28	-	35	-	0.05%	91950	69195	At 22:48 Hrs / 29-05-2024, R phase CT burst of 220/132 kV Chhattarpur-ICT-1 resulted in busbar protection operation in 220 kV Chhattarpur-Main Bus and tripping of all connected elements. Load loss of around 35 MW occurred at Chhattarpur area due to the event.	Tripping of following Elements: 1. 220 kV Chhattarpur-Main Bus 2. 220 kV Chhattarpur-Satna(MP) 3. 220 kV Chhattarpur-Satna(PG) 4. 220 kV Chhattarpur-Tikamgarh 5. 220/132 kV Chhattarpur-ICT-1&2 (160 MVA)
16	GD-1	WR	18:20 / 30-05-2024	18:48 / 30-05-2024	00:28	420	-	0.50%	-	84168	66176	At 18:20 Hrs / 30-05-2024, R phase CT of 220 kV Birsingpur-Jabalpur-2 at Birsinghpur burst resulting in multiple trippings at 220 kV Birsinghpur. Busbar protection module was out of service since 17-11-2021,which was not intimated to WRLDC. 220 kV Birsingpur-Jabalpur-1 tripped from Birsingpur on Z-1 operation which is maloperation and tripped on Z-2 operation from remote end. 220 kV Birsingpur-Sanjay Gandhi TPS(IC)-2,3&4 tripped correctly on Z-4 operation and tripped on Z-1 operation from remote end. 220 kV Birsingpur-Sanjay Gandhi TPS(IC)-1 hand tripped at Birsinghpur end and tripped on Z-1 operation from remote end. 220 kV Birsingpur-Nowrozabad Traction-1&2 and 220 kV Birsingpur-Hydel Feeder hand tripped at Birsinghpur end and were holding from remote end. Generation loss of 420 MW occurred due to tripping of Sanjay Gandhi-Unit-1&2 (210 MW) on heavy voltage dip.	Tripping of following Elements: 1. 220 kV Birsingpur-Jabalpur-1&2 2. 220 kV Birsingpur-Nowrozabad Traction-1&2 3. 220 kV Birsingpur-Hydel Feeder 4. 220 kV Birsingpur-Sanjay Gandhi TPS(IC)-1,2,3&4 5. 220/132 kV Birsinghpur-ICT-1 (160 MVA) 6. Sanjay Gandhi-Unit-1&2 (210 MW)
17	GD-1	WR	22:25 / 31-05-2024	22:57 / 31-05-2024	00:32	-	700	-	1.02%	93111	68738	At 22:25 Hrs / 31-05-2024, Flashover of Y phase CT of Tie bus resulted in tripping of all connected lines on Z-2 from remote ends only. 220/132 kV Suhela-ICT-1 tripped on earth fault protection operation. Non operation of Busbar protection is under investigation by CSPTCL. Load loss of 700 MW occurred at 220 kV Suhela, 220 kV Paraswani, 220 kV Bemetara and other down stream network.	Tripping of following Elements: 1. 220 kV Suhela-Paraswani-1&2 2. 220 kV Suhela-Bemetara-1&2 3. 220 kV Suhela-Bhatapara(CG) 4. 220 kV Suhela-Bhatapara(PG)-1,2&3 5. 220 kV Suhela-DSPM 6. 220 kV Suhela-Banari 7. 220/132 kV Suhela-ICT-1

20.Details of Grid Events during the Month of May 2024 in Southern Region

Sl No.	Category of Grid Event (GI 1or GI 2/ GD-1 to GD-5)	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
						Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
1	GD-1	Andhra Pradesh	02-05-2024 23:22	03-05-2024 00:00	0:38	0	270	0.00%	0.46%	47410	58405	Complete Outage of 220kV/132kV AP Carbides SS of APTRANSCO:220kV/132kV AP Carbides SS is being radially fed from 400kV/220kV Narnoor SS. As per the reports submitted, the triggering incident was Y-phase CT failure in 220kV/132kV AP carbides. The fault was cleared by remote lines (220kV Narnoor Ap Carbide Line-1&2) on zone-2. Tripping of both lines resulted in complete outage of 220kV/132kV AP Carbides station.	1. 220kV Narnoor AP Carbide Line-1&2
2	GD-1	Andhra Pradesh	02-05-2024 14:25	02-05-2024 15:02	0:37	0	160	0.00%	0.24%	53399	65330	Complete Outage of 220kV/132kV Nandyal SS of APTRANSCO: 220kV/132kV Nandyal is being radially fed from 220kV Somayajulapally Pooling station. As per the reports submitted, the triggering incident was R-N faults in 220kV Somayajulapally Nandyal Line-2 and the line tripped. Subsequently, 220kV Somayajulapally Nandyal Line-1 also tripped on R-N fault at 14:15hrs. Tripping of both lines resulted in complete outage of 220kV/132kV Nandyal SS.	1. 220kV Somayajulapally Nandyal Line-1&2
3	GD-1	Kerala	03-05-2024 17:19	03-05-2024 18:12	0:53	0	0	0.00%	0.00%	47418	55902	Complete Outage of 220kV Kanniyampeta of KSEB: 220kV Kanniyampeta is connected with two 220kV feeders (220kV Kadakola-Kanniyampeta and 220kV Kaniyampeta-Kunnamangalam). In the antecedent condition 220kV Kadakola-Kanniyampeta was idle charged from Kanniyampeta end. The triggering incident is the R-N fault in the line in Kanniyampeta Kadakola line and the line tripped. While charging the line at 17:55hrs, R-N fault due to the insulator decapping was again observed. R-N fault during the charging persisted without the operation of SOTF at Kanniyampeta end and after around 1sec 220kV Kaniyampeta-Kunnamangalam tripped at Kunnamangalam end on IN>1. Tripping of the only connected line led to the Complete Outage of 220kV Kanniyampeta of KSEB.	1. 220kV Kadakola-Kanniyampeta 2. 220kV Kanniyampeta-Kunnamangalam
4	GD-1	Andhra Pradesh	07-05-2024 19:29	07-05-2024 22:06	2:37	0	0	0.00%	0.00%	41357	47368	Complete Outage of 220kV Tallapally Switching Station of APTRANSCO: As per the reports submitted the triggering incident was R-phase LA flashover in 220kV Tallapally Nagarjuna Sagar Line-1 at Tallapally end. At the same time, LBB of 220kV Tallapally ICT-1(connected to bus-2) & ICT-2(connected to Bus-1) operated and all the elements connected to 220kV Bus-1 and Bus-2 tripped. Tripping of both buses led to complete outage of 220kV Tallapally Switching Station.	1. 220kV Tallapalli NagarjunaSagar-1,2&3 2. 220kV Tallapalli Srisailam-1&2 3. 220kV Tallapalli Rentachintala 4. 220kV Tallapalli Inupurajupalli-1&2 5. 220kV Tallapalli Chalakurthy 6. 220kV Tallapalli ICT-1,2&3
5	GD-1	Tamil Nadu	08-05-2024 11:14	08-05-2024 11:40	0:26	0	183	0.00%	0.34%	49396	53621	Complete Outage of 230kV/110kV Gummudipoondi SS of TANTRANSCO, 230kV Cauvery SS, 230kV Kamachi SS, 230kV Thervoi Kandigai SS, and 230kV Mychelin SS: 230kV Cauvery SS, 230kV Kamachi SS, 230kV Thervoi Kandigai SS, and 230kV Mychelin SS are being radially connected to 230kV/110kV Gummudipoondi SS. During antecedent conditions, 230kV Gummudipoondi NCTPS line was idle charged from Gummudipoondi end and 230kV Gummudipoondi Suryadev line was under outage. 230kV Gummudipoondi Sullurpet is the only source to 230kV/110kV Gummudipoondi SS. As per the reports submitted the triggering incident was B-phase jumper failure in 230kV Gummidipoondi Sullurpet line at Gummudipoondi station causing a bus fault in 230kV Bus-1. At Gummidipoondi station, BBP of Bus-1 operated and all the elements connected for bus tripped. Tripping of the only source led to complete outage of 230kV/110kV Gummudipoondi SS.	1. 230kV Gummudipoondi Sullurpet line
6	GD-1	Taamil Nadu	10-05-2024 15:12	10-05-2024 19:07	3:55	0	0	0.00%	0.00%	47159	59083	Complete Outage of 230kV Ettayapuram Solar Plant: 230kV Ettayapuram Solar Plant is being radially connected through 230kV TTGS Ettayapuram line. The triggering incident was R-N fault in 230kV TTGS Ettayapuram line. Tripping of only connected line led to complete outage of 230kV Ettayapuram Solar Plant.	1. 230kV TTGS Ettayapuram line.
7	GD-1	Karnataka	10-05-2024 23:32	11-05-2024 01:03	1:31	0	92	0.00%	0.18%	40566	50922	Complete Outage of 220kV/110kV NRS SS of KPTCL: 220kV/110kV NRS SS is operating with single bus at 220kV Level. As per the reports submitted, the triggering incident was 220kV BBP maloperation at NRS and all the elements connected to the bus tripped. This led to complete outage of 220kV/110kV NRS SS.	1. 220kV Peenya NRS 2. 220kv Vrushabhavati NRS 3. 220kV AR circle NRS line, 4. 100 MVA NRS Transformer-1&2
8	GD-1	Karnataka	11-05-2024 16:39	11-05-2024 20:18	3:39	32	226	0.07%	0.43%	43037	52329	Complete Outage of 220kV/110kV Ambewadi SS of KPTCL: 220kV/110kV Ambewadi SS is operating with single bus at 220kV level. As per the reports submitted, the triggering incident was B-phase jumper failure in 220kV Ambewadi Ponda line at Ambewadi SS causing a bus fault. Immediately, BBP operated and all the lines connected to the bus tripped. This led to complete outage of 220kV/110kV Ambewadi SS.	1. 210kV Ambewadi Ponda
9	GD-1	Karnataka	11-05-2024 15:13	11-05-2024 16:36	1:23	30	250	0.06%	0.45%	52329	55675	Complete Outage of 400kV/220kV Koppal SS of WKTL, 400kV Kudgi Generating station of NTPC_Kudgi, 220kV Nadhihal Pooling Station, 220kV GM Navar SS, 220kV/110kV Bijapur SS, 220kV/110kV Indi SS, and 220kV/110kV Aheri220kV RSOPL, 220kV RSRPL, 220kV Ayana: As per the reports submitted, the triggering incident was B-N fault in 400kV Koppal Kudgi_PG Line-1&2 due to lightening near Kudgi_PG station and both lines tripped. Tripping of both lines led to complete outage of 400kV/220kV Koppal, 220kV RSRPL, 220kV RSOPL, 220kV Ayana. At the same time under frequency relay envisaged for Bangalore islanding scheme operated tripping 400kV Kudgi_NTPC Kudgi_PG Line-1,2,3&4. This led to formation of island with 400kV Kudgi_NTPC, 220kV Nadhihal Pooling Station, 220kV GM Navar SS, 220kV/110kV Bijapur SS, 220kV/110kV Indi SS, and 220kV/110kV Aheri which later collapsed due to load generation imbalance.	1. 400kV Koppal Kudgi_PG Line-1&2 2. 400kV Kudgi_NTPC Kudgi_PG Line-1,2,3&4 3. 220kV Vajramathi Kudgi Line-1&2
10	GD-1	Tamil Nadu	13-05-2024 12:40	13-05-2024 16:53	4:13	77	0	0.17%	0.00%	46324	49425	Complete Outage of 230kV JSW_Vilathikulam_Wind: 230kV JSW_Vilathikulam_Wind is radially connected through 230kV TTGS JSW_Vilathikulam_Wind line. As per the reports submitted, the triggering incident was B-N fault in the line and the line tripped. Tripping of the only connected line resulted in complete outage of 230kV JSW_Vilathikulam_Wind.	1. 230kV TTGS JSW_Vilathikulam_Wind line
11	GD-1	Tamil Nadu	17-05-2024 09:01	17-05-2024 09:59	0:58	0	147	0.00%	0.31%	41052	46801	Complete Outage of 230kV Tondiarpet SS, 230kV Basin Bridge SS, 400kV/230kV Puliyanthope SS, 230kV Koratturi SS, 230kV CMRL SS, Vyasarpadi SS and 230kV TNEB HQ of TANTRANSCO and 230kV NCTPS Generating station of TANGEDCO: 230kV Tondiarpet SS, 230kV Basin Bridge SS, 400kV/230kV Puliyanthope SS, 230kV Koratturi SS, 230kV CMRL SS, Vyasarpadi SS and 230kV TNEB HQ are being radially connected through 230kV NCTPS Tondiarpet Line-1&2, 230kV Manali Korattur, 400kV Manali Alamathy Line. These source feeders tripped on different faults from 09:01hrs leading to complete outage of 230kV Tondiarpet SS, 230kV Basin Bridge SS, 400kV/230kV Puliyanthope SS, 230kV Koratturi SS, 230kV CMRL SS, Vyasarpadi SS and 230kV TNEB HQ and 230kV NCTPS Generating station.	1. 230kV NCTPS Tondiarpet Line-1&2 2. 230kV Manali Korattur 3. 400kV Manali Alamathy Line
12	GD-1	Karnataka	18-05-2024 20:10	18-05-2024 22:03	1:53	40	0	0.09%	0.00%	43115	45951	Complete outage of 220kV Ostro Kannada: 220kV Ostro Kannada is being radially connected through 220kV Hiriyur Ostro kannada line. As per the reports submitted, the triggering incident was R-N fault in the line and the line tripped. Tripping of the only connected line resulted in complete outage of 220kV Ostro Kannada.	i. 220kV Hiriyur Ostro kannada line
13	GD-1	Karnataka	20-05-2024 03:45	20-05-2024 06:04	2:19	32	0	0.10%	0.00%	33607	36045	Complete outage of 220kV Ostro Kannada: 220kV Ostro Kannada is being radially connected through 220kV Hiriyur Ostro kannada line. As per the reports submitted, the triggering incident was operation of over voltage stage-1 protection at Ostro Kannada end and DT was sent to Hiriyur end in the line and the line tripped. Tripping of the only connected line resulted in complete outage of 220kV Ostro Kannada.	i. 220kV Hiriyur Ostro kannada line

20.Details of Grid Events during the Month of May 2024 in Southern Region

Sl No.	Category of Grid Event	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
	(GI 1or GI 2/ GD-1 to GD-5)					Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
14	GD-1	Karnataka	20-05-2024 23:45	21-05-2024 05:09	5:24	30	0	0.07%	0.00%	45869	50758	Complete outage of 220kV Ostro Kannada: 220kV Ostro Kannada is being radially connected through 220kV Hiriyur Ostro kannada line. As per the reports submitted, the triggering incident was R-N fault in the line and the line tripped. Tripping of the only connected line resulted in complete outage of 220kV Ostro Kannada.	1. 220kV Hiriyur Ostro kannada line
15	GD-1	Karnataka	22-05-2024 05:47	22-05-2024 06:43	0:56	100	450	0.25%	1.12%	40671	40060	Complete Outage of 220kV/110kV Shimoga SS, 220kV/110kV Heggunje SS,220kV/110kV Puttur SS, 220kV Bus-2 of 220kV/110kV Kemar SS of KPTCL and 220kV Varahi PH of KPCL: 220kV/110kV Heggunje SS,220kV/110kV Puttur SS, 220kV Bus-2 of220kV/110kV Kemar SS, of KPTCL and 220kV Varahi PH of KPCL are being radially connected to 220kV/110kV Shimoga SS. 220kV/110kV Shimoga SS was operating with single bus operation at 220kV level. As per the reports submitted, the triggering incident was failure of R-phase CVT of bus coupler through which 220kV Shimoga Arasikere Line was charged. Immediately, BBP operated and all elements connected to the bus tripped resulting in complete outage of 220kV/110kV Shimoga SS leading to outage of the radial connected stations.	1.220kV Shimoga Arasikere Line 2. 220kV Shimoga Varahi Line-1,2&3 3. 220kV Shimoga Chikmanagalore 4. 220kV Shimoga Shantipura 5. 220kV Shimoga Bettedaverekere 6. 220kV Shimoga Honnali Line
16	GD-1	Karnataka	24-05-2024 16:50	24-05-2024 17:20	0:30	0	259	0.00%	0.56%	46560	48363	Complete Outage of 220kV/66kV Manyata SS, 220kV/66kV HBR Layout SS 220kV Bus-1 of 220kV Yelahanka SS and 220kV Bus-1 of 220kV/66kV Shakarinagar SS of KPTCL: 220kV/66kV Manyata SS, 220kV/66kV HBR Layout SS are radially connected to 220kV Bus-1 of 220kV/66kV Shakarinagar SS which was being radially fed from 220kV Bus-1 of 220kV Yelahanka SS. As per the reports submitted, the triggering incident was LBB operation in 220kV Yelahanka Sahakarinagar Line-1. Immediately all lines connected to 220kV Yelahanka Bus-1 tripped. This led to complete outage of all radial connected stations.	1. 220kV Yelahanka_PG Yelahanka Line-1 2. 220kV/66kV Yelahanka Transformer-1 3. 220kV Yelahanka Shakarinagar Line-1 4. 220kV Yelahanka YCCP Line-1
17	GD-1	Telangana	26-05-2024 14:48	26-05-2024 19:16	4:28	0	0	0.00%	0.00%	42970	47960	Complete Outage of 765kV/400kV Warangal (New) SS of WKTL: As per the reports submitted, the triggering incident was B-N fault in 765kV Maheshwaram Warangal (New) Line-1 and 400kV Warangal Warangal (New) Line-2 and the lines tripped. At the same time, 765kV/400kV Warangal (New ICT-1&2 tripped on earth fault stage-2 protection and 400kV Warangal Warangal (New) Line-1 tripped due to spurious DT receipt at Warangal_PG end. Tripping of all these lines led to complete outage of 765kV/400kV Warangal (New) SS.	1. 765kV Maheshwaram Warangal(New) Line-1 2. 400kV Warangal Warangal(New) Line-1&2 3. 765kV/400kV Warangal(New ICT-1&2
18	GD-1	Tamil Nadu	30-05-2024 16:44	30-05-2024 17:45	1:01	0	0	0.00%	0.00%	52677	56542	Complete Outage of 230kV/110kV Checkanurani SS of TANTRANSCO: As per the reports submitted, the triggering incident was R-phase CT failure in 230kV Checkanurani Renganathapuram line at Checkanurani end causing a Bus fault in 230kV Bus-1. Immediately, Bus-1 BBP operated and all lines connected to the bus tripped. Subsequently, Bus-2 BBP operated as other CTs connected to 230kV Bus-2 were damaged due to R-phase CT failure in 230kV Checkanurani Renganathapuram line. All elements connected to 230kV Bus-2 tripped. This resulted in complete outage of 230kV/110kV Checkanurani SS.	1. 230kV Checkanurani Amuthapuram 2. 230kV Checkanurani Sempatty 3. 230kV Checkanurani Theni 4. 230kV Checkanurani Pasumalai 5. 230kV Checkanurani Kinnimangalam 6.230kV Checkanurani Savaspuram 7. 230kV Checkanurani Nallamanakkayanpatty 8. 230kV Checkanurani Renganathapuram 9. 400kV/230kV Madhurai ICT-1,2&3

20.Details of Grid Events during the Month of May 2024 in Eastern Region

Sl No.	Category of Grid Event	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
	(GI 1or GI 2/ GD-1 to GD-5)					Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
1	GD-1	Garaul	09.05.2024 08:02	09.05.2024 11:02	03:00	0	15	0.00%	0.07%	24825	20809	At 08:02 hrs on 09.05.2024, 220kV Muzaffarpur-Garaul-2 tripped due to R-N fault which led to total power failure at Garaul as 220kV Muzaffarpur-Garaul Ckt#1 was under breakdown. Total load loss of around 15 MW at Garaul was reported. Power was extended through 132kV system via Hazipur, Jandaha and Manhar link.	220 kV Muzaffarpur-Garaul-2
2	GI-2	Jharsuguda, Darlipalli, OPGC	21.05.2024 17:02	21.05.2024 18:50	01:48	1900 (750+575+ 575)	0	6.25%	0.00%	30391	25500	At 17:02 Hrs on 21.05.2024, 765 kV Bus Reacotr-1 & 1500 MVA 765/400 kV ICT-1 at Jharsuguda tripped due to failure of tie bay CT at Jharsuguda. At the same time, 800 MW U#1 at Darlipalli tripped due to tripping of VFDs used for pulverisers which led to loss of fuel. Subsequently, U#3 and U#4 at OPGC also tripped one by one on low forward power. Total generation loss of around 1900 MW occurred within a span of 3 minutes.	240 MVAr Bus Reactor-1 at Jharsuguda 1500 MVA 765/400 kV ICT-1 at Jharsuguda 765 kV Jhasuguda-Darlipalli-2 800 MW U#2 at Darlipalli 660 MW U#3 at OPGC 660 MW U#4 at OPGC
3	GD-1	Barkot, Chandiposh	21.05.2024 18:09	21.05.2024 18:25	00:16	0	40	0.00%	0.16%	28931	24949	At 18:09 Hrs on 21.05.2024, Y_ph BPI of 220 kV Rengali-Rengali (PH)-1 punctured at Rengali end. At the same time, 220 kV Rengali-Rengali (PG) D/c, 220 kV Rengali-Chandiposh and 220 kV Rengali-Barkot also tripped. This led to total power failure at Chandiposh and Barkot. Load loss of around 40 MW occurred.	220 kV Rengali- Chandiposh 220 kV Rengali- Rengali(PG)-I 220 kV Rengali- Rengali(PG)-II 220 kV Rengali- Barkot 220 kV Rengali- Renagli(PH)-I
4	GD-1	Haldia	29.05.2024 12:38	29.05.2024 12:57	00:19	566	0	2.19%	0.00%	25886	28256	At 12:34 hrs.on 29.05.2024, R_ph bushing of GT-1 at Haldia caught fire which led to tripping of Unit-1 due to operation of GT-1 Transformer Differential protection. At the same time, 400 kV Haldia-Subhashgram-2tripped due to fault in R phase LA & 400 kV Haldia-Subahshgram-1 tripped due to E/F. Consequently, U#2 also tripped due to loss of evacuation path. Total generation loss of around 566 MW occurred.	400 kV Haldia-Subhashgram D/c 2*300 MW Units at Haldia
5	GD-1	Tenughat	29.05.2024 12:57	29.05.2024 13:14	00:17	333	0	1.30%	0.00%	25657	28796	At 12:57 hrs on 29.05.2024, 220 kV Tenughat-Govindpur-1 tripped due to Y-B Fault. 220 kV Tenughat Govindpur-2 also tripped at the same time from Tenughat only. 220 kV Tenughat-Biharsharif was already was already under breakdown. Consequently, both units of Tenughat tripped due to loss of evacuation path and around 333 MW generation loss occurred..	220 Kv Tenughat -Govindpur-1 220 KV Tenught - Govindpur-2 210 MW U#1 at Tenughat 210 MW U#2 at Tenughat
6	GD-1	Dalkhola	31.05.2024 02:42	31.05.2024 04:32	00:10	0	4	0.00%	0.02%	28426	22896	At 02:42 Hrs on 31.05.2024, 132 kV Bus PT burst at Dalkhola(WB) and 132 kV Bus became dead. 220 kV Dalkhola(PG)-Kishanganj D/c also tripped from Ksihanganj in Zone-3. Consequently, 220 kV Dalkhola (WB) S/s became dead. Load loss of 4 MW occurred at Dalkhola as other areas remained on alternate sources. All load was restored within 10 minutes.	220 KV Kishanganj-Dalkhola (PG)-D/c 132 KV Baisi-Dalkhola(WB)-1

20.Details of Grid Events during the Month of May 2024 in North-East Region

Sl No.	Category of Grid Event (GI 1or GI 2/ GD-1 to GD-5)	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
						Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
1	GD I	Rhongkhon Area of Meghalaya	02-05-2024 01:12	02-05-2024 01:58	00:46:00	0	40	0.00%	1.67%	2291	2390	Rongkhon, Ampati, Phulbari and Ganol areas of Meghalaya Power System are connected to the rest of NER Grid through 132 kV Nangalbibra-Rongkhon line. At 01:12 Hrs of 02.05.2024, 132 kV Nangalbibra-Rongkhon line tripped. Due to tripping of this line, Rongkhon, Ampati, Phulbari and Ganol areas of Meghalaya Power System were isolated from NER Grid and collapsed due to no source available in these areas. Power supply was extended to Rongkhon area of Meghalaya Power System by charging 132kV Nangalbibra –Rhonghkon at 01:58 hrs of 02- 05-2024.	132kV Nangalbibra –Rhonghkon
2	GD I	Lumshnong S/S of Meghalaya	02-05-2024 07:01	02-05-2024 07:26	00:25:00	0	14	0.00%	0.62%	2234	2241	Lumshnong area of Meghalaya Power System is connected to the rest of NER Grid through 132 kV Lumshnong-Panchgram and 132 kV Lumshnong-Khliehriat lines. Prior to the event, 132 kV Lumshnong-Khleihriat line tripped at 06:10 Hrs of 02.05.2024. At 07:01 Hrs of 02.05.2024, 132 kV Lumshnong-Panchgram line tripped. Due to tripping of this element, Lumshnong area of Meghalaya Power System got isolated from NER Grid and collapsed due to no source available in this area. Power supply was extended to Lumshnong S/S of Meghalaya Power System by charging 132 kV Khliehriat – Lumshnong line at 07:26 hrs of 02- 05-2024.	132kV Lumshnong - Panchgram
3	GD I	Monarchak Generating station (NEEPCO) and Rabindranagar of Tripura Power	02-05-2024 18:48	02-05-2024 21:09	02:21:00	69	5	2.38%	0.18%	2904	2844	Monarchak Generating station of NEEPCO and Rabindranagar area of Tripura Power System is connected to the rest of NER Grid through 132 kV Monarchak-Rokhia line and 132 kV Monarchak-Udaipur line. Prior to the event, 132 kV Monarchak-Rokhia line tripped at 18:42 Hrs of 02.05.2024. At 18:48 Hrs of 02.05.2024, 132 kV Monarchak-Udaipur line tripped. Due to tripping of this element, Monarchak generation and Rabindranagar area of Tripura power system got isolated from NER Grid and collapsed due to load generation mismatch in this area. Power supply was extended to Rabindranagar S/S of Tripura Power System by charging 132 kV Monarchak-Udaipur line at 21:09 Hrs of 02-05-2024. Monarchak Generation was synchronised at 00:01 Hrs of 03-05-2024.	132 kV Monarchak-Udaipur line, Monarchak GT & ST
4	GD I	Leshka HEP of Meghalaya Power System	02-05-2024 00:45	02-05-2024 00:56	00:11:00	42	0	1.92%	0.00%	2189	2209	Leshka HEP of Meghalaya Power System were connected with rest of NER Grid through 132 kV Leshka-Khleihriat D/C lines. At 00:45 Hrs of 02-05-2024, 132 kV Leshka-Khleihriat D/C lines tripped. Due to tripping of these elements, Leshka HEP of Meghalaya power system got isolated from NER Grid and collapsed due to no source available in this area. Power supply was extended to Leshka HEP of Meghalaya Power System by charging 132 kV Leshka - Khleihriat 1 line at 00:56 hrs of 02-05-2024. Subsequently, 132 kV Leshka - Khleihriat 2 line.was charged at 01:08 hrs of 02-05-2024.	132 kV Leshka-Khleihriat D/C
5	GD I	Leshka HEP of Meghalaya Power System	02-05-2024 04:11	02-05-2024 04:51	00:40:00	42	0	1.96%	0.00%	2142	1921	Leshka HEP of Meghalaya Power System were connected with rest of NER Grid through 132 kV Leshka-Khleihriat D/C lines. Prior to the event, 132 kV Leshka-Khleihriat I Line tripped at 04:10 Hrs of 02.05.2024. At 04:11 Hrs of 02-05-2024, 132 kV Leshka-Khleihriat II lines tripped. Due to tripping of these elements, Leshka HEP of Meghalaya power system got isolated from NER Grid and collapsed due to no source available in this area. Power supply was extended to Leshka HEP of Meghalaya Power System by charging 132 kV Leshka - Khleihriat 1 line at 04:51 hrs of 02-05-2024. Subsequently, 132 kV Leshka - Khleihriat 2 line.was charged at 04:53 hrs of 02-05-2024.	132 kV Leshka-Khleihriat D/C
6	GD I	Cherapunji Area of Meghalaya Power System	02-05-2024 02:51	02-05-2024 04:11	01:20:00	0	2	0.00%	0.09%	2260	2290	Cherapunji area of Meghalaya Power System is connected to the rest of NER Grid through 132 kV Mawlai-Cherapunji line. At 02:51 Hrs of 02.05.2024, 132 kV Mawlai-Cherapunji line tripped. Due to tripping of this element, Cherapunji area of Meghalaya Power System got isolated from NER Grid and collapsed due to no source available in this area. Power supply was extended to Cherapunji S/S of Meghalaya Power System by charging 132 kV Mawlai-Cherapunji line at 04:11 hrs of 02-05-2024.	132kV Mawali-Cherapunji line
7	GD I	Pasighat Area of Arunachal Pradesh	02-05-2024 19:26	02-05-2024 19:56	00:30:00	0	5	0.00%	0.19%	2867	2702	Pasighat Area of Arunachal Pradesh Power System were connected with rest of NER Grid through 132 kV Roing - Pasighat Line. 132kV Along – Pasighat tripped at 18:36 hrs of 02.05.2024 At 15:46 Hrs of 02.05.2024, 132kV Along – Pasighat and 132kV Pasighat - Roing lines tripped. Due to tripping of these lines, Pasighat area of Arunachal Pradesh Power System was isolated from NER Grid and collapsed due to no source available in this area. Power supply was extended to Pasighat area of Arunachal Pradesh Power System by charging 132kV Pasighat - Roing at 19:56 Hrs of 02.05.2024.	132 kV Roing - Pasighat Line
8	GD I	Ampati and Pulbari substation of Meghalaya	05-05-2024 12:27	05-05-2024 12:46	00:19:00	0	14	0.00%	0.78%	1619	1794	Ampati and Pulbari substation of Meghalaya is connected to NER Power system via 132kV Rongkhong - Ampati line At 12:27 Hrs of 05-05-2024, 132kV Rongkhong - Ampati line tripped. Due to tripping of this line, Ampati and Pulbari areas of Meghalaya Power System was isolated from NER Grid and collapsed due to no source available in this area Line was charged at 12:46 and the power was restored.	132kV Rongkhong - Ampati line
9	GD I	Zuangtui, saitual, Vankal and Khawzal area of Mizoram	05-05-2024 04:13	05-05-2024 04:44	00:31:00	0	15	0.00%	0.93%	1939	1614	Zuangtui substation and radially connected Saitual, Vankal, Khawzawl and Serchhip substations are connected to the rest of the grid via 132 kV Melriat(PG)-Zuangtui line. 132 kV Serchhip-Lunglei line is kept open due to system requirement. At 04:13 Hrs of 05-05-2024, 132 kV Melriat- Zuangtui line tripped. Due to tripping of this element, Zuangtui and radially connected substations of Mizoram power system got separated from rest of the grid due to no source available in these areas. Power was extended to Zuangtui and radially connected substations by charging 132 kV Melriat-Zuangtui Line at 04:44 Hrs of 05-05-2024.	132 kV Melriat- Zunagtui Line
10	GD I	Tipaimukh area of Manipur	05-05-2024 23:39	06-05-2024 00:19	00:40:00	0	0	0.00%	0.00%	2291	1600	Tipaimukh area of Manipur Power System was connected with rest of NER Grid through 132 kV Jiribam- Tipaimukh line & 132 kV Aizawl-Tipaimukh. At 21:54 Hrs of 05-05-2024, 132 kV Aizawl-Tipaimukh tripped. At 23:39 Hrs of 05-05-2024, during the charging attempt of 132 kV Aizawl-Tipaimukh line, 132 kV Jiribam- Tipaimukh line also tripped and Tipaimukh area was isolated from NER Grid and collapsed due to no source available in this area. Power was extended to Tipaimukh area of Manipur Power System, by charging 132 kV Jiribam- Tipaimukh line at 00:19 Hrs.	132 kV Jiribam- Tipaimukh line

20.Detals of Grid Events during the Month of May 2024 in North-East Region

Sl No.	Category of Grid Event (GI 1or GI 2/ GD-1 to GD-5)	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
						Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
11	GD I	Leshka HEP of Meghalaya Power System	05-05-2024 16:06	05-05-2024 16:29	00:23:00	42	0	2.42%	0.00%	1735	1658	Leshka HEP of Meghalaya Power System was connected with rest of NER Grid via 132 kV Myntdu Leshka - Khleihriat D/C lines. At 16:06 Hrs of 05-05-2024, 132 kV Myntdu Leshka - Khleihriat D/C lines tripped. Due to tripping of these lines, Leshka HEP of Meghalaya Power System was isolated from NER Grid and collapsed due to loss of evacuation path. Power supply was extended to Leshka HEP of Meghalaya Power System by charging 132 kV Myntdu Leshka - Khleihriat D/C at 16:29 hrs of 05.05.2024	132 kV Leshka-Khleihriat D/C
12	GD I	Nangalbibra, Ronkhon and Ampati areas of Meghalaya Power System	06-05-2024 05:21	06-05-2024 05:56	00:35:00	0	14	0.00%	1.19%	1996	1173	Nangalbibra substation of Meghalaya is connected to NER Power system via 132kV Agia- Nangalbibra line, 132kV Nangalbibra- Mendipathar line and 132kV Nangalbibra – Nongstoin line. Prior to the event, 132kV Nangalbibra – Nongstoin line tripped at 05:18hrs; 132kV Agia- Nangalbibra line tripped at 05:20 hrs. At 05:21 hrs Hrs of 06-05-2024, 132 kV Nangalbibra- Mendipathar line tripped. Due to tripping of this line, Nangalbibra, Rongkhon and Ampati areas of Meghalaya Power System was isolated from NER Grid and collapsed due to no source available in this area. Power supply was extended to Nangalbibra, Ronkhon and Ampati areas of Meghalaya Power System by charging 132 kV Nangalbibra-Nongstoin line at 05:56 hrs of 06.05.2024.	132 kV Nangalbibra- Mendipathar line
13	GD I	Kongba, Thoubal New, Thoubal old areas of Manipur Power System	12-05-2024 12:32	12-05-2024 15:46	03:14:00	0	15	0.00%	0.62%	2164	2420	Kongba, Thoubal New, Thoubal old areas of Manipur Power were connected with rest of NER Grid through 132 kV Yiangangpokpi-Kongba D/C line, 132 kV Thoubal New Kakching and 132 kV Thoubal old-Kakching line. 400 kV-Imphal(PG)-Thoubal-1 line was under outage since 13:32 Hrs of 18.10.2021, 400 kV-Imphal(PG)-Thoubal-2 line was under outage since 15:05 Hrs of 24.04.2024. At 12:32 Hrs of 12-05-2024, 132 kV Yiangangpokpi-Kongba D/C line, 132 kV Thoubal New-Kakching and 132 kV Thoubal old-Kakching lines tripped. Due to tripping of these elements, Kongba, Thoubal New, Thoubal old areas of Manipur Power System were isolated from NER Grid and collapsed due to no source available in these areas.	132kV Yiangangpokpi-Kongba D/C line, 132 kV Thoubal New-Kakching and 132 kV Thoubal old-Kakching lines
14	GD I	Dhemaji and Silapathar areas of Assam Power System	14-05-2024 13:36	14-05-2024 13:57	00:21:00	0	16	0.00%	0.59%	1734	2696	Dhemaji and Silapthar areas of Assam power system were connected with rest of NER Grid through 132kV North Lakhimpur - Dhemaji line. At 13:36 Hrs of 14-05-2024, 132 kV North Lakhimpur - Dhemaji tripped. Due to tripping of this line,Dhemaji and Silapthar areas of Assam Power System were isolated from NER Grid and collapsed due to no source available in these areas. Power was extended to Dhemaji and Silapathar by charging 132 kV North Lakhimpur – Dhemaji line at 13:57 Hrs of 14-05-2024.	132 kV North Lakhimpur - Dhemaji line
15	GD I	NEIGRIHMS area of Meghalaya Power system	16-05-2024 15:37	16-05-2024 15:45	00:08:00	0	2	0.00%	0.07%	2087	2991	NEIGRIHMS area of Meghalaya Power System is connected to the rest of NER Grid through 132 kV NEIGRIHMS-NEHU line and 132 kV NEIGRIHMS-Khliehriat line. Prior to the event, 132 kV NEIGRIHMS-Khliehriat line tripped at 14:45 Hrs of 16.05.2024. At 15:37 Hrs of 16.05.2024, 132 kV NEIGRIHMS-NEHU line tripped. Due to tripping of this element, NEIGRIHMS area of Meghalaya Power System got isolated from NER Grid and collapsed due to no source available in this area. Power supply was extended to NEIGRIHMS S/S of Meghalaya Power System by charging 132 kV NEIGRIHMS-NEHU line at 15:45 Hrs of 16-05-2024.	132 kV NEIGRIHMS-NEHU Line
16	GD I	Nathkuchi area of Assam Power System	17-05-2024 23:34	18-05-2024 00:13	00:39:00	0	4	0.00%	0.16%	2385	2525	Nathkuchi area of Assam Power System is connected to NER Power system via 132 kV Bornagar-Nathkuchi line. 132 kV Rangia - Nathkuchi line was under outage since 22:54 Hrs of 17.05.2024. At 23:34 Hrs of 17-05-2024, 132 kV Bornagar-Nathkuchi line tripped. Due to tripping of this line, Nathkuchi area of Assam Power System was isolated from NER Grid and collapsed due to no source available in this area. Power was extended to Nathkuchi area by charging 132 kV Bornagar-Nathkuchi line at 13:57 Hrs of 14-05-2024.	132 kV Bornagar-Nathkuchi line
17	GD I	Barpeta area of Assam Power System	17-05-2024 23:44	17-05-2024 23:58	00:14:00	0	14	0.00%	0.56%	2402	2485	Barpeta area of Assam Power System is connected to NER Power system via 132 kV Dhaligaon - Barpeta line. 132 kV Nalbari – Barpeta line was kept open for Load segregation. At 23:44 Hrs of 17-05-2024, 132 kV Dhaligaon - Barpeta line tripped. Due to tripping of thisline, Barpeta area of Assam Power System was isolated from NER Grid and collapsed due to no source available in this area. Power was extended to Barpeta area by shifting Barpeta load to Nalbari side at 23:58 Hrs of 17-05-2024.	132 kV Dhaligaon - Barpeta line
18	GD I	Gossaigaon area of Assam Power System	18-05-2024 01:17	18-05-2024 01:39	00:22:00	0	0	0.00%	0.00%	2105	2235	Gossaigaon area of Assam Power System is connected to NER Power system via 132 kV Dhaligaon - Gossaigaon line. 132 kV Gossaigaon-Gauripur line was kept open on system requirement. At 01:17 Hrs of 18-05-2024, 132 kV Dhaligaon - Gossaigaon line tripped. Due to tripping of this line, Gossaigaon area of Assam Power System was isolated from NER Grid and collapsed due to no source available in this area. Power was extended to Gossaigaon area by charging 132 kV Gossaigaon-Gauripur line at 01:39 Hrs of 18-05-2024.	132 kV Dhaligaon - Gossaigaon line
19	GD I	New Shillong area of Meghalaya Power System	21-05-2024 12:52	21-05-2024 14:11	01:19:00	0	1	0.00%	0.04%	2070	2783	New Shillong area of Meghalaya Power System is connected to NER Power system via 220kV Mawngap-New Shillong D/C lines. At 12:52 hrs Hrs of 21-05-2024, 220kV Mawngap-New Shillong D/C tripped. Due to tripping of this line, New Shillong area of Meghalaya Power System was isolated from NER Grid and collapsed due to no source available in this area. Power was restored to New Shillong area by charging 220kV Mawngap-New Shillong 1 at 14:11 hrs of 21-05-2024.	220kV Mawngap-New Shillong D/C lines
20	GD I	Serchhip area of Mizoram Power System	21-05-2024 16:06	21-05-2024 17:02	00:56:00	0	7	0.00%	0.25%	2126	2854	Serchhip substation of Mizoram power system are connected to the rest of the grid via 132 kV Zuangtui-Serchhip and 132 kV Serchhip-Lunglei lines. 132 kV Serchhip-Lunglei line is kept open due to system requirement. At 16:06 Hrs of 21-05-2024, 132 kV Zuangtui-Serchhip line tripped. Due to tripping of this element, Serchhip area of Mizoram power system got separated from rest of the grid due to no source available in this area. Power supply was extended to Serchhip area by charging 132 kV Zuangtui-Serchhip Line at 17:02 Hrs of 21-05-2024.	132 kV Zuangtui-Serchhip line

20.Details of Grid Events during the Month of May 2024 in North-East Region

Sl No.	Category of Grid Event	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
	(GI 1or GI 2/ GD-1 to GD-5)					Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
21	GD I	Kohima area of Nagaland Power System	21-05-2024 16:42	21-05-2024 17:15	00:33:00	0	15	0.00%	0.53%	2165	2826	Kohima area of Nagaland Power System were connected with rest of NER Grid through 132 kV Dimapur-Kohima line, 132 kV Karong-Kohima line, 132 kV Kohima-Zadima line and 132 kV Kohima-Meluri line. Prior to the event, 132 kV Karong-Kohima line and 132 kV Kohima-Meluri line was under outage. At 16:42 Hrs of 21.05.2024, 132 kV Dimapur(PG) - Kohima and 132 kV Kohima-Zadima Lines tripped. Due to tripping of these lines, Kohima area of Nagaland Power System got isolated from NER Grid and collapsed due to no source available in this area. Power supply was extended to Kohima area of Nagaland Power System by charging 132 kV Kohima-Zadima Line at 17:15 Hrs of 21.05.2024.	132 kV Dimapur(PG) - Kohima and 132 kV Kohima-Zadima Lines
22	GD I	Rengpang area of Manipur Power System	21-05-2024 20:22	21-05-2024 20:54	00:32:00	0	3	0.00%	0.09%	3133	3346	Rengpang area of Manipur power system is connected to the rest of the grid via 132 kV Loktak-Rengpang line and 132 kV Jiribam-Rengpang line. 132 kV Jiribam-Rengpang line is under outage since 18:18 Hrs of 17.11.2023. At 20:22 Hrs of 21-05-2024, 132 kV Loktak-Rengpang line tripped. Due to tripping of this element, Rengpang area of Manipur power system got separated from rest of the grid due to no source available in this area. Power was extended to Rengpang area of Manipur power system by charging 132 kV Loktak-Rengpang Line at 20:54 Hrs of 21-05-2024.	132 kV Loktak-Rengpang line
23	GD I	Leshka HEP of Meghalaya Power System	23-05-2024 14:05	23-05-2024 15:01	00:56:00	0	0	0.00%	0.00%	1840	2877	Leshka HEP of Meghalaya Power System was connected with rest of NER Grid via 132 kV Myntdu Leshka - Khleihriat D/C lines. At 14:05 Hrs of 23-05-2024, 132 kV Myntdu Leshka - Khleihriat D/C lines tripped. Due to tripping of these lines, Leshka HEP of Meghalaya Power System was isolated from NER Grid and collapsed due to loss of evacuation path. Power supply was extended to Leshka HEP of Meghalaya Power System by charging 132 kV Myntdu Leshka - Khleihriat 1 line at 15:01 Hrsof 23.05.2024. Subsequently, 132 kV Myntdu Leshka - Khleihriat 2 line was charged at 15:03 hrs of 23.05.2024.	132 kV Myntdu Leshka - Khleihriat D/C lines
24	GD I	MUSTEM area of Meghalaya Power System	23-05-2024 14:47	23-05-2024 15:00	00:13:00	0	0	0.00%	0.00%	1989	2960	MUSTEM area of Meghalaya Power System was connected with rest of NER Grid via 132 kV Mawlyndep – Mustem line. 132 kV Khliehriat – Mustem line was under outage since 14:37 Hrs on 23.05.2024. At 14:47 Hrs of 23-05-2024, 132 kV Mawlyndep – Mustem line tripped. Due to tripping of these line, MUSTEM area of Meghalaya Power System was isolated from NER Grid and collapsed due to load generation mismatch in this area. Power supply was extended to MUSTEM area of Meghalaya Power System by charging 132 kV Khliehriat – Mustem line at 15:00 Hrs of 23.05.2024.	132 kV Mawlyndep – Mustem line
25	GD I	Kongba, Thoubal New, Thoubal old, Chandel, Kakching, Elangkangpokpi, Thanlon, Churachandpur and Moreh areas of Manipur Power System and Tamu load of Myanmar power system	24-05-2024 14:44	24-05-2024 14:48	00:04:00	0	33	0.00%	1.11%	2288	2975	Kongba, Thoubal New, Thoubal old, Chandel, Kakching, Elangkangpokpi, Thanlon, Churachandpur and Moreh areas of Manipur Power System and Tamu load of Myanmar power system were connected with rest of NER Grid through 132kV Yiangangpokpi - Kongba 2 line. 132kV Yiangangpokpi - Kongba 1 line was under outage since 20:37 Hrs of 16-05-2024 due to hotspot observed at Yiangangpokpi end for this line. 132kV Ningthoukhong-Churachandpur 1 line was under outage since 21:14 Hrs of 29-04-2024 and 132kV Ningthoukhong-Churachandpur 2 line was under ESD since 13:10 Hrs of 24.05.24 for insulator Replacement of Ningthoukhong-Churachandpur 1 line. At 14:44 Hrs of 24-05-2024, 132kV Yiangangpokpi - Kongba 2 line tripped. Due to tripping of this element, Kongba, Thoubal New, Thoubal old, Chandel, Kakching, Elangkangpokpi, Thanlon, Churachandpur and Moreh areas of Manipur Power System and Tamu load of Myanmar power system were isolated from NER Grid and collapsed due to no source available in these areas. Power was extended to Kongba, Thoubal New, Thoubal old, Chandel, Kakching, Elangkangpokpi, Thanlon, Churachandpur and Moreh areas of Manipur Power System and Tamu load of Myanmar power system by charging 132kV Yiangangpokpi -Kongba 2 line at 14:48 hrs.	132kV Yiangangpokpi - Kongba 2 line
26	GD I	Rengpang area of Manipur Power System	24-05-2024 16:23	24-05-2024 20:59	04:36:00	0	1	0.00%	0.03%	2258	2986	Rengpang area of Manipur power system is connected to the rest of the grid via 132 kV Loktak-Rengpang line and 132 kV Jiribam-Rengpang line. 132 kV Jiribam-Rengpang line is under outage since 18:18 Hrs of 17.11.2023. At 16:23 Hrs of 24-05-2024, 132 kV Loktak-Rengpang line tripped. Due to tripping of this element, Rengpang area of Manipur power system got separated from rest of the grid due to no source available in this area. Power was extended to Rengpang area of Manipur power system by charging 132 kV Loktak-Rengpang Line at 20:59 Hrs of 24-05-2024.	132kV Loktak-Rengpang line
27	GD I	Wokha area of Nagaland Power System	24-05-2024 14:54	24-05-2024 16:17	01:23:00	0	1	0.00%	0.03%	2258	2986	Wokha area of Nagaland Power System was connected with rest of NER Grid through 132kV Sanis-Wokha and 132kV Wokha-Chiephebozou lines. At 14:54 Hrs of 24-05-2024, 132kV Sanis-Wokha and 132kV Wokha-Chiephebozou lines tripped. Due to tripping of these elements, Wokha area of Nagaland Power System was isolated from NER Grid and collapsed due to no source available in this area. Power was extended to Wokha by charging 132kV Sanis-Wokha at 16:17 hrs of 24.05.2024.	132kV Sanis-Wokha and 132kV Wokha-Chiephebozou lines
28	GD I	Serchhip area of Mizoram Power System	26-05-2024 09:34	-	-	0	2	0.00%	0.08%	2705	2638	Serchhip area of Mizoram Power System was connected with rest of NER Grid through 132 kV Zuangtui - Serchhip line. 132 KV Serchhip – Lunglei kept open for Load segregation purpose. At 22:45 Hrs of 26-05-2024, 132 kV Zuangtui - Serchhip line tripped. Due to tripping of thiselement, Serchhip area of Mizoram Power System was isolated from NER Grid and collapsed due to no source available in this area.	132 kV Zuangtui - Serchhip line
29	GD I	Daporijo, Basar & Along areas of Arunachal Pradesh Power System	24-05-2024 09:53	24-05-2024 11:18	01:25:00	0	5	0.00%	0.22%	1784	2272	Daporijo, Basar & Along areas of Arunachal Pradesh Power System were connected with rest of NER Grid through 132 kV Ziro-Daporijo line. (132 kV Along-Pasighat was under tripped condition since 21-05-2024) At 09:53 Hrs of 24-05-2024, 132 kV Ziro-Daporijo line tripped. Due to tripping of this element, Daporijo, Basar & Along areas of Arunachal Pradesh Power System were isolatedfrom NER Grid and collapsed due to no source available in this area. Power was extended to Daporijo by charging 132 kV Ziro-Daporijo at 11:18 hrs of 24.05.2024.	132 kV Ziro-Daporijo line
30	GD I	Wokha area of Nagaland Power System	26-05-2024 20:26	26-05-2024 21:19	00:53:00	0	5	0.00%	0.15%	2811	3233	Wokha area of Nagaland Power System was connected to rest of NER Grid through 132 kV Sanis-Wokha line and 132 kV Wokha- Chiephobozou line. At 20:26 Hrs of 26-05-2024, 132 kV Sanis-Wokha and 132 kV Wokha- Chiephobozou lines tripped. Due to tripping of these elements, Wokha area of Nagaland Power System was isolated from NER Grid and collapsed due to no source available in this area. Power was extended to Wokha by charging 132 kV Wokha-Sanis at 21:19 hrs of 26.05.2024.	132 kV Sanis-Wokha and 132 kV Wokha- Chiephobozou lines
31	GD I	Serchhip area of Mizoram Power System	26-05-2024 22:45	-	-	0	2	0.00%	0.08%	2705	2638	Serchhip area of Mizoram Power System was connected with rest of NER Grid through 132 kVZuangtui - Serchhip line. 132 KV Serchhip – Lunglei kept open for Load segregation purpose. At 22:45 Hrs of 26-05-2024, 132 kV Zuangtui - Serchhip line tripped. Due to tripping of thiselement, Serchhip area of Mizoram Power System was isolated from NER Grid and collapsed due to no source available in this area.	132 kV Zuangtui - Serchhip line
32	GD I	Bornagar area of Assam Power System	27-05-2024 01:04	27-05-2024 01:19	00:15:00	0	11	0.00%	0.53%	2007	2084	Bornagar area of Assam Power System is connected with rest of NER Grid via 132 kV Dhaligaon-Bornagar and 132 kV Bornagar-Nathkuchi Lines. Due to bus split at Bornagar, part load of Bornagar is fed from Dhaligaon end and part load is fed from Nathkuchi end. At 01:04 Hrs of 27-05-2024, 132 kV Dhaligaon-Bornagar line tripped. Due to tripping of this element, part load of Bornagar area of Assam Power System got isolated from NER Grid. Part load of Bornagar area of Assam Power System was restored by charging 132 kV Dhaligaon-Bornagar line at 01:19 Hrs of 27.05.2024.	132 kV Dhaligaon-Bornagar Line

20.Details of Grid Events during the Month of May 2024 in North-East Region

Sl No.	Category of Grid Event (GI 1or GI 2/ GD-1 to GD-5)	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
						Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
33	GD I	Karong area of Manipur Power system	28-05-2024 08:44	28-05-2024 15:04	06:20:00	0	13	0.00%	1.87%	1917	694	Karong area of Manipur Power System is connected with rest of NER Grid through 132 kV Imphal (MSPCL)-Karong and 132 kV Karong-Kohima lines. At 08:44 Hrs of 28.05.2024, 132 kV Imphal (MSPCL)-Karong and 132 kV Karong-Kohima Lines tripped. Due to tripping of these lines, Karong area of Manipur Power System got isolated from NER Grid and collapsed due to no source available in this area. Power supply was extended to Karong area of Manipur Power System by charging 132 kV Imphal(MSPCL)-Karong Line at 15:04 Hrs of 28.05.2024.	132 kV Imphal (MSPCL)-Karong and 132 kV Karong-Kohima Lines
34	Near Miss	Sarusajai, Kahelipara, Dispur, Narengi,Chandrapur,Jagiroad,Sonapur and Jawaharnagar areas of Assam Power System	28-05-2024 02:50	-	-	0	0	0.00%	0.00%	2522	854	The following tripping occured before near miss event:• 220kV Samaguri – Sonapur tripped at 01:45 Hrs of 28.05.2024 and declared faulty. 220/132kV ICT-3 at Sarusajai kept as Spare transformer by SLDC Assam • 220/132kV 100MVA ICT-2 at Sarusajai handtripped at 02:10 Hrs due to overvoltage • 220kV Mirza – Sarusajai I & II tripped at 02:27 Hrs and 02:50 Hrs respectively. • Karbi-Langpi generation was out as per water availability and low demand.Sarusajai, Kahelipara,Dispur, Narengi, Chandrapur, Jagiroad, Sonapur and Jawaharnagar of Assam Power System were connected with rest of the NER grid through 220kV Sarusajai –Mirza I & II, 220kV Samaguri – Jawaharnagar and 220kV Samaguri –Sonapur. Due to inclement weather in the prevailing cyclonic situation (Cyclone-"Remal"), multiple lines at downstream 33kV feeders tripped. Along with that the connecting lines mainly 220kV Samaguri – Sonapur line tripped at 01:45 Hrs, then 220kV Mirza-Sarusajai I & II tripped at 02:27 Hrs and 02:50 Hrs respectively. Power to above areas were supplying through 220kV Samaguri-Jawaharnagar line only. Tripping of 220kV Samaguri – Jawaharnagar might have led to complete blackout of above-mentioned areas of Assam Power System with load loss of 16 MW(as per SCADA). As reported via mail by SLDC Assam, load crash of 1143.05 MW observed in entire Assam. Restoration: Karbi-Langpi generation Unit-2 was black started and synced at Sarusajai at 12:37 Hrs, there was some reduction in voltage observed in Sarusajai. Subsequently 220/132kV 100 MVA ICT-3 at Sarusajai was brought into service at 13:27 Hrs. Thereafter other ICTs at Sarusajai were restored. Then 220kV Mirza Sarusajai I and II lines were restored at 14:42 Hrs and 14:43 Hrs. Later on Karbi Langpi Unit-1 was also taken into service at 14:57 Hrs.	220kV Samaguri -Sonapur, 220/132kV 100 MVA ICT-2 at Sarusajai, 220kV Mirza – Sarusajai I & II, 132kV Sarusajai – Kahelipara I, II & III, 132 kV Sarusajai-Kamakhya
35	GD I	Gauripur area of Assam Power system	29-05-2024 02:49	29-05-2024 04:07	01:18:00	0	15	0.00%	1.05%	2583	1425	Gauripur area of Assam Power System is connected with rest of NER Grid via 132 kV Bilasipara-Gauripur and 132 kV Gossaigaon-Gauripur lines. 132 kV Gossaigaon-Gauripur line is kept open to control overloading of 132 kV Kokrajhar-Bilasipara D/C lines. At 02:49 Hrs of 29-05-2024, 132 kV Bilasipara-Gauripur line tripped. Due to tripping of this element, Gauripur area of Assam Power System was isolated from NER Grid and collapsed due to no source available in this area. Power supply was extended to Gauripur area of Assam Power System by charging 132 kV Bilasipara-Gauripur line at 04:07 Hrs of 29.05.2024.	132 kV Bilasipara-Gauripur Line
36	GD I	Kokrajhar area of Assam Power System	29-05-2024 03:18	29-05-2024 03:44	00:26:00	0	10	0.00%	0.72%	2262	1390	Kokrajhar area of Assam Power System is connected with rest of NER Grid via 132 kV BTPS-Kokrajhar D/C lines. Prior to the event, 132 kV Bilasipara-Kokrajhar D/C lines and 132 kV BTPS-Kokrajhar I line was under outage since 03:04 Hrs and 03:11 Hrs of 29.05.2024 respectively. At 03:18 Hrs of 29-05-2024, 132 kV BTPS-Kokrajhar II line tripped. Due to tripping of this element, Kokrajhar area of Assam Power System was isolated from NER Grid and collapsed due to no source available in this area. Power supply was extended to Kokrajhar area of Assam Power System by charging 132 kV BTPS-Kokrajhar II line at 03:44 Hrs of 29.05.2024.	132 kV BTPS-Kokrajhar II Line
37	GD I	New Shillong and Mawngap areas of Meghalaya Power System	29-05-2024 15:17	29-05-2024 17:29	02:12:00	0	0.1	0.00%	0.00%	3211	2002	New Shillong and Mawngap areas of Meghalaya Power System are connected with the rest of NER grid via 220kV Mawngap-Killing D/C lines and 220/132 kV ICT-I & II at Mawngap. At 15:17 hrs Hrs of 29-05-2024, 220 kV Mawngap - Killing D/C, 220 kV Mawngap - New Shillong D/C & 220/132 Kk ICT-1 & 2 at Mawngap tripped. Due to tripping of these elements, New Shillong and Mawngap areas of Meghalaya Power System were isolated from NER Grid and collapsed due to no source available in these areas. Power supply was extended to 220 kV Mawngap S/S by charging 220 kV Mawngap – Killing I line at 17:19 Hrs of 29.05.2024 and to New Shillong by charging 220 kV Mawngap-New Shillong D/C lines at 17:29 Hrs of 29.05.2024.	220 kV Mawngap-Killing D/C Line, 220 kV Mawngap-New Shillong D/C Line, 220/132 kV ICT-I & II at Mawngap
38	GD I	Lumshnong area of Meghalaya Power system	30-05-2024 06:39	30-05-2024 06:52	00:13:00	0	2	0.00%	0.12%	2809	1732	Lumshnong area of Meghalaya Power System is connected to the rest of NER Grid through 132 kV Lumshnong-Panchgram and 132 kV Lumshnong-Khliehriat lines. Prior to the event, 132 kV Lumshnong-Panchgram line tripped at 03:02 Hrs of 30.05.2024. At 06:39 Hrs of 30.05.2024, 132 kV Lumshnong-Khliehriat line tripped. Due to tripping of this element, Lumshnong area of Meghalaya Power System got isolated from NER Grid and collapsed due to no source available in this area. Power supply was extended to Lumshnong S/S of Meghalaya Power System by charging 132 kV Khliehriat – Lumshnong line at 06:52 hrs of 30-05-2024.	132 kV Lumshnong-Khliehriat Line
39	GD I	Phulbari area of Meghalaya Power System	31-05-2024 13:44	31-05-2024 14:10	00:26:00	0	1	0.00%	0.04%	2819	2224	Phulbari area of Meghalaya Power System are connected to the rest of NER Grid through 132 kV Ampati-Phulbari line. At 13:44 Hrs of 31-05-2024, 132 kV Ampati-Phulbari line tripped. Due to tripping of this line, Phulbari area of Meghalaya Power System was isolated from NER Grid and collapsed due to no source available in this area. Power supply was extended to Phulbari area by charging 132 kV Ampati-Phulbari line at 14:10 Hrs of 31-05-2024	132 kV Ampati-Phulbari Line

21. NEW ELEMENTS COMMISSIONED* IN MAY 2024

COMMISSIONING OF NEW GENERATING UNITS

REGION	S.NO.	ENTITY	UNIT NAME / LOCATION	ADD. CAPACITY (MW)	UNIT NO.	INS. CAPACITY (MW)	DATE@
NR	1	RENEW SURYA VIHAAN PRIVATE LIMITED	Renew Surya Vihaan Private Limited	2	-	100	03-05-2024
	2	RENEW SURYA PRATAP PRIVATE LIMITED	Renew Surya Pratap Private Limited	3	-	200	03-05-2025
	3	AYANA RENEWABLE POWER THREE PVT LTD (ARP3PL)	Ayana Renewable Power Three Pvt Limited	97.16	-	272	11-05-2024
	4	RENEW SURYA AAYAN PRIVATE LIMITED	Renew Surya Aayan Private Limited	10	-	300	07-05-2024
	5	ANTA SOLAR POWER PLANT NTPC LIMITED	Anta Solar Power Plant NTPC Limited	33	-	90	28-05-2024
WR	6	AEPL	Apraava Energy Private Limited (Wind)/Gujarat	12.6	I	222.6	01-05-2024
	7	ADANI	Adani Wind Energy Kutchh Four Ltd(Wind)/Gujarat	67.2	I	270.9	04-05-2024
SR	8	JSW RENEW ENERGY TWO LIMITED	JSW Renew Energy Two Limited(Wind)/Tamilnadu	13.5	LOT-XIII		30-05-2024
	9	JSW RENEW ENERGY LTD	JSW Renew Energy Ltd(Wind)/Tamilnadu	21.6	LOT-V	170.1	30-05-2024
	10	RENEW SURYA ROSHINI PRIVATE LIMITED	Renew Surya Roshini Private Limited(Wind)	17.325	LOT-VI	429.66	02-05-2024
	11	SERENTIC RENEWABLES INDIA PRIVATE LIMITED	Serentic Renewables India Private Limited(Solar)/Karnataka	13.2	Block-I	13.2	26-05-2024
	12	SERENTIC RENEWABLES INDIA PRIVATE LIMITED	Serentic Renewables India Private Limited(Solar)/Karnataka	13.2	Block-II	26.4	24-05-2024
	13	SERENTIC RENEWABLES INDIA PRIVATE LIMITED	Serentic Renewables India Private Limited(Solar)/Karnataka	13.2	Block-III	39.6	24-05-2024
	14	SERENTIC RENEWABLES INDIA PRIVATE LIMITED	Serentic Renewables India Private Limited(Solar)/Karnataka	13.2	Block-IV	52.8	25-05-2024
	15	SERENTIC RENEWABLES INDIA PRIVATE LIMITED	Serentic Renewables India Private Limited(Solar)/Karnataka	13.2	Block-V	66	25-05-2024
	16	SERENTIC RENEWABLES INDIA PRIVATE LIMITED	Serentic Renewables India Private Limited(Solar)/Karnataka	13.2	Block-VI	79.2	25-05-2024
	17	SERENTIC RENEWABLES INDIA PRIVATE LIMITED	Serentic Renewables India Private Limited(Solar)/Karnataka	13.2	Block-VII	92.4	25-05-2024
	18	SERENTIC RENEWABLES INDIA PRIVATE LIMITED	Serentic Renewables India Private Limited(Solar)/Karnataka	13.2	Block-VIII	105.6	25-05-2024
	19	SERENTIC RENEWABLES INDIA PRIVATE LIMITED	Serentic Renewables India Private Limited(Solar)/Karnataka	13.2	Block-IX	118.8	25-05-2024
	20	SERENTIC RENEWABLES INDIA PRIVATE LIMITED	Serentic Renewables India Private Limited(Solar)/Karnataka	13.2	Block-X	132	25-05-2024
	21	SERENTIC RENEWABLES INDIA PRIVATE LIMITED	Serentic Renewables India Private Limited(Solar)/Karnataka	13.2	Block-XI	145.2	25-05-2024
	22	SERENTIC RENEWABLES INDIA PRIVATE LIMITED	Serentic Renewables India Private Limited(Solar)/Karnataka	13.2	Block-XII	158.4	25-05-2024
NER	23	NEEPCO	Kopili(Hydro)/Assam	50	-	200	24-05-2024

@RE (Solar & Wind) generation FTC for complete capacity is in phases. Date mentioned is the date by which complete (Added) capacity synch./COD

COMMISSIONING OF NEW ICTs

REGION	S.NO.	ENTITY	SUB-STATION	VOLTAGE (kV)	ICT NO.	CAPACITY (MVA)	DATE
NR	1	UPPTCL	Jaunpur	400/220/33	III	315	14-05-2024
	2	PBTSL	Bikaner_II	400/220/33	III	500	21-05-2024
SR	3	TNTRANSCO	Sriperumbdur	400/220/33	VI	500	14-05-2024
	4	TNTRANSCO	Salem	400/110/33	IV	200	20-05-2024
	5	TNTRANSCO	Thiruvalam	400/230/33	III	500	21-05-2024

COMMISSIONING OF NEW TRANSMISSION LINES

REGION	S.NO.	ENTITY	LINE NAME	CKT NO.	VOLTAGE (kV)	DATE
SR	1	TSTRANSCO	400kV QMDC YTPP- Dindi - Q1	I	400	03-05-2024
	2	TSTRANSCO	400kV QMDC YTPP- Dindi - Q2	II	400	03-05-2024
ER	3	BSPTCL	400kV Buxar-Naubatpur	II	400	11-05-2024
NER	4	AEGCL	220kV Rangia - Amingaon	II	220	27-05-2024

LILO OF TRANSMISSION LINES

REGION	S.NO.	ENTITY	LINE NAME (LILOed LINE)	LILO AT	VOLTAGE (kV)	DATE
NR-ER	1	PGCIL	400kV Biharshriff-Sahupuri-II (400kV Varanasi-Biharsharif-II)	Sahupuri	400	18-05-2024
NR-ER	2	UPPTCL	400kV Varanasi-Sahupuri-II (400kV Varanasi-Biharsharif-II)	Sahupuri	400	18-05-2024

COMMISSIONING OF BUS/LINE REACTORS

REGION	S.NO.	ENTITY	SUB-STATION / LINE (@SUB-STATION)	VOLTAGE (kV)	CAPACITY (MVAR)	DATE
NR	1	UPPTCL	Biharshriff-Sahupuri-II @Sahupuri	400	63	18-05-2024
SR	2	TSTRANSCO	Bus Reactor-I @Jangaon	400	125	20-05-2024
	3	TSTRANSCO	Bus Reactor-I @Asupaka	400	80	08-05-2024

COMMISSIONING OF HVDC/FACTS DEVICE/SYSTEM

REGION	S.NO.	ELEMENT NAME	SUB-STATION / CONSTITUENT	RATINGS	VOLTAGE (kV)	DATE

*Includes first time charging (FTC) as well

22. Voltage Profile

विद्युत विभव रूपरेखा माह - मई 2024 VOLTAGE PROFILE - MAY 2024

Date of reporting: 05-May-24

क्र.स. Sl. No.	क्षेत्र REGION	उपकेंद्र SUBSTATION	VOLTAGE < V(lower)* (V=380,728 kV)	V(lower) < VOLTAGE < V(upper)*	VOLTAGE > V(upper)* (V=420,800 kV)	Voltage Deviation Index (%age of time voltage is outside range)	Number of hours voltage was outside IEGC band during month	उच्चतम (कि.वो.) Maximum(kV)	निम्नतम(कि.वो.) Minimum(kV)	औसत (कि.वो.) Average(kV)
1	पूर्वी क्षेत्र ER	अंगुल ANGUL	0%	100%	0%	0%	0	799	754	774
2		दालिपली DARLIPALI	0%	100%	0%	0%	0	792	767	778
3		गया GAYA	0%	100%	0%	0%	0	784	752	766
4		जीरत JEERAT	0%	100%	0%	0%	0	800	740	769
5		झारसुगुड़ा JHARSUGUDA	0%	100%	0%	0%	1	803	775	788
6		मिदनापुर MEDINIPUR	0%	100%	0%	0%	0	800	749	774
7		राँची RANCHI	0%	100%	0%	0%	0	794	763	781
8		सासाराम SASARAM	0%	100%	0%	0%	1	805	743	763
1	उत्तरी क्षेत्र NR	आगरा AGRA	0%	100%	0%	0%	1	802	760	781
2		आगरा (फ़तेहाबाद) AGRA (FATEHABAD)	0%	100%	0%	0%	0	789	752	770
3		अजमेर AJMER	0%	99%	1%	1%	5	806	756	784
4		अलीगढ़ ALIGARH	0%	100%	0%	0%	0	797	758	777
5		अनपरा सी ANPARA-C	0%	100%	0%	0%	0	780	757	766
6		अनपरा डी ANPARA-D	0%	100%	0%	0%	0	778	757	764
7		अंता ANTA	0%	100%	0%	0%	0	798	762	779
8		बलिया BALLIA	0%	100%	0%	0%	0	784	742	760
9		बारा BARA	0%	100%	0%	0%	0	784	757	768
10		भाड़ला BHADLA	0%	100%	0%	0%	0	799	735	778
11		भाड़ला-2 BHADLA-2	0%	100%	0%	0%	0	801	743	779
12		भिवानी BHIWANI	0%	100%	0%	0%	1	804	765	783
13		बीकानेर BIKANER	0%	100%	0%	0%	0	798	747	779
14		बरेली BAREILLY	0%	100%	0%	0%	1	803	749	772
15		चित्तौड़गढ़ CHITTORGARH	0%	99%	1%	1%	7	806	760	784
16		फ़तेहगढ़-2 FATEHGARH-2	0%	100%	0%	0%	0	801	717	777
17		फ़तेहपुर FATEHPUR	0%	100%	0%	0%	0	794	741	772
18		ग्रेटर नोएडा GREATER NOIDA	0%	100%	0%	0%	0	800	759	780
19		हापुड़ HAPUR	0%	100%	0%	0%	0	793	750	772
20		जवाहरपुर JAWAHARPUR	0%	100%	0%	0%	1	802	748	773
21		झटिकरा JHATIKARA	0%	100%	0%	0%	0	797	753	777
22		कानपुर जीआईएस KANPUR GIS	0%	100%	0%	0%	0	796	737	773
23		खेतरी KHETRI	0%	100%	0%	0%	0	798	747	779
24		कोटेश्वर KOTESHWAR	0%	100%	0%	0%	0	796	753	775
25		ललितपुर LALITPUR	0%	100%	0%	0%	0	774	747	759
26		लखनऊ LUCKNOW	0%	100%	0%	0%	0	795	744	767
27		मैनपुरी MAINPURI	0%	100%	0%	0%	0	793	742	767
28		मेरठ MEERUT	0%	100%	0%	0%	0	798	755	779
29		मेरठ(UP) MEERUT (UP)	0%	100%	0%	0%	0	795	751	773
30		मोगा MOGA	0%	100%	0%	0%	1	806	751	779
31		ओबरा सी OBRA C	0%	100%	0%	0%	0	776	749	759
32		उरई ORAI	0%	100%	0%	0%	3	806	772	788
33		फागी PHAGI	0%	100%	0%	0%	1	804	759	782
34		रामपुर RAMPUR	0%	100%	0%	0%	1	805	746	773
35		उन्नाव UNNAO	0%	100%	0%	0%	0	792	738	758
36		वाराणसी VARANASI	0%	100%	0%	0%	0	793	760	774
1	पश्चिमी क्षेत्र WR	अकोला AKOLA	0%	100%	0%	0%	0	792	751	771
2		औरंगाबाद AURANGABAD	0%	100%	0%	0%	2	805	756	781
3		भोपाल (बीडीटीसीएल) BHOPAL (BDTCL)	0%	100%	0%	0%	0	789	751	770
4		भुज BHUJ	0%	100%	0%	0%	0	793	750	778
5		भुज-2 BHUJ-II	0%	100%	0%	0%	0	785	750	772
6		बिलासपुर BILASPUR	0%	100%	0%	0%	0	782	758	770
7		बीना BINA	0%	100%	0%	0%	0	798	772	784
8		बनसकेंठा BANASKANTHA	0%	100%	0%	0%	0	800	760	785
9		चांपा CHAMPA	0%	99%	1%	1%	9	804	774	790
10		धूले (बीडीटीसीएल) DHULE (BDTCL)	0%	100%	0%	0%	0	797	744	776
11		धर्मजयगढ़ DHARAMJAIGARH	0%	100%	0%	0%	0	784	761	771
12		रायपुर पूलिंग RAIPUR POOLING	0%	100%	0%	0%	0	793	766	782
13		एकतुनी EKTUNI	0%	100%	0%	0%	0	791	744	770
14		गदरवाड़ा GADARWARA	0%	100%	0%	0%	0	799	767	784
15		ग्वालियर GWALIOR	0%	100%	0%	0%	1	804	746	784
16		इंदौर INDORE	0%	100%	0%	0%	0	791	746	772
17		जबलपुर JABALPUR	0%	100%	0%	0%	0	802	776	788
18		खंडवा KHANDWA	0%	100%	0%	0%	0	792	749	773
19		कोरडी KORADI	0%	100%	0%	0%	0	780	749	764
20		लकाडिया LAKADIYA	0%	100%	0%	0%	3	804	757	785
21		रायगढ़ पूलिंग RAIGARH POOLING	0%	100%	0%	0%	0	796	773	785
22		पड़घे PADGHE	0%	100%	0%	0%	1	806	743	774
23		परली PARLI	0%	99%	1%	1%	6	808	760	784
24		पुणे PUNE	0%	100%	0%	0%	1	805	744	774
25		राजनन्दगांव RAJNANDGAON	0%	100%	0%	0%	1	803	770	787
26		सासन SASAN	0%	100%	0%	0%	0	787	769	778
27		सतना SATNA	0%	100%	0%	0%	0	799	774	785
28		सिवनी SEONI	0%	100%	0%	0%	0	793	761	777
29		सीपत SIPAT	0%	100%	0%	0%	0	783	760	771
30		सोलापुर SOLAPUR	0%	99%	1%	1%	8	807	738	785
31		तिरोड़ा TIRORA	0%	100%	0%	0%	0	775	753	763
32		तामनार TAMNAR	0%	100%	0%	0%	0	796	773	784
33		वड़ादरा VADODARA	0%	98%	2%	2%	12	804	746	782
34		विंध्याचल पूलिंग VINDHYACHAL PS	0%	100%	0%	0%	0	791	772	780
35		वर्धा WARDHA	0%	100%	0%	0%	0	803	763	783
36		वरोरा WARORA	0%	100%	0%	0%	0	801	757	779
1	दक्षिणी क्षेत्र SR	अरियलूर ARIYALUR	0%	100%	0%	0%	0	801	744	773
2		कड़पा CUDDAPAH	0%	98%	2%	2%	17	810	760	785
3		चिल्कलूरीपेट CHILAKALURIPETA	0%	99%	1%	1%	7	812	763	785
4		कनूल KURNOOL	0%	100%	0%	0%	2	808	754	779
5		महेश्वरम MAHESWARAM	0%	98%	2%	2%	12	815	761	785
6		निज़ामाबाद NIZAMABAD	0%	96%	4%	4%	30	817	761	788
7		नेल्लूर पूलिंग NELLORE PS	0%	100%	0%	0%	0	805	755	774
8		नॉर्थ चेन्नै पूलिंग NORTH CHENNAI PS	0%	100%	0%	0%	1	806	730	769
9		रायचूर RAICHUR	0%	100%	0%	0%	1	805	755	778
10		श्रीकाकुलम SRIKAKULAM	0%	100%	0%	0%	2	810	756	782
11		तिरुवलम THIRUVALEM	0%	100%	0%	0%	0	796	745	773
12		वेमगिरी VEMAGIRI	0%	99%	1%	1%	10	813	757	785
13		वारंगल WARANGAL	0%	99%	1%	1%	7	811	759	784
1	पूर्वोत्तर क्षेत्र NER	बिस्वनाथ चरिआलि BISWANATH CHARIALI (400 kV)	0%	100%	0%	0%	0	419	392	405
2		बोंगाईगांव BONGAIGAON (400 kV)	0%	100%	0%	0%	0	414	396	407
3		बोंगाईगांव टीपीएस BONGAIGAON TPS (400 kV)	0%	100%	0%	0%	0	415	398	408
4		इम्फाल IMPHAL (400 kV)	0%	100%	0%	0%	0	420	359	405
5		बर्निहाट BYRNIHAT (400 kV)	0%	100%	0%	0%	0	418	406	412
6		कामेंग KAMENG (400 kV)	0%	100%	0%	0%	0	416	391	403
7		अज़ारा AZARA (400 kV)	0%	100%	0%	0%	0	412	399	407
8		मीसा MISA (400 kV)	0%	100%	0%	0%	0	414	392	404
9		न्यू मरियानी NEW MARIANI (400 kV)	0%	100%	0%	0%	0	417	394	405
10		न्यू कोहिमा NEW KOHIMA (400 kV)	0%	100%	0%	0%	0	420	393	407
11		पालाटाना PALATANA (400 kV)	0%	100%	0%	0%	0	418	399	407
12		पूर्वेकंचनबारी PK BARI (400 kV)	0%	100%	0%	0%	0	418	361	403
13		रंगानदी RANGANADI (400 kV)	0%	100%	0%	0%	0	419	388	409
14		सिलचर SILCHAR (400 kV)	0%	100%	0%	0%	0	419	398	408
15		सूर्यामणिनगर SURJYAMANINAGAR (400 kV)	0%	100%	0%	0%	0	418	361	402
16		थौबाल THOUBAL (400 kV)	0%	100%	0%	0%	0	415	384	401

All listed stations are 765 kV stations unless otherwise mentioned

***Percentage is calculated w.r.t. Time of one month.**

23.ALL TIME HIGHEST

31-05-2024

	Maximum Demand Met during the day (MW)	Demand Met during Evening Peak hrs(MW)	Energy Met (MU)	Hydro Gen. (MU)	Wind Gen. (MU)	Solar Gen. (MU)
NR	86785 30-05-2024	78585 27-05-2024	1890 29-05-2024	443 01-08-2023	86 07-08-2023	186 01-05-2024
WR	74914 31-05-2024	67482 30-05-2024	1648 06-05-2024	167 18-12-2014	310 28-05-2024	102.85 30-04-2024
SR	68735 13-03-2024	54798 29-04-2024	1416 05-04-2024	208 31-08-2018	298 21-07-2023	148.7 31-03-2024
ER	31895 30-04-2024	29695 29-05-2024	670 29-05-2024	157 14-09-2022	-	7 23-03-2023
NER	3712 04-09-2023	3544 24-05-2024	72 15-09-2023	41 29-08-2023	-	2 22-06-2022
All India	250070 30-05-2024	227354 29-05-2024	5466 30-05-2024	877 30-08-2022	619 28-05-2024	426 30-04-2024

Regions	States	Max. Demand Met during the day (MW)		Energy Consumption (MU)	
			As on date		As on date
NR	Punjab	15256	24-06-2023	343.5	24-06-2023
	Haryana	12692	18-08-2023	273.1	18-08-2023
	Rajasthan	17949	20-01-2024	379.1	30-05-2024
	Delhi	8188	29-05-2024	163.8	31-05-2024
	UP	29219	31-05-2024	642.3	27-05-2024
	Uttarakhand	2781	29-05-2024	60.7	31-05-2024
	HP	2235	20-01-2024	40.7	22-08-2023
	J&K(UT) and Ladakh(UT)	3636	29-06-2023	66.8	26-01-2024
	Chandigarh	432	30-05-2024	8.6	30-05-2024
	Railways_NR_ISTS	-	-	-	-
WR	Chhattisgarh	6347	05-04-2024	144.4	06-04-2024
	Gujarat	24922	25-05-2024	511.6	02-09-2023
	MP	17819	29-12-2023	335.9	10-11-2023
	Maharashtra	28969	07-02-2024	648.3	30-04-2024
	Goa	803	19-04-2024	17.6	18-05-2024
	DD & DNH	1348	30-05-2024	31.2	30-05-2024
	AMNSIL	1083	10-01-2024	21.0	31-05-2022
	Balco	-	-	-	-
SR	Andhra Pradesh*	13712	04-05-2024	263.8	16-06-2023
	Telangana*	15623	08-03-2024	311.0	13-03-2024
	Karnataka	17220	12-03-2024	336.9	05-04-2024
	Kerala	5797	02-05-2024	116.1	03-05-2024
	Tamil Nadu	20830	02-05-2024	443.6	30-04-2024
	Pondy	530	31-05-2024	11.8	31-05-2024
ER	Bihar	7578	27-07-2023	161.3	28-07-2023
	DVC	3751	18-04-2023	81.2	22-04-2022
	Jharkhand	2324	29-05-2024	50.1	29-05-2024
	Odisha	7104	13-07-2023	148.5	19-04-2024
	West Bengal	12756	30-04-2024	264.0	30-04-2024
	Sikkim	137	03-02-2022	2.5	28-01-2020
	Railways_ER_ISTS	-	-	-	-
NER	Arunachal Pradesh	198	26-05-2024	3.4	25-03-2024
	Assam	2498	16-09-2023	50.0	15-09-2023
	Manipur	258	31-01-2022	3.7	10-02-2022
	Meghalaya	415	09-02-2022	7.8	31-01-2022
	Mizoram	162	27-01-2024	2.6	02-03-2024
	Nagaland	179	28-09-2023	3.29	05-09-2023
	Tripura	386	04-05-2024	7.0	06-06-2023
*SR	Andhra Pradesh (Undivided)	13162	23-03-2014	284.8	22-03-2014

25. System Reliability Indices Report for the month of MAY 2024

Percentage (%) of times ATC was violated

S.No.	Corridor	Number of Blocks Violated	Number of Hours Violated	%Violation
1	WR-NR	207	51.75	6.96
2	ER-NR	0	0.00	0.00
3	Import of NR	11	2.75	0.37
4	NEW-SR	0	0.00	0.00
5	NER Import	0	0.00	0.00

Percentage(%) of times (N-1) Criteria was violated

S.No.	Corridor	Number of Blocks Violated	Number of Hours Violated	%Violation
1	WR-NR	130	32.50	4.37
2	ER-NR	0	0.00	0.00
3	Import of NR	0	0.00	0.00
4	NEW-SR	0	0.00	0.00
4	NER Import	38	9.50	1.28

Remarks: Flows crossing Total Transfer Capability (TTC) on interregional corridors has been worked out as a proxy for (N-1) violation.



POWER SYSTEM OPERATION CORPORATION LIMITED

पॉवर सिस्टम ऑपरेशन कारपोरेशन लिमिटेड

NATIONAL LOAD DESPATCH CENTRE – राष्ट्रीय भार प्रेषण केन्द्र

First floor, B-9, Qutab Institutional Area,
Katwaria Sarai, New Delhi-110016.